S-125

Marine Aids to Navigation (AtoN) — Annex A: Data Classification and Encoding Guide

Edition 1.0.0 – April 2025

IHO

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Organization

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1 Metadata Features

1.1 Data Coverage

Table 1-1 — Data Coverage

IHO Definition: A geographical area that describes the coverage and extent of spatial objects.				
S-10x Metadata Feature: Data Coverage (M_COVR, M_CSCL)				
Super Type:				
Primitives: surface				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Maximum Display Scale			IN	1, 1
Minimum Display Scale			IN	1, 1
INT 1 Reference: Remarks: Distinction:				

1.2 Local Direction of Buoyage

Table 1-2 — Local Direction of Buoyage

IHO Definition: An area within which the navigational system of marks has been established in relation to a specific direction.				
S-10x Metadata Feature: Local Direction of Buoyage (M_NSYS)				
Super Type:				
Primitives: surface				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Orientation			C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value	(ORIENT)		(S) RE	1, 1
INT 1 Reference: Remarks: Distinction:				

1.3 Navigational System of Marks

Table 1-3 — Navigational System of Marks

IHO Definition: An area within which the navigational system of marks has been established in relation to a specific direction.				
S-10x Metadata Feature: Navigational System of Marks (M_NSYS)				
Super Type:				
Primitives: surface				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Marks Navigational — System Of	(MARSYS)	1 : IALA A 2 : IALA B 9 : No System 10 : Other System 11 : CEVNI 12 : Russian Inland Waterway Regulations 13 : Brazilian National Inland Waterway Regulations — Two Sides 15 : Paraguay-Parana Waterway — Brazilian Complementary Aids	EN	1, 1
INT 1 Reference: Remarks: Distinction:				

1.4 Sounding Datum

Table 1-4 — Sounding Datum

IHO Definition: The horizontal plane or tidal datum to which soundings have been reduced. Also called datum for sounding reduction.				
S-10x Metadata Feature: Sounding Datum (M_SDAT)				
Super Type:				
Primitives: surface				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs	EN	1, 1

	(Datum for Heights)	7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GlW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level			

		41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : 47 : 48 : 49 :		
INT 1 Reference: Remarks: Distinction:				

1.5 Vertical Datum of Data

Table 1-5 — Vertical Datum of Data

IHO Definition: Any level surface (for example Mean Sea Level) taken as a surface of reference to which the elevations within a data set are reduced. Also called datum level, reference level, reference plane, levelling datum, datum for heights.				
S-10x Metadata Feature: Vertical Datum of Data (M_VDAT)				
Super Type:				
Primitives: surface				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water	EN	1, 1

		13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GlW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level			

		40 : Russian Normal Backwater Level 41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : 47 : 48 : 49 :		
INT 1 Reference: Remarks: Distinction:				

2 Geo Features

2.1 Aids to Navigation

Table 2-1 — Aids to Navigation

IHO Definition: A visual, acoustical, or radio device, external to a ship, designed to assist in determining a safe course or a vessel's position, or to warn of dangers and/or obstructions. Aids to navigation usually include buoys, beacons, fog signals, lights, radio beacons, leading marks, radio position fixing systems and GNSS which are chart-related and assist safe navigation.				
S-10x Geo Feature: Aids to Navigation				
Super Type:				
Primitives: noGeometry				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
ID Code	(Identification Number) (Identification Code)		TE	0, 1
Interoperability Identifier			UN	0, 1
Information	(INFORM)		C	0, *
File Locator			(S) TE	0, 1
File Reference	(TXTDSC) (NTXTDS)		(S) TE	0, 1
Headline			(S) TE	0, 1
Language			(S) TE	1, 1

Text	(INFORM) (NINFOM)		(S) TE	0, 1
Feature Name			C	0, *
Display Name			(S) BO	0, 1
Language			(S) TE	0, 1
Name	(OBJNAM)		(S) TE	1, 1
Scale Minimum	(SCAMIN)		IN	0, 1
Source Date	(SORDAT)		DA	0, 1
Source			TE	0, 1
Pictorial Representation	(PICREP)		TE	0, 1
AtoN Number			UI	1, 1
Installation Date			DA	0, 1
Fixed Date Range			C	0, 1
Date End	(DATEND)		(S) TD	0, 1
Date Start	(DATSTA)		(S) TD	0, 1
Periodic Date Range			C	0, 1
Date End	(DATEND)		(S) TD	1, 1
Date Start	(DATSTA)		(S) TD	1, 1
Seasonal Action Required			TE	0, *
INT 1 Reference: Remarks: Distinction:				

Feature/Information associations				
Type	**Association Name**	**Association Ends**		
		Class	**Role**	**Mult**
association	Aton Status	**AtonStatusInformation**	Statuspart	0, 1
Asso	Aton Aggregations	**AtonAggregation**	peer	0, *
Asso	Aton Associations	**AtonAssociation**	peer	0, *

2.2 Structure Object

Table 2-2 — Structure Object

IHO Definition: Something (such as a house, tower, bridge, etc.) that is built by putting parts together and that usually stands on its own.
S-10x Geo Feature: Structure Object

<table>
<tr><td colspan="5"><u>Super Type:</u> AidsToNavigation</td></tr>
<tr><td colspan="5"><u>Primitives:</u> noGeometry</td></tr>
<tr><td>Real World</td><td colspan="2">Paper Chart Symbol</td><td colspan="2">ECDIS Symbol</td></tr>
<tr><td>S-10x Attribute</td><td>S-57 Acronym</td><td>Allowable Encoding Value</td><td>Type</td><td>Multiplicity</td></tr>
<tr><td colspan="5"><u>INT 1 Reference:</u>
<u>Remarks:</u>
<u>Distinction:</u></td></tr>
</table>

<table>
<tr><td colspan="5"><u>Feature/Information associations</u></td></tr>
<tr><td rowspan="2">Type</td><td rowspan="2">Association Name</td><td colspan="3">Association Ends</td></tr>
<tr><td>Class</td><td>Role</td><td>Mult</td></tr>
<tr><td>Asso</td><td>Structure Equipment</td><td>Equipment</td><td>child</td><td>0, *</td></tr>
</table>

2.3 Equipment

Table 2-3 — Equipment

<table>
<tr><td colspan="5"><u>IHO Definition:</u> The implements used in an operation or activity.</td></tr>
<tr><td colspan="5"><u>S-10x Geo Feature:</u> Equipment</td></tr>
<tr><td colspan="5"><u>Super Type:</u> AidsToNavigation</td></tr>
<tr><td colspan="5"><u>Primitives:</u> noGeometry</td></tr>
<tr><td>Real World</td><td colspan="2">Paper Chart Symbol</td><td colspan="2">ECDIS Symbol</td></tr>
<tr><td>S-10x Attribute</td><td>S-57 Acronym</td><td>Allowable Encoding Value</td><td>Type</td><td>Multiplicity</td></tr>
<tr><td colspan="5"><u>INT 1 Reference:</u>
<u>Remarks:</u>
<u>Distinction:</u></td></tr>
</table>

<table>
<tr><td colspan="5"><u>Feature/Information associations</u></td></tr>
<tr><td rowspan="2">Type</td><td rowspan="2">Association Name</td><td colspan="3">Association Ends</td></tr>
<tr><td>Class</td><td>Role</td><td>Mult</td></tr>
<tr><td>Asso</td><td>Structure Equipment</td><td>StructureObject</td><td>parent</td><td>1, 1</td></tr>
</table>

2.4 Generic Beacon

Table 2-4 — Generic Beacon

<table>
<tr><td><u>IHO Definition:</u> A fixed artificial navigation mark that can be recognized by its shape, colour, pattern, topmark or light character, or a combination of these. It may carry various additional aids to navigation.</td></tr>
<tr><td><u>S-10x Geo Feature:</u> Generic Beacon</td></tr>
</table>

Super Type: StructureObject				
Primitives: noGeometry				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Beacon Shape	(BCNSHP)	1 : Stake, Pole, Perch, Post 2 : Withy 3 : Beacon Tower 4 : Lattice Beacon 5 : Pile Beacon 6 : Cairn 7 : Buoyant Beacon	EN	1, 1
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	1, * (ordered)
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Height	(HEIGHT)		RE	0, 1
Marks Navigational — System Of	(MARSYS)	1 : IALA A 2 : IALA B 9 : No System 10 : Other System 11 : CEVNI	EN	0, 1

		12 : Russian Inland Waterway Regulations 13 : Brazilian National Inland Waterway Regulations — Two Sides 14 : Brazilian National Inland Waterway Regulations — Side Independent 15 : Paraguay-Parana Waterway — Brazilian Complementary Aids		
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request	EN	0, *

		20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Length	(VERLEN)		RE	0, 1
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	0, 1
Vertical Accuracy			RE	0, 1
INT 1 Reference: Remarks: Distinction:				

2.5 Generic Buoy

Table 2-5 — Generic Buoy

IHO Definition: A floating object moored to the bottom in a particular (charted) place, as an aid to navigation or for other specific purposes.
S-10x Geo Feature: Generic Buoy
Super Type: StructureObject
Primitives: noGeometry

Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Buoy Shape	(BOYSHP)	1 : Conical 2 : Can 3 : Spherical 4 : Pillar 5 : Spar 6 : Barrel 7 : Superbuoy 8 : Ice Buoy	EN	1, 1
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	1, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Marks Navigational — System Of	(MARSYS)	1 : IALA A 2 : IALA B 9 : No System 10 : Other System 11 : CEVNI 12 : Russian Inland Waterway Regulations 13 : Brazilian National Inland Waterway Regulations — Two Sides	EN	0, 1

		14 : Brazilian National Inland Waterway Regulations — Side Independent 15 : Paraguay-Parana Waterway — Brazilian Complementary Aids		
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing	EN	0, *

		24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Type of Buoy			TE	0, 1
INT 1 Reference: Remarks: Distinction:				

Feature/Information associations				
Type	**Association Name**	**Association Ends**		
		Class	**Role**	**Mult**
Asso	Structure Equipment	**Topmark**	equipment	0, *

2.6 Generic Light

Table 2-6 — Generic Light

IHO Definition: -				
S-10x Geo Feature: Generic Light				
Super Type: Equipment				
Primitives: noGeometry				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Colour	(COLOUR)	1 : White	EN	1, *

		2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink		
Height	(HEIGHT)		RE	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational	EN	0, *

		31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide	EN	0, 1

		31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GlW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level 41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW)		
Vertical Length	(VERLEN)		RE	0, 1
Candela			RE	0, 1
INT 1 Reference: Remarks: Distinction:				

2.7 Landmark

Table 2-7 — Landmark

IHO Definition: A prominent object at a fixed location on land which can be used in determining a location or a direction.				
S-10x Geo Feature: Landmark (LNDMRK)				
Super Type: StructureObject				
Primitives: point, curve, surface				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Landmark	(CATLMK)	1 : Cairn 2 : Cemetery 3 : Chimney 4 : Dish Aerial 5 : Flagstaff 6 : Flare Stack 7 : Mast	EN	1, *

		8 : Windsock 9 : Monument 10 : Column/Pillar 11 : Memorial Plaque 12 : Obelisk 13 : Statue 14 : Cross 15 : Dome 16 : Radar Scanner 17 : Tower 18 : Windmill 19 : Windmotor 20 : Spire/Minaret 21 : Large Rock or Boulder on Land 22 : Triangulation Mark 23 : Boundary Mark 24 : Observation Wheel 25 : Torii 26 : Bridge 27 : Dam		
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	0, * (ordered)
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle	EN	0, *

		9 : Triangle		
Function	(FUNCTN)	2 : Harbour-Masters Office 3 : Customs Office 4 : Health Office 5 : Hospital 6 : Post Office 7 : Hotel 8 : Railway Station 9 : Police Station 10 : Water-Police Station 11 : Pilot Office 12 : Pilot Lookout 13 : Bank Office 14 : Headquarters for District Control 15 : Transit Shed/Warehouse 16 : Factory 17 : Power Station 18 : Administrative 19 : Educational Facility 20 : Church 21 : Chapel 22 : Temple 23 : Pagoda 24 : Shinto Shrine 25 : Buddhist Temple 26 : Mosque 27 : Marabout 28 : Lookout 29 : Communication 30 : Television 31 : Radio 32 : Radar 33 : Light Support 34 : Microwave 35 : Cooling 36 : Observation 37 : Timeball 38 : Clock 39 : Control 40 : Airship Mooring 41 : Stadium 42 : Bus Station	EN	0, *

		43 : Passenger Terminal Building 44 : Sea Rescue Control 45 : Observatory 46 : Ore Crusher 47 : Boathouse 48 : Pumping Station		
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising	EN	0, *

		22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	1, 1
Height	(HEIGHT)		RE	0, 1
INT 1 Reference: Remarks: Distinction:				

2.8 Lateral Beacon

Table 2-8 — Lateral Beacon

IHO Definition: A beacon is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey (IHO Dictionary, S-32, 5th Edition, 420). A lateral beacon is used to indicate the port or starboard hand side of the route to be followed. They are generally used for well defined channels and are used in conjunction with a conventional direction of buoyage. (UKHO NP 735, 5th Edition)		
S-10x Geo Feature: Lateral Beacon		
Super Type: GenericBeacon		
Primitives: point		
Real World	*Paper Chart Symbol*	*ECDIS Symbol*

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Lateral Mark	(CATLAM)	1 : Port-Hand Lateral Mark 2 : Starboard-Hand Lateral Mark 3 : Preferred Channel to Starboard Lateral Mark 4 : Preferred Channel to Port Lateral Mark 5 : Right-Hand Side of the Waterway 6 : Left-Hand Side of the Waterway 7 : Right-Hand Side of the Channel 8 : Left-Hand Side of the Channel 9 : Bifurcation of the Waterway 10 : Bifurcation of the Channel 11 : Channel Near the Right Bank 12 : Channel Near the Left Bank 13 : Channel Cross-Over to the Right Bank 14 : Channel Cross-Over to the Left Bank 15 : Danger Point or Obstacles at the Right-Hand Side 16 : Danger Point or Obstacles at the Left-Hand Side 17 : Turn Off at the Right-Hand Side 18 : Turn Off at the Left-Hand Side 19 : Junction at the Right-Hand Side 20 : Junction at the Left-Hand Side 21 : Harbour Entry at the Right-Hand Side 22 : Harbour Entry at the Left-Hand Side	EN	1, 1

		23 : Bridge Pier Mark 24 : Entry From a Lake to a Narrower Waterway, Right Bank 25 : Entry From a Lake to a Narrower Waterway, Left Bank 26 : Change Bank 27 : Continue Along Bank		
INT 1 Reference: Remarks: Distinction:				

2.9 Lateral Buoy

Table 2-9 — Lateral Buoy

IHO Definition: A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary, S-32, 5th Edition, 565). A lateral buoy is used to indicate the port or starboard hand side of the route to be followed. They are generally used for well defined channels and are used in conjunction with a conventional direction of buoyage. (UKHO NP 735, 5th Edition)				
S-10x Geo Feature: Lateral Buoy				
Super Type: GenericBuoy				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Lateral Mark	(CATLAM)	1 : Port-Hand Lateral Mark 2 : Starboard-Hand Lateral Mark 3 : Preferred Channel to Starboard Lateral Mark 4 : Preferred Channel to Port Lateral Mark 5 : Right-Hand Side of the Waterway 6 : Left-Hand Side of the Waterway 7 : Right-Hand Side of the Channel 8 : Left-Hand Side of the Channel 9 : Bifurcation of the Waterway	EN	1, 1

		10 : Bifurcation of the Channel 11 : Channel Near the Right Bank 12 : Channel Near the Left Bank 13 : Channel Cross-Over to the Right Bank 14 : Channel Cross-Over to the Left Bank 15 : Danger Point or Obstacles at the Right-Hand Side 16 : Danger Point or Obstacles at the Left-Hand Side 17 : Turn Off at the Right-Hand Side 18 : Turn Off at the Left-Hand Side 19 : Junction at the Right-Hand Side 20 : Junction at the Left-Hand Side 21 : Harbour Entry at the Right-Hand Side 22 : Harbour Entry at the Left-Hand Side 23 : Bridge Pier Mark 24 : Entry From a Lake to a Narrower Waterway, Right Bank 25 : Entry From a Lake to a Narrower Waterway, Left Bank 26 : Change Bank 27 : Continue Along Bank			
<u>INT 1 Reference:</u> <u>Remarks:</u> <u>Distinction:</u>					

2.10 Navigation Line

Table 2-10 — Navigation Line

<u>IHO Definition:</u> A straight line extending towards an area of navigational interest and generally generated by two navigational aids or one navigational aid and a bearing.

S-10x Geo Feature: Navigation Line (NAVLNE)				
Super Type: AidsToNavigation				
Primitives: curve				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Navigation Line	(CATNAV)	1 : Clearing Line 2 : Transit Line 3 : Leading Line Bearing a Recommended Track	EN	1, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued	EN	0, *

		35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Orientation			C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value	(ORIENT)		(S) RE	1, 1
INT 1 Reference: Remarks: Distinction:				

Feature/Information associations				
Type	**Association Name**	**Association Ends**		
		Class	**Role**	**Mult**
Asso	Range System	**RecommendedTrack**	navigableTrack	0, *

2.11 Recommended Track

Table 2-11 — Recommended Track

IHO Definition: A route which has been specially examined to ensure so far as possible that it is free of dangers and along which ships are advised to navigate.				
S-10x Geo Feature: Recommended Track (RECTRC)				
Super Type: AidsToNavigation				
Primitives: curve				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Based On Fixed Marks	(CATTRK=1)		BO	1, 1
Depth Range Minimum Value	(DRVAL1)		RE	0, 1
Maximal Permitted Draught	(lg_drt)		RE	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended	EN	0, *

		4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water	EN	0, 1

	(Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GlW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission			

		37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level 41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW)		
Orientation			C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value	(ORIENT)		(S) RE	1, 1
Vertical Uncertainty	(VERACC) (SOUACC)		C	0, 1
Uncertainty Fixed	(POSACC) (SOUACC) (VERACC)		(S) RE	1, 1
Uncertainty Variable Factor			(S) RE	0, 1
Quality of Vertical Measurement	(QUASOU)	1 : Depth Known 2 : Depth or Least Depth Unknown 3 : Doubtful Sounding 4 : Unreliable Sounding 5 : No Bottom Found at Value Shown 6 : Least Depth Known 7 : Least Depth Unknown, Safe Clearance at Value Shown 8 : Value Reported (Not Surveyed) 9 : Value Reported (Not Confirmed) 10 : Maintained Depth 11 : Not Regularly Maintained	EN	0, *
Technique of Vertical Measurement	(TECSOU)	1 : Found by Echo Sounder 2 : Found by Side Scan Sonar 3 : Found by Multi Beam 4 : Found by Diver	EN	0, *

		5 : Found by Lead Line 6 : Swept by Wire-Drag 7 : Found by Laser 8 : Swept by Vertical Acoustic System 9 : Found by Electromagnetic Sensor 10 : Photogrammetry 11 : Satellite Imagery 12 : Found by Levelling 13 : Swept by Side Scan Sonar 14 : Computer Generated 15 : Found by LIDAR 16 : Synthetic Aperture Radar 17 : Hyperspectral Imagery		
Traffic Flow	(TRAFIC)	1 : Inbound 2 : Outbound 3 : One-Way 4 : Two-Way	EN	1, 1
INT 1 Reference: Remarks: Distinction:				

Feature/Information associations				
Type	**Association Name**	**Association Ends**		
		Class	**Role**	**Mult**
Asso	Range System	**NavigationLine**	navigationLine	1, *

2.12 Light Sectored

Table 2-12 — Light Sectored

IHO Definition: A light presenting different appearances (in particular, different colours) over various parts of the horizon of interest to maritime navigation.				
S-10x Geo Feature: Light Sectored (LIGHTS, Sector Light)				
Super Type: GenericLight				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Light	(CATLIT)	1 : Directional Function 4 : Leading Light 5 : Aero Light	EN	0, *

		6 : Air Obstruction Light 8 : Flood Light 9 : Strip Light 10 : Subsidiary Light 11 : Spotlight 12 : Front 13 : Rear 14 : Lower 15 : Upper 17 : Emergency 18 : Bearing Light 19 : Horizontally Disposed 20 : Vertically Disposed		
Exhibition Condition of Light	(EXCLIT)	1 : Light Shown Without Change of Character 2 : Daytime Light 3 : Fog Light 4 : Night Light	EN	0, 1
Marks Navigational — System Of	(MARSYS)	1 : IALA A 2 : IALA B 9 : No System 10 : Other System 11 : CEVNI 12 : Russian Inland Waterway Regulations 13 : Brazilian National Inland Waterway Regulations — Two Sides 14 : Brazilian National Inland Waterway Regulations — Side Independent 15 : Paraguay-Parana Waterway — Brazilian Complementary Aids	EN	0, 1
Signal Generation	(SIGGEN)	1 : Automatically 2 : By Wave Action 3 : By Hand 4 : By Wind 5 : Radio Activated 6 : Call Activated	EN	0, 1
Obscured Sector			C	0, *
Sector Limit			(S) C	1, 1
Sector Limit One	(SECTR1)		(S) C	1, 1
Sector Bearing	(SECTR1)		(S) RE	1, 1

	(SECTR2)			
Sector Line Length			(S) IN	0, 1
Sector Limit Two	(SECTR2)		(S) C	1, 1
Sector Bearing	(SECTR1) (SECTR2)		(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Sector Information			(S) C	0, 1
Language			(S) TE	0, 1
Text	(INFORM) (NINFOM)		(S) TE	1, 1
Sector Characteristics			C	1, *
Light Characteristic	(LITCHR) (Character of Light)	1 : Fixed 2 : Flashing 3 : Long-Flashing 4 : Quick-Flashing 5 : Very Quick-Flashing 6 : Continuous Ultra Quick-Flashing 7 : Isophased 8 : Occulting 9 : Interrupted Quick Flashing 10 : Interrupted Very Quick Flashing 11 : Interrupted Ultra Quick-Flashing 12 : Morse 13 : Fixed and Flash 14 : Flash and Long-Flash 15 : Occulting and Flash 16 : Fixed and Long-Flash 17 : Occulting Alternating 18 : Long-Flash Alternating 19 : Flash Alternating 20 : Group Alternating 25 : Quick-Flash Plus Long-Flash 26 : Very Quick-Flash Plus Long-Flash 27 : Ultra Quick-Flash Plus Long-Flash 28 : Alternating 29 : Fixed and Alternating Flashing	(S) EN	1, 1

Light Sector			(S) C	1, 1
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	(S) EN	1, * (ordered)
Directional Character			(S) C	0, 1
Moire Effect			(S) BO	0, 1
Orientation			(S) C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value	(ORIENT)		(S) RE	1, 1
Light Visibility	(LITVIS)	1 : High Intensity 2 : Low Intensity 3 : Faint 4 : Intensified 5 : Unintensified 6 : Visibility Deliberately Restricted 7 : Obscured 8 : Partially Obscured 9 : Visible in Line of Range	(S) EN	0, *
Sector Limit			(S) C	0, 1
Sector Limit One	(SECTR1))		(S) C	1, 1
Sector Bearing	(SECTR1) (SECTR2)		(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Sector Limit Two	(SECTR2)		(S) C	1, 1
Sector Bearing	(SECTR1) (SECTR2)		(S) RE	1, 1
Sector Line Length			(S) IN	0, 1

Value of Nominal Range	(VALNMR)		(S) RE	0, 1
Sector Information			(S) C	0, *
Language			(S) TE	0, 1
Text	(INFORM) (NINFOM)		(S) TE	1, 1
Sector Extension			(S) BO	0, 1
Signal Group	(SIGGRP)		(S) TE	0, 1 (ordered)
Signal Period	(SIGPER)		(S) RE	0, 1
Signal Sequence	(SIGSEQ)		(S) C	0, 1 (ordered)
Signal Duration			(S) RE	1, 1
Signal Status		1 : Lit/Sound 2 : Eclipsed/Silent	(S) EN	1, 1
INT 1 Reference: Remarks: Distinction:				

2.13 Light All Around

Table 2-13 — Light All Around

IHO Definition: An all around light is a light that is visible over the whole horizon of interest to marine navigation and having no change in the characteristics of the light.				
S-10x Geo Feature: Light All Around (LIGHTS)				
Super Type: GenericLight				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Light	(CATLIT)	1 : Directional Function 4 : Leading Light 5 : Aero Light 6 : Air Obstruction Light 8 : Flood Light 9 : Strip Light 10 : Subsidiary Light 11 : Spotlight 12 : Front 13 : Rear 14 : Lower 15 : Upper 17 : Emergency	EN	0, *

		18 : Bearing Light 19 : Horizontally Disposed 20 : Vertically Disposed		
Exhibition Condition of Light	(EXCLIT)	1 : Light Shown Without Change of Character 2 : Daytime Light 3 : Fog Light 4 : Night Light	EN	0, 1
Light Visibility	(LITVIS)	1 : High Intensity 2 : Low Intensity 3 : Faint 4 : Intensified 5 : Unintensified 6 : Visibility Deliberately Restricted 7 : Obscured 8 : Partially Obscured 9 : Visible in Line of Range	EN	0, 1
Major Light			BO	0, 1
Marks Navigational — System Of	(MARSYS)	1 : IALA A 2 : IALA B 9 : No System 10 : Other System 11 : CEVNI 12 : Russian Inland Waterway Regulations 13 : Brazilian National Inland Waterway Regulations — Two Sides 14 : Brazilian National Inland Waterway Regulations — Side Independent 15 : Paraguay-Parana Waterway — Brazilian Complementary Aids	EN	0, 1
Signal Generation	(SIGGEN)	1 : Automatically 2 : By Wave Action 3 : By Hand 4 : By Wind 5 : Radio Activated 6 : Call Activated	EN	0, 1
Value of Geographic Range			RE	0, 1
Value of Luminous Range			RE	0, 1

Value of Nominal Range	(VALNMR)		RE	0, 1
Multiplicity of Features			C	0, 1
Multiplicity Known			(S) BO	1, 1
Number of Features			(S) IN	0, 1
<u>INT 1 Reference:</u> <u>Remarks:</u> <u>Distinction:</u>				

2.14 Light Air Obstruction

Table 2-14 — Light Air Obstruction

<u>IHO Definition:</u> An air obstruction light is a light marking an obstacle which constitutes a danger to air navigation.				
<u>S-10x Geo Feature:</u> Light Air Obstruction (LIGHTS)				
<u>Super Type:</u> GenericLight				
<u>Primitives:</u> point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Exhibition Condition of Light	(EXCLIT)	1 : Light Shown Without Change of Character 2 : Daytime Light 3 : Fog Light 4 : Night Light	EN	0, 1
Light Visibility	(LITVIS)	1 : High Intensity 2 : Low Intensity 3 : Faint 4 : Intensified 5 : Unintensified 6 : Visibility Deliberately Restricted 7 : Obscured 8 : Partially Obscured 9 : Visible in Line of Range	EN	0, *
Value of Geographic Range			RE	0, 1
Value of Luminous Range			RE	0, 1
Value of Nominal Range	(VALNMR)		RE	0, 1

Multiplicity of Features			C	0, 1
Multiplicity Known			(S) BO	1, 1
Number of Features			(S) IN	0, 1
INT 1 Reference: Remarks: Distinction:				

2.15 Light Fog Detector

Table 2-15 — Light Fog Detector

IHO Definition: A fog detector light is a light used to automatically determine conditions of visibility which warrant the turning on or off of a sound signal.				
S-10x Geo Feature: Light Fog Detector				
Super Type: GenericLight				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Signal Generation	(SIGGEN)	1 : Automatically 2 : By Wave Action 3 : By Hand 4 : By Wind 5 : Radio Activated 6 : Call Activated	EN	0, 1
INT 1 Reference: Remarks: Distinction:				

2.16 Radar Reflector

Table 2-16 — Radar Reflector

IHO Definition: A device capable of, or intended for, reflecting radar signals.				
S-10x Geo Feature: Radar Reflector (RADRFL)				
Super Type: Equipment				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Height	(HEIGHT)		RE	0, 1
Status	(STATUS)	1 : Permanent	EN	0, *

		2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		

INT 1 Reference:

Remarks:

Distinction:

2.17 Fog Signal

Table 2-17 — Fog Signal

IHO Definition: A warning signal transmitted by a vessel, or aid to navigation, during periods of low visibility. Also, the device producing such a signal.				
S-10x Geo Feature: Fog Signal (FOGSIG)				
Super Type: Equipment				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Signal Sequence	(SIGSEQ)		C	0, 1 (ordered)
Signal Duration			(S) RE	1, 1
Signal Status		1 : Lit/Sound 2 : Eclipsed/Silent	(S) EN	1, 1
Category of Fog Signal	(CATFOG)	1 : Explosive 2 : Diaphone 3 : Siren 4 : Nautophone 5 : Reed 6 : Tyfon 7 : Bell 8 : Whistle 9 : Gong 10 : Horn	EN	1, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized	EN	0, *

		16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
INT 1 Reference: Remarks: Distinction:				

2.18 Radio Station

Table 2-18 — Radio Station

IHO Definition: A place equipped to transmit radio waves. Such a station may be either stationary or mobile, and may also be provided with a radio receiver.
S-10x Geo Feature: Radio Station (RDOSTA, W/T Station)
Super Type: Equipment
Primitives: point

Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Radio Station	(CATROS)	1 : Circular (Non-Directional) Marine or Aero-Marine Radiobeacon 2 : Directional Radiobeacon 3 : Rotating Pattern Radiobeacon 4 : Consol Beacon 5 : Radio Direction-Finding Station 6 : Coast Radio Station Providing QTG Service 7 : Aeronautical Radiobeacon 8 : Decca 9 : Loran C 10 : Differential GNSS 11 : Toran 12 : Omega 13 : Syledis 14 : Chaika 19 : Radio Telephone Station 20 : AIS Base Station	EN	1, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing	EN	0, 1

		24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
INT 1 Reference: Remarks: Distinction:				

Feature/Information associations				
Type	**Association Name**	**Association Ends**		
		Class	**Role**	**Mult**
Asso	Virtual AIS	**PhysicalAISAidToNavigation, SyntheticAISAidToNavigation, VirtualAISAidToNavigation**	broadcastBy	0, *

2.19 Daymark

Table 2-19 — Daymark

IHO Definition: 1) The identifying characteristics of an aid to navigation which serve to facilitate its recognition against a daylight viewing background. On those structures that do not by themselves present an adequate viewing area to be seen at the required distance, the aid is made more visible by affixing a daymark to the structure. A daymark so affixed has a distinctive colour and shape depending on the purpose of the aid. 2) An unlighted navigational mark.		
S-10x Geo Feature: Daymark (DAYMAR)		
Super Type: Equipment		
Primitives: point		
Real World	*Paper Chart Symbol*	*ECDIS Symbol*

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Special Purpose Mark	(CATSPM)	1 : Firing Danger Mark 2 : Target Mark 3 : Marker Ship Mark 4 : Degaussing Range Mark 5 : Barge Mark 6 : Cable Mark 7 : Spoil Ground Mark 8 : Outfall Mark 9 : ODAS 10 : Recording Mark 11 : Seaplane Anchorage Mark 12 : Recreation Zone Mark 13 : Private Mark 14 : Mooring Mark 15 : LANBY 16 : Leading Mark 17 : Measured Distance Mark 18 : Notice Mark 19 : TSS Mark 20 : Anchoring Prohibited Mark 21 : Berthing Prohibited Mark 22 : Overtaking Prohibited Mark 23 : Two-Way Traffic Prohibited Mark 24 : Reduced Wake Mark 25 : Speed Limit Mark 26 : Stop Mark 27 : General Warning Mark 28 : Sound Ship's Siren Mark 29 : Restricted Vertical Clearance Mark 30 : Maximum Vessel's Draught Mark 31 : Restricted Horizontal Clearance Mark	EN	0, 1

		32 : Strong Current Warning Mark 33 : Berthing Permitted Mark 34 : Overhead Power Cable Mark 35 : Channel Edge Gradient Mark 36 : Telephone Mark 37 : Ferry Crossing Mark 39 : Pipeline Mark 40 : Anchorage Mark 41 : Clearing Mark 42 : Control Mark 43 : Diving Mark 44 : Refuge Beacon 45 : Foul Ground Mark 46 : Yachting Mark 47 : Heliport Mark 48 : GNSS Mark 49 : Seaplane Landing Mark 50 : Entry Prohibited Mark 51 : Work in Progress Mark 52 : Mark With Unknown Purpose 53 : Wellhead Mark 54 : Channel Separation Mark 55 : Marine Farm Mark 56 : Artificial Reef Mark 57 : Ice Mark 58 : Nature Reserve Mark 59 : Fish Aggregating Device 60 : Wreck Mark 61 : Customs Mark 62 : Causeway Mark 63 : Wave Recorder 64 : Jetski Prohibited		
Colour	(COLOUR)	1 : White 2 : Black 3 : Red	EN	1, *

		4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink		
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Height	(HEIGHT)		RE	0, 1
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Orientation Value	(ORIENT)		RE	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent	EN	0, *

		6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Topmark/Daymark Shape	(TOPSHP)	1 : Cone (Point Up) 2 : Cone (Point Down) 3 : Sphere 4 : 2 Spheres	EN	1, 1

		5 : Cylinder 6 : Board 7 : X-Shaped 8 : Upright Cross 9 : Cube (Point Up) 10 : 2 Cones (Point to Point) 11 : 2 Cones (Base to Base) 12 : Rhombus 13 : 2 Cones (Points Upward) 14 : 2 Cones (Points Downward) 15 : Besom (Point Up) 16 : Besom (Point Down) 17 : Flag 18 : Sphere Over a Rhombus 19 : Square 20 : Rectangle (Horizontal) 21 : Rectangle (Vertical) 22 : Trapezium (Up) 23 : Trapezium (Down) 24 : Triangle (Point Up) 25 : Triangle (Point Down) 26 : Circle 27 : Two Upright Crosses (One Over the Other) 28 : T-Shape 29 : Triangle Pointing Up Over a Circle 30 : Upright Cross Over a Circle 31 : Rhombus Over a Circle 32 : Circle Over a Triangle Pointing Up 33 : Other Shape (See Shape Information) 34 : Tubular		
IsSlatted			BO	1, 1
<u>INT 1 Reference:</u>				

Remarks:
Distinction:

2.20 Retroreflector

Table 2-20 — Retroreflector

IHO Definition: A means of distinguishing unlighted marks at night. Retro-reflective material is secured to the mark in a particular pattern to reflect back light.				
S-10x Geo Feature: Retroreflector (RETRFL)				
Super Type: Equipment				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	0, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Marks Navigational — System Of	(MARSYS)	1 : IALA A 2 : IALA B 9 : No System 10 : Other System 11 : CEVNI 12 : Russian Inland Waterway Regulations	EN	0, 1

		13 : Brazilian National Inland Waterway Regulations — Two Sides 14 : Brazilian National Inland Waterway Regulations — Side Independent 15 : Paraguay-Parana Waterway — Brazilian Complementary Aids		
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed	EN	0, *

		38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
INT 1 Reference: Remarks: Distinction:				

2.21 Radar Transponder Beacon

Table 2-21 — Radar Transponder Beacon

IHO Definition: A transponder beacon transmitting a coded signal on radar frequency, permitting an interrogating craft to determine the bearing and range of the transponder.				
S-10x Geo Feature: Radar Transponder Beacon (RTPBCN, Radar Beacon, RACON)				
Super Type: Equipment				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Radar Transponder Beacon	(CATRTB)	1 : Ramark, Radar Beacon Transmitting Continuously 2 : Racon, Radar Transponder Beacon 3 : Leading Racon/Radar Transponder Beacon	EN	1, 1
Radar Wave Length			C	0, 1
Radar Band			(S) TE	1, 1
Wave Length Value	(RadarWaveLength)		(S) RE	1, 1
Signal Group	(SIGGRP)		TE	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated	EN	0, *

		13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Value of Nominal Range	(VALNMR)		RE	0, 1
Sector Limit One	(SECTR1)		C	0, 1
Sector Bearing	(SECTR1) (SECTR2)		(S) RE	1, 1
Sector Line Length			(S) IN	0, 1
Sector Limit Two	(SECTR2)		C	0, 1
Sector Bearing	(SECTR1) (SECTR2)		(S) RE	1, 1
Sector Line Length			(S) IN	0, 1

Signal Sequence	(SIGSEQ)		C	0, 1
Signal Duration			(S) RE	1, 1
Signal Status		1 : Lit/Sound 2 : Eclipsed/Silent	(S) EN	1, 1
INT 1 Reference: Remarks: Distinction:				

2.22 Virtual AIS Aid to Navigation

Table 2-22 — Virtual AIS Aid to Navigation

IHO Definition: An Automatic Identification System (AIS) message 21 transmitted from an AIS station to simulate on navigation systems an Aid to Navigation which does not physically exist.				
S-10x Geo Feature: Virtual AIS Aid to Navigation (NEWOBJ)				
Super Type: Equipment				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Estimated Range of Transmission	(ESTRNG)		RE	0, 1
MMSI Code			TE	1, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/ Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request	EN	0, *

		20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Virtual AIS Aid to Navigation Type		1 : North Cardinal 2 : East Cardinal 3 : South Cardinal 4 : West Cardinal 5 : Port Lateral 6 : Starboard Lateral 7 : Preferred Channel to Port 8 : Preferred Channel to Starboard 9 : Isolated Danger 10 : Safe Water 11 : Special Purpose 12 : New Danger Marking	EN	1, 1
<u>INT 1 Reference:</u>				

Remarks:
Distinction:

Feature/Information associations				
Type	**Association Name**	**Association Ends**		
		Class	**Role**	**Mult**
Asso	Virtual AIS	**RadioStation**	broadcasts	1, *

2.23 Physical AIS Aid to Navigation

Table 2-23 — Physical AIS Aid to Navigation

IHO Definition: An Automatic Identification System (AIS) message 21 transmitted from a physical Aid to Navigation, or transmitted from an AIS station for an Aid to Navigation which physically exists.				
S-10x Geo Feature: Physical AIS Aid to Navigation				
Super Type: Equipment				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category Of Physical AIS Aid To Navigation		1 : Physical AIS Type 1 2 : Physical AIS Type 2 3 : Physical AIS Type 3	EN	1, 1
Estimated Range of Transmission	(ESTRNG)		RE	0, 1
MMSI Code			TE	1, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/ Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public	EN	0, *

		15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Virtual AIS Aid to Navigation Type		1 : North Cardinal 2 : East Cardinal 3 : South Cardinal 4 : West Cardinal 5 : Port Lateral 6 : Starboard Lateral 7 : Preferred Channel to Port 8 : Preferred Channel to Starboard 9 : Isolated Danger	EN	1, 1

		10 : Safe Water 11 : Special Purpose 12 : New Danger Marking		
INT 1 Reference: Remarks: Distinction:				

Feature/Information associations				
Type	**Association Name**	**Association Ends**		
		Class	**Role**	**Mult**
Asso	Virtual AIS	**RadioStation**	broadcasts	1, *

2.24 Synthetic AIS Aid To Navigation

Table 2-24 — Synthetic AIS Aid To Navigation

IHO Definition: -				
S-10x Geo Feature: Synthetic AIS Aid To Navigation				
Super Type: Equipment				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category Of Synthetic AIS Aid To Navigation		1 : predicted 2 : monitored	EN	1, 1
Estimated Range of Transmission	(ESTRNG)		RE	0, 1
MMSI Code			TE	1, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public	EN	0, *

		15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Virtual AIS Aid to Navigation Type		1 : North Cardinal 2 : East Cardinal 3 : South Cardinal 4 : West Cardinal 5 : Port Lateral 6 : Starboard Lateral 7 : Preferred Channel to Port 8 : Preferred Channel to Starboard 9 : Isolated Danger 10 : Safe Water 11 : Special Purpose 12 : New Danger Marking	EN	1, 1

INT 1 Reference:
Remarks:
Distinction:

Feature/Information associations					
Type	**Association Name**	**Association Ends**		
		Class	**Role**	**Mult**
Asso	Virtual AIS	**RadioStation**	broadcasts	1, *

2.25 Isolated Danger Beacon

Table 2-25 — Isolated Danger Beacon

IHO Definition: A beacon is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey (IHO Dictionary, S-32, 5th Edition, 420). An isolated danger beacon is a beacon erected on an isolated danger of limited extent, which has navigable water all around it. (UKHO NP735, 5th Edition)				
S-10x Geo Feature: Isolated Danger Beacon				
Super Type: GenericBeacon				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
INT 1 Reference: Remarks: Distinction:				

2.26 Cardinal Beacon

Table 2-26 — Cardinal Beacon

IHO Definition: A cardinal beacon is used in conjunction with the compass to indicate where the mariner may find the best navigable water. It is placed in one of the four quadrants (North, East, South and West), bounded by inter-cardinal bearings from the point marked.				
S-10x Geo Feature: Cardinal Beacon				
Super Type: GenericBeacon				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Cardinal Mark	(CATCAM)	1 : North Cardinal Mark 2 : East Cardinal Mark	EN	1, 1

		3 : South Cardinal Mark 4 : West Cardinal Mark		
INT 1 Reference: Remarks: Distinction:				

2.27 Isolated Danger Buoy

Table 2-27 — Isolated Danger Buoy

IHO Definition: A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary, S-32, 5th Edition, 565). A isolated danger buoy is a buoy moored on or above an isolated danger of limited extent, which has navigable water all around it. (UKHO NP735, 5th Edition)				
S-10x Geo Feature: Isolated Danger Buoy				
Super Type: GenericBuoy				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
INT 1 Reference: Remarks: Distinction:				

2.28 Cardinal Buoy

Table 2-28 — Cardinal Buoy

IHO Definition: A cardinal buoy is used in conjunction with the compass to indicate where the mariner may find the best navigable water. It is placed in one of the four quadrants (North, East, South and West), bounded by inter-cardinal bearings from the point marked.				
S-10x Geo Feature: Cardinal Buoy				
Super Type: GenericBuoy				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Cardinal Mark	(CATCAM)	1 : North Cardinal Mark 2 : East Cardinal Mark 3 : South Cardinal Mark	EN	1, 1

		4 : West Cardinal Mark		
INT 1 Reference: Remarks: Distinction:				

2.29 Installation Buoy

Table 2-29 — Installation Buoy

IHO Definition: A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary, S-32, 5th Edition, 565). An installation buoy is a buoy used for loading tankers with gas or oil. (IHO Chart Specifications, M-4)				
S-10x Geo Feature: Installation Buoy				
Super Type: GenericBuoy				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category Of Installation Buoy		1 : catenary anchor leg mooring (CALM) 2 : catenary anchor leg mooring (CALM)	EN	1, 1
INT 1 Reference: Remarks: Distinction:				

2.30 Mooring Buoy

Table 2-30 — Mooring Buoy

IHO Definition: The equipment or structure used to secure a vessel. (IHO Registry)				
S-10x Geo Feature: Mooring Buoy				
Super Type: GenericBuoy				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
INT 1 Reference: Remarks: Distinction:				

2.31 Emergency Wreck Marking Buoy

Table 2-31 — Emergency Wreck Marking Buoy

IHO Definition: An emergency wreck marking buoy is a buoy moored on or above a new wreck, designed to provide a prominent (both visual and radio) and easily identifiable temporary (24-72 hours) first response. (IHO Registry)				
S-10x Geo Feature: Emergency Wreck Marking Buoy				
Super Type: GenericBuoy				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
INT 1 Reference: Remarks: Distinction:				

2.32 Lighthouse

Table 2-32 — Lighthouse

IHO Definition: A distinctive structure on or off a coast exhibiting a major light designed to serve as an aid to navigation.				
S-10x Geo Feature: Lighthouse				
Super Type: Landmark				
Primitives: point, surface				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
INT 1 Reference: Remarks: Distinction:				

2.33 Light Float

Table 2-33 — Light Float

IHO Definition: A boat-like structure used instead of a light buoy in waters where strong streams or currents are experienced, or when a greater elevation than that of a light buoy is necessary.		
S-10x Geo Feature: Light Float (LITFLT)		
Super Type: StructureObject		
Primitives: point		
Real World	*Paper Chart Symbol*	*ECDIS Symbol*

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	1, * (ordered)
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent	EN	0, *

		2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/ Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion			

		42 : Removed / Deleted 43 : Candidate for Modification		
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	0, 1
INT 1 Reference: Remarks: Distinction:				

2.34 Light Vessel

Table 2-34 — Light Vessel

IHO Definition: A distinctively marked vessel anchored or moored at a charted point, to serve as an aid to navigation. By night, it displays a characteristic light(s) and is usually equipped with other devices, such as fog signal, submarine sound signal, and radio-beacon, to assist navigation.				
S-10x Geo Feature: Light Vessel (LITVES, Lightship)				
Super Type: StructureObject				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	1, * (ordered)
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown)	EN	0, *

		6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle		
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/ Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing	EN	0, *

		23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	0, 1
INT 1 Reference: Remarks: Distinction:				

2.35 Offshore Platform

Table 2-35 — Offshore Platform

IHO Definition: A permanent offshore structure, either fixed or floating.		
S-10x Geo Feature: Offshore Platform (OFSPLF)		
Super Type: StructureObject		
Primitives: point, surface		
Real World	*Paper Chart Symbol*	*ECDIS Symbol*

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Offshore Platform	(CATOFP)	1 : Oil Rig 2 : Production Platform 3 : Observation/Research Platform 4 : Articulated Loading Platform 5 : Single Anchor Leg Mooring 6 : Mooring Tower 7 : Artificial Island 8 : Floating Production, Storage and Off-Loading Vessel 9 : Accommodation Platform 10 : Navigation, Communication and Control Buoy 11 : Floating Oil Tank	EN	0, *
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	0, * (ordered)
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Condition	(CONDTN)	1 : Under Construction 2 : Ruined 3 : Under Reclamation 4 : Wingless 5 : Planned Construction	EN	0, 1

Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed	EN	0, *

		29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	0, 1
INT 1 Reference: Remarks: Distinction:				

2.36 Silo/Tank

Table 2-36 — Silo/Tank

IHO Definition: A large storage structure used for storing loose materials, liquids and/or gases.				
S-10x Geo Feature: Silo/Tank (SILTNK)				
Super Type: StructureObject				
Primitives: point, surface				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Building Shape	(BUISHP)	5 : High-Rise Building 6 : Pyramid 7 : Cylindrical 8 : Spherical 9 : Cubic	EN	0, 1
Category of Silo/Tank	(CATSIL)	1 : Silo in General 2 : Tank in General 3 : Grain Elevator 4 : Water Tower	EN	0, 1
Colour	(COLOUR)	1 : White 2 : Black	EN	0, * (ordered)

		3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink		
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Height	(HEIGHT)		RE	0, 1
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent	EN	0, *

		6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	0, 1
INT 1 Reference: Remarks:				

Distinction:

2.37 Pile

Table 2-37 — Pile

IHO Definition: A long heavy timber or section of steel, wood, concrete, etc., forced into the earth or sea floor to serve as a support, as for a pier, or to resist lateral pressure; or as a free standing pole within a marine environment.				
S-10x Geo Feature: Pile (PILPNT)				
Super Type: StructureObject				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Pile	(CATPLE)	1 : Stake 3 : Post 4 : Tripodal 5 : Piling 6 : Area of Piles 7 : Pipe	EN	0, 1
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	0, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle	EN	0, *

		9 : Triangle		
Height	(HEIGHT)		RE	0, 1
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	0, 1
INT 1 Reference: Remarks: Distinction:				

2.38 Building

Table 2-38 — Building

IHO Definition: A free-standing self-supporting construction that is roofed, usually walled, and is intended for human occupancy (for example: a place of work or recreation) and/or habitation.				
S-10x Geo Feature: Building (BUISGL)				
Super Type: StructureObject				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
INT 1 Reference: Remarks: Distinction:				

2.39 Bridge

Table 2-39 — Bridge

IHO Definition: (1) An elevated structure extending across or over the weather deck of a vessel, or part of such a structure. The term is sometimes modified to indicate the intended use, such as navigating bridge or signal bridge. (2) A structure erected over a depression or an obstacle such as a body of water, railroad, etc., to provide a roadway for vehicles or pedestrians.				
S-10x Geo Feature: Bridge (BRIDGE)				
Super Type: StructureObject				
Primitives: noGeometry				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
INT 1 Reference: Remarks: Distinction:				

2.40 Topmark

Table 2-40 — Topmark

IHO Definition: A characteristic shape secured at the top of a buoy or beacon to aid in its identification. (IHO Dictionary, S-32, 5th Edition, 5548)				
S-10x Geo Feature: Topmark				
Super Type: AidsToNavigation				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	0, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory	EN	0, *

		11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Topmark/ Daymark Shape	(TOPSHP)	1 : Cone (Point Up) 2 : Cone (Point Down) 3 : Sphere 4 : 2 Spheres 5 : Cylinder 6 : Board 7 : X-Shaped 8 : Upright Cross 9 : Cube (Point Up) 10 : 2 Cones (Point to Point)	EN	1, 1

		11 : 2 Cones (Base to Base) 12 : Rhombus 13 : 2 Cones (Points Upward) 14 : 2 Cones (Points Downward) 15 : Besom (Point Up) 16 : Besom (Point Down) 17 : Flag 18 : Sphere Over a Rhombus 19 : Square 20 : Rectangle (Horizontal) 21 : Rectangle (Vertical) 22 : Trapezium (Up) 23 : Trapezium (Down) 24 : Triangle (Point Up) 25 : Triangle (Point Down) 26 : Circle 27 : Two Upright Crosses (One Over the Other) 28 : T-Shape 29 : Triangle Pointing Up Over a Circle 30 : Upright Cross Over a Circle 31 : Rhombus Over a Circle 32 : Circle Over a Triangle Pointing Up 33 : Other Shape (See Shape Information) 34 : Tubular		
INT 1 Reference: Remarks: Distinction:				

Feature/Information associations				
Type	**Association Name**	**Association Ends**		
		Class	**Role**	**Mult**
Asso	Structure Equipment	**GenericBuoy**	structure	1, 1

2.41 Safe Water Beacon

Table 2-41 — Safe Water Beacon

IHO Definition: A safe water beacon is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey (IHO Dictionary, S-32, 5th Edition, 420). A safe

water beacon may be used to indicate that there is navigable water around the mark. (UKHO NP735, 5th Edition)				
S-10x Geo Feature: Safe Water Beacon				
Super Type: GenericBeacon				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
INT 1 Reference: Remarks: Distinction:				

2.42 Special Purpose General Beacon

Table 2-42 — Special Purpose General Beacon

IHO Definition: A beacon is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey (IHO Dictionary, S-32, 5th Edition, 420). A special purpose beacon is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notices to Mariners. (UKHO NP 735, 5th Edition) Beacon in general: A beacon whose appearance or purpose is not adequately known.				
S-10x Geo Feature: Special Purpose General Beacon				
Super Type: GenericBeacon				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Special Purpose Mark	(CATSPM)	1 : Firing Danger Mark 2 : Target Mark 3 : Marker Ship Mark 4 : Degaussing Range Mark 5 : Barge Mark 6 : Cable Mark 7 : Spoil Ground Mark 8 : Outfall Mark 9 : ODAS 10 : Recording Mark 11 : Seaplane Anchorage Mark 12 : Recreation Zone Mark 13 : Private Mark 14 : Mooring Mark	EN	1, *

		15 : LANBY			
		16 : Leading Mark			
		17 : Measured Distance Mark			
		18 : Notice Mark			
		19 : TSS Mark			
		20 : Anchoring Prohibited Mark			
		21 : Berthing Prohibited Mark			
		22 : Overtaking Prohibited Mark			
		23 : Two-Way Traffic Prohibited Mark			
		24 : Reduced Wake Mark			
		25 : Speed Limit Mark			
		26 : Stop Mark			
		27 : General Warning Mark			
		28 : Sound Ship's Siren Mark			
		29 : Restricted Vertical Clearance Mark			
		30 : Maximum Vessel's Draught Mark			
		31 : Restricted Horizontal Clearance Mark			
		32 : Strong Current Warning Mark			
		33 : Berthing Permitted Mark			
		34 : Overhead Power Cable Mark			
		35 : Channel Edge Gradient Mark			
		36 : Telephone Mark			
		37 : Ferry Crossing Mark			
		39 : Pipeline Mark			
		40 : Anchorage Mark			
		41 : Clearing Mark			
		42 : Control Mark			
		43 : Diving Mark			
		44 : Refuge Beacon			
		45 : Foul Ground Mark			

		46 : Yachting Mark 47 : Heliport Mark 48 : GNSS Mark 49 : Seaplane Landing Mark 50 : Entry Prohibited Mark 51 : Work in Progress Mark 52 : Mark With Unknown Purpose 53 : Wellhead Mark 54 : Channel Separation Mark 55 : Marine Farm Mark 56 : Artificial Reef Mark 57 : Ice Mark 58 : Nature Reserve Mark 59 : Fish Aggregating Device 60 : Wreck Mark 61 : Customs Mark 62 : Causeway Mark 63 : Wave Recorder 64 : Jetski Prohibited		
INT 1 Reference: Remarks: Distinction:				

2.43 Safe Water Buoy

Table 2-43 — Safe Water Buoy

IHO Definition: A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary, S-32, 5th Edition, 565). A safe water buoy is used to indicate that there is navigable water around the mark. (UKHO NP735, 5th Edition)				
S-10x Geo Feature: Safe Water Buoy				
Super Type: GenericBuoy				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
INT 1 Reference:				

Remarks:
Distinction:

2.44 Special Purpose General Buoy

Table 2-44 — Special Purpose General Buoy

IHO Definition: A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary, S-32, 5th Edition, 565). A special purpose buoy is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notices to Mariners. (UKHO NP 735, 5th Edition) Buoy in general: A buoy whose appearance or purpose is not adequately known.				
S-10x Geo Feature: Special Purpose General Buoy				
Super Type: GenericBuoy				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category of Special Purpose Mark	(CATSPM)	1 : Firing Danger Mark 2 : Target Mark 3 : Marker Ship Mark 4 : Degaussing Range Mark 5 : Barge Mark 6 : Cable Mark 7 : Spoil Ground Mark 8 : Outfall Mark 9 : ODAS 10 : Recording Mark 11 : Seaplane Anchorage Mark 12 : Recreation Zone Mark 13 : Private Mark 14 : Mooring Mark 15 : LANBY 16 : Leading Mark 17 : Measured Distance Mark 18 : Notice Mark 19 : TSS Mark 20 : Anchoring Prohibited Mark 21 : Berthing Prohibited Mark	EN	1, *

		22 : Overtaking Prohibited Mark 23 : Two-Way Traffic Prohibited Mark 24 : Reduced Wake Mark 25 : Speed Limit Mark 26 : Stop Mark 27 : General Warning Mark 28 : Sound Ship’s Siren Mark 29 : Restricted Vertical Clearance Mark 30 : Maximum Vessel’s Draught Mark 31 : Restricted Horizontal Clearance Mark 32 : Strong Current Warning Mark 33 : Berthing Permitted Mark 34 : Overhead Power Cable Mark 35 : Channel Edge Gradient Mark 36 : Telephone Mark 37 : Ferry Crossing Mark 39 : Pipeline Mark 40 : Anchorage Mark 41 : Clearing Mark 42 : Control Mark 43 : Diving Mark 44 : Refuge Beacon 45 : Foul Ground Mark 46 : Yachting Mark 47 : Heliport Mark 48 : GNSS Mark 49 : Seaplane Landing Mark 50 : Entry Prohibited Mark 51 : Work in Progress Mark			

		52 : Mark With Unknown Purpose 53 : Wellhead Mark 54 : Channel Separation Mark 55 : Marine Farm Mark 56 : Artificial Reef Mark 57 : Ice Mark 58 : Nature Reserve Mark 59 : Fish Aggregating Device 60 : Wreck Mark 61 : Customs Mark 62 : Causeway Mark 63 : Wave Recorder 64 : Jetski Prohibited		
INT 1 Reference: Remarks: Distinction:				

2.45 Dangerous Feature

Table 2-45 — Dangerous Feature

IHO Definition: -				
S-10x Geo Feature: Dangerous Feature				
Super Type:				
Primitives: point				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Information	(INFORM)		C	0, *
File Locator			(S) TE	0, 1
File Reference	(TXTDSC) (NTXTDS)		(S) TE	0, 1
Headline			(S) TE	0, 1
Language			(S) TE	1, 1
Text	(INFORM) (NINFOM)		(S) TE	0, 1
Interoperability Identifier			UN	0, 1
INT 1 Reference:				

Remarks: Distinction:

Feature/Information associations				
Type	**Association Name**	**Association Ends**		
		Class	**Role**	**Mult**
Asso	Dangerous Feature Association	**AtonAssociation**	markingAton	1, *

2.46 Aton Aggregation

Table 2-46 — Aton Aggregation

IHO Definition: Used to identify an aggregation of two or more objects. This aggregation may be named content of categoryOfAggregation should be put in information attribute when converting to S-57.				
S-10x Geo Feature: Aton Aggregation				
Super Type:				
Primitives: noGeometry				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category Of Aggregation		1 : leading line 3 : measured distance 2 : range system	CL	1, 1
INT 1 Reference: Remarks: Distinction:				

Feature/Information associations				
Type	**Association Name**	**Association Ends**		
		Class	**Role**	**Mult**
Asso	Aton Aggregations	**AidsToNavigation**	peer	0, *

2.47 Aton Association

Table 2-47 — Aton Association

IHO Definition: Used to identify an association between two or more objects. The association may be named content of categoryOfAssociation should be put in information attribute when converting to S-57
S-10x Geo Feature: Aton Association
Super Type:
Primitives: noGeometry

Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Category Of Association		1 : channel markings 2 : danger markings	CL	1, 1
INT 1 Reference: Remarks: Distinction:				

Feature/Information associations				
Type	**Association Name**	**Association Ends**		
		Class	**Role**	**Mult**
Asso	Dangerous Feature Association	**DangerousFeature**	danger	0, *
Asso	Aton Associations	**AidsToNavigation**	peer	0, *

3 Information types

3.1 AtoN Fixing Method

Table 3-1 — AtoN Fixing Method

IHO Definition: Method used for fixing the position of an aid to navigation.				
S-10x Information Type: AtoN Fixing Method				
Super Type:				
Primitives: None				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Reference Point			TE	0, 1
Horizontal Datum	(HORDAT)	1 : WGS 72 2 : WGS 84 3 : European 1950 4 : Potsdam Datum 5 : Adindan 6 : Afgooye 7 : Ain el Abd 1970 8 : Anna 1 Astro 1965 9 : Antigua Island Astro 1943 10 : Arc 1950 11 : Arc 1960 12 : Ascension Island 1958	EN	0, 1

		13 : Astro Beacon ‘E’ 1945 14 : Astro DOS 71/4 15 : Astro Tern Island (FRIG) 1961 16 : Astronomical Station 1952 17 : Australian Geodetic 1966 18 : Australian Geodetic 1984 19 : Ayabelle Lighthouse 20 : Bellevue (IGN) 21 : Bermuda 1957 22 : Bissau 23 : Bogota Observatory 24 : Bukit Rimpah 25 : Camp Area Astro 26 : Campo Inchauspe 1969 27 : Canton Astro 1966 28 : Cape Datum 29 : Cape Canaveral 30 : Carthage 31 : Chatam Island Astro 1971 32 : Chua Astro 33 : Corrego Alegre 34 : Dabola 35 : Djakarta (Batavia) 36 : DOS 1968 37 : Easter Island 1967 38 : European 1979 39 : Fort Thomas 1955 40 : Gan 1970 41 : Geodetic Datum 1949 42 : Graciosa Base SW 1948 43 : Guam 1963 44 : Gunung Segara 45 : GUX 1 Astro 46 : Herat North 47 : Hjorsey 1955 48 : Hong Kong 1963 49 : Hu-Tzu-Shan 50 : Indian 51 : Indian 1954 52 : Indian 1975 53 : Ireland 1965 54 : ISTS 061 Astro 1968 55 : ISTS 073 Astro 1969			

		56 : Johnston Island 1961 57 : Kandawala 58 : Kerguelen Island 1949 59 : Kertau 1968 60 : Kusaie Astro 1951 61 : L. C. 5 Astro 1961 62 : Leigon 63 : Liberia 1964 64 : Luzon 65 : Mahe 1971 66 : Massawa 67 : Merchich 68 : Midway Astro 1961 69 : Minna 70 : Montserrat Island Astro 1958 71 : M’poraloko 72 : Nahrwan 73 : Naparima, BWI 74 : North American 1927 75 : North American 1983 76 : Observatorio Meteorologico 1939 77 : Old Egyptian 1907 78 : Old Hawaiian 79 : Oman 80 : Ordnance Survey of Great Britain 1936 81 : Pico de las Nieves 82 : Pitcairn Astro 1967 83 : Point 58 84 : Pointe Noire 1948 85 : Porto Santo 1936 86 : Provisional South American 1956 87 : Provisional South Chilean 1963 88 : Puerto Rico 89 : Qatar National 90 : Qornoq 91 : Reunion 92 : Rome 1940 93 : Santo (DOS) 1965 94 : Sao Braz 95 : Sapper Hill 1943 96 : Schwarzeck 97 : Selvagem Grande 1938 98 : South American 1969			

		99 : South Asia 100 : Tananarive Observatory 1925 101 : Timbalai 1948 102 : Tokyo 103 : Tristan Astro 1968 104 : Viti Levu 1916 105 : Wake-Eniwetok 1960 106 : Wake Island Astro 1952 107 : Yacare 108 : Zanderij 109 : American Samoa 1962 110 : Deception Island 111 : Indian 1960 112 : Indonesian 1974 113 : North Sahara 1959 114 : Pulkovo 1942 116 : S-JTSK 117 : Voirol 1950 118 : Average Terrestrial System 1977 119 : Compensation Geodesique du Quebec 1977 120 : Finnish (KKJ) 121 : Ordnance Survey of Ireland 122 : Revised Kertau 123 : Revised Nahrwan 124 : GGRS 76 (Greece) 125 : Nouvelle Triangulation de France 126 : RT 90 (Sweden) 127 : Geocentric Datum of Australia 128 : BJZ54 (A954 Beijing Coordinates) 129 : Modified BJZ54 130 : GDZ80 131 : Local Datum		
Source Date	(SORDAT)		DA	1, 1
Positioning Procedure			TE	1, 1
INT 1 Reference: Remarks: Distinction:				

3.2 Aton Status Information

Table 3-2 — Aton Status Information

IHO Definition: -				
S-10x Information Type: Aton Status Information				
Super Type:				
Primitives: None				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Change Details			C	1, 1
Aton Commissioning		1 : Buoy establishment 2 : Light establishment 3 : Beacon establishment 4 : Audible signal establishment 5 : Fog signal establishment 6 : AIS transmitter establishment 7 : V-AIS establishment 8 : RACON establishment 9 : DGPS station establishment 10 : eLORAN station establishment 11 : DGLONASS station establishment 12 : e-Chayka station establishment 13 : EGNOS establishment	(S) EN	0, 1
Aton Removal		1 : Buoy removal 2 : Buoy temporary removal 3 : Light removal 4 : Light temporary removal 5 : Beacon removal 6 : Beacon temporary removal 7 : Fog signal removal 8 : Fog signal temporary removal 9 : Audible signal removal 10 : Audible signal temporary removal 11 : V-AIS removal 12 : V-AIS temporary removal 13 : RACON signal removal 14 : RACON temporary removal 15 : DGPS removal 16 : DGPS temporary removal 17 : EGNOS removal 18 : EGNOS temporary removal	(S) EN	0, 1

		19 : LORAN C station removal 20 : LORAN C station temporary removal 21 : eLORAN removal 22 : eLORAN temporary removal 23 : Chayka station removal 24 : Chayka station temporary removal 25 : e-Chayka station removal 26 : e-Chayka station temporary removal 27 :		
Aton Replacement		1 : Buoy change 2 : Buoy temporary change 3 : Light change 4 : Light temporary change 5 : Sector light change 6 : Sector light temporary change 7 : Beacon change 8 : Beacon temporary change 9 : Fog signal change 10 : Fog signal temporary change 11 : Audible signal change 12 : Audible signal temporary change 13 : V-AIS change 14 : V-AIS temporary change 15 : RACON signal change 16 : RACON temporary change	(S) EN	0, 1
Fixed Aton Change		1 : Beacon missing 2 : Beacon damaged 3 : Light beacon Unlit 4 : Light beacon Unreliable 5 : Light beacon Not synchronized 6 : Light beacon damaged 7 : Beacon topmark missing 8 : Beacon topmark damaged 9 : Beacon daymark unreliable 10 : Floodlit beacon Unlit 11 : Beacon restored to normal	(S) EN	0, 1
Floating Aton Change		1 : Buoy adrift 2 : Buoy damaged 3 : Buoy daymark unreliable 4 : Buoy destroyed 5 : Buoy missing 6 : Buoy move	(S) EN	0, 1

		7 : Buoy off position 8 : Buoy re-establishment 9 : Buoy restored to normal 10 : Buoy topmark damaged 11 : Buoy topmark missing 12 : Buoy will be withdrawn 13 : Buoy withdrawn 14 : Decommissioned for winter 15 : Lifted for Winter 16 : Light buoy Light damaged 17 : Light buoy Light not synchronized 18 : Light buoy Light unlit 19 : Light buoy Light unreliable 20 : Marine Aids to Navigation unreliable 21 : Recommissioned for navigation season 22 : Replaced by Winter Spar 23 : Seasonal decommissioning complete 24 : Seasonal decommissioning in progress 25 : Seasonal recommissioning complete 26 : Seasonal recommissioning in progress		
Audible Signal Aton Change		1 : Audible signal out of service 2 : Fog signal out of service 3 : Audible signal operating properly 4 : Fog signal operating properly	(S) EN	0, 1
Lighted Aton Change		1 : Light unlit 2 : Light unreliable 3 : Light re-establishment 4 : Light range reduced 5 : Light without rhythm 6 : Light out of synchronization 7 : Light daymark unreliable 8 : Light operating properly 9 : Sector light Sector obscured 10 : Front leading/range light Unlit 11 : Rear leading/range light Unlit 12 : Front leading/range light Unreliable 13 : Rear leading/range light Unreliable 14 : Front leading/range light Light range reduced 15 : Rear leading/range light Light range reduced	(S) EN	0, 1

		16 : Front leading/range light without rhythm 17 : Rear leading/range light without rhythm 18 : Leading/range lights out of synchronization 19 : Front leading/range beacon Unreliable 20 : Rear leading/range beacon Unreliable 21 : Front leading/range light is operating properly 22 : Rear leading/range light is operating properly 23 : Front leading/range beacon restored to normal 24 : Rear leading/range beacon restored to normal		
Electronic Aton Change		1 : AIS transmitter out of service 2 : AIS transmitter unreliable 3 : AIS transmitter operating properly 4 : V-AIS out of service 5 : V-AIS unreliable 6 : V-AIS operating properly 7 : RACON out of service 8 : RACON unreliable 9 : RACON operating properly 10 : DGPS out of service 11 : DGPS operating properly 12 : DGPS unreliable 13 : LORAN C operating properly 14 : LORAN C unreliable 15 : LORAN C out of service 16 : eLORAN operating properly 17 : eLORAN unreliable 18 : eLORAN out of service 19 : DGLOANSS operating properly 20 : DGLOANSS unreliable 21 : DGLOANSS out of service 22 : Chayka operating properly 23 : Chayka unreliable 24 : Chayka out of service 25 : e-Chayka operating properly 26 : e-Chayka unreliable 27 : e-Chayka out of service 28 : EGNOS operating properly 29 : EGNOS unreliable	(S) EN	0, 1

		30 : EGNOS out of service		
Change Types		1 : Advanced notice of changes 2 : Discrepancy 3 : Proposed changes 4 : Temporary changes	EN	0, 1
INT 1 Reference: Remarks: Distinction:				

3.3 Positioning Information

Table 3-3 — Positioning Information

IHO Definition: Information about how a position was obtained. (proposed by CCG)				
S-10x Information Type: Positioning Information				
Super Type:				
Primitives: None				
Real World	*Paper Chart Symbol*		*ECDIS Symbol*	
S-10x Attribute	**S-57 Acronym**	**Allowable Encoding Value**	**Type**	**Multiplicity**
Positioning Device			TE	1, 1
Positioning Method			C	0, 1
Positioning Equipment		1 : DGPS Receiver 2 : GLONASS Receiver 3 : GPS Receiver 4 : GPS/WAAS Receiver	(S) EN	1, 1
NMEAString			(S) TE	1, 1
INT 1 Reference: Remarks: Distinction:				

3.4 Spatial Quality

Table 3-4 — Spatial Quality

IHO Definition: The indication of the quality of the locational information for features in a dataset.		
S-10x Information Type: Spatial Quality		
Super Type:		
Primitives: None		
Real World	*Paper Chart Symbol*	*ECDIS Symbol*

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Quality of Horizontal Measurement	(QUAPOS)	1 : Surveyed 2 : Unsurveyed 3 : Inadequately Surveyed 4 : Approximate 5 : Position Doubtful 6 : Unreliable 7 : Reported (Not Surveyed) 8 : Reported (Not Confirmed) 9 : Estimated 10 : Precisely Known 11 : Calculated	EN	0, 1
Spatial Accuracy			C	0, 1
Fixed Date Range			(S) C	0, 1
Date End	(DATEND)		(S) TD	0, 1
Date Start	(DATSTA)		(S) TD	0, 1
Horizontal Position Uncertainty	(POSACC)		(S) C	0, 1
Uncertainty Fixed	(POSACC) (SOUACC) (VERACC)		(S) RE	1, 1
Uncertainty Variable Factor			(S) RE	0, 1
Vertical Uncertainty	(VERACC) (SOUACC)		(S) C	0, 1
Uncertainty Fixed	(POSACC) (SOUACC) (VERACC)		(S) RE	1, 1
Uncertainty Variable Factor			(S) RE	0, 1
INT 1 Reference: Remarks: Distinction:				

4 Association Names

4.1 Aton Status

Table 4-1 — Aton Status

IHO Definition:

Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
Association	Status Part		0, 1
	Aton Part	**Aids to Navigation**	

4.2 Fixing Status

Table 4-2 — Fixing Status

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
	Fixed By		
	Fixing Method		

4.3 Aton Status

Table 4-3 — Aton Status

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
	Status Part		
	Aton Part		

4.4 Dangerous Feature Association

Table 4-4 — Dangerous Feature Association

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
Association	Marking Aton	**Aton Association**	1, *
	Danger	**Dangerous Feature**	0, *

4.5 Aton Aggregations

Table 4-5 — Aton Aggregations

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
Association	Peer	**Aids to Navigation, Aton Aggregation**	

	Peer		0, *

4.6 Aton Associations

Table 4-6 — Aton Associations

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
Association	Peer	**Aids to Navigation, Aton Association**	
	Peer		0, *

4.7 Swivel Connection

Table 4-7 — Swivel Connection

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
	Holds		
	Hangs		

4.8 Cable Connection

Table 4-8 — Cable Connection

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
	Holds		
	Attached		

4.9 Shackle Connection

Table 4-9 — Shackle Connection

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
	Connected To		
	Connected		

4.10 Bridle Connection

Table 4-10 — Bridle Connection

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
	Holds		
	Hangs		

4.11 Structure Equipment

Table 4-11 — Structure Equipment

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
Association	Child	**Equipment**	0, *
	Parent	**Structure Object**	1, 1

4.12 Buoy Counter Weight

Table 4-12 — Buoy Counter Weight

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
	Holds		
	Attached		

4.13 Virtual AIS

Table 4-13 — Virtual AIS

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
Association	Broadcasts	**Radio Station**	1, *
	Broadcast By	**Virtual AIS Aid to Navigation, Physical AIS Aid to Navigation, Synthetic AIS Aid To Navigation**	0, *

4.14 Range System

Table 4-14 — Range System

IHO Definition: Remarks: -			
Role Type	**Role**	**Associated With**	**Multiplicity**
Association	Navigation Line	**Navigation Line**	1, *
	Navigable Track	**Recommended Track**	0, *

5 Association Roles

5.1 Structure

Table 5-1 — Structure

IHO Definition: -

5.2 Equipment

Table 5-2 — Equipment

IHO Definition: -

5.3 Danger

Table 5-3 — Danger

IHO Definition: -

5.4 Marking Aton

Table 5-4 — Marking Aton

IHO Definition: -

5.5 Peer

Table 5-5 — Peer

IHO Definition: The role given to an object of equal status in the relationship.

5.6 Aton Part

Table 5-6 — Aton Part

IHO Definition: -

5.7 Hangs

Table 5-7 — Hangs

IHO Definition: -

5.8 Attached

Table 5-8 — Attached

IHO Definition: -

5.9 Connected

Table 5-9 — Connected

IHO Definition: -

5.10 Parent

Table 5-10 — Parent

IHO Definition: -

5.11 Fixing Method

Table 5-11 — Fixing Method

IHO Definition: -

5.12 Broadcast By

Table 5-12 — Broadcast By

IHO Definition: -

5.13 Navigable Track

Table 5-13 — Navigable Track

IHO Definition: The role given to the navigable part of the navigation line.

5.14 Status Part

Table 5-14 — Status Part

IHO Definition: -

5.15 Holds

Table 5-15 — Holds

IHO Definition: -

5.16 Connected To

Table 5-16 — Connected To

IHO Definition: -

5.17 Child

Table 5-17 — Child

IHO Definition: -

5.18 Fixed By

Table 5-18 — Fixed By

IHO Definition: -

5.19 Broadcasts

Table 5-19 — Broadcasts

IHO Definition: -

5.20 Navigation Line

Table 5-20 — Navigation Line

IHO Definition: The role given to the navigation line(s) that is generally formed between two or more objects, or by one object and a bearing.

6 Attribute and Enumerate Descriptions

6.1 Category Of Association

Table 6-1 — Category Of Association

IHO Definition: Named associations between two or more aids to navigation and/or navigationally relevant features 1) **channel markings** IHO Definition: - 2) **danger markings** IHO Definition: - Remarks:

6.2 Seasonal Action Required

Table 6-2 — Seasonal Action Required

IHO Definition: - Remarks:

6.3 Counter Weight Type

Table 6-3 — Counter Weight Type

IHO Definition: - Remarks:

6.4 Swivel Type

Table 6-4 — Swivel Type

IHO Definition: - Remarks:

6.5 Legs Details

Table 6-5 — Legs Details

IHO Definition: - Remarks:

6.6 BridleLink Type

Table 6-6 — BridleLink Type

IHO Definition: - Remarks:

6.7 Sinker Type

Table 6-7 — Sinker Type

IHO Definition: - Remarks:

6.8 Weight

Table 6-8 — Weight

IHO Definition: - Remarks:

6.9 Cable Length

Table 6-9 — Cable Length

IHO Definition: Total length of a cable.
Remarks:

6.10 Diameter

Table 6-10 — Diameter

IHO Definition: -
Remarks:

6.11 Horizontal Accuracy

Table 6-11 — Horizontal Accuracy

IHO Definition: -
Remarks:

6.12 Manufactorer

Table 6-12 — Manufactorer

IHO Definition: -
Remarks:

6.13 IsSlatted

Table 6-13 — IsSlatted

IHO Definition: -
Remarks:

6.14 Candela

Table 6-14 — Candela

IHO Definition: -
Remarks:

6.15 Vertical Accuracy

Table 6-15 — Vertical Accuracy

IHO Definition: -
Remarks:

6.16 Category Of Installation Buoy

Table 6-16 — Category Of Installation Buoy

IHO Definition: Classification of fixed installation buoy 1) **catenary anchor leg mooring (CALM)** IHO Definition: Incorporates a large buoy which remains on the surface at all times and is moored by 4 or more anchors. Mooring hawsers and cargo hoses lead from a turntable on top of the buoy, so that the buoy does not turn as the ship swings to wind and stream. 2) **catenary anchor leg mooring (CALM)** IHO Definition: A mooring structure used by tankers to load and unload in port approaches or in offshore oil and gas fields. The size of the structure can vary between a large mooring buoy and a manned floating structure. Also known as single point mooring (SPM) Remarks: No remarks.

6.17 Quality of Horizontal Measurement

Table 6-17 — Quality of Horizontal Measurement

IHO Definition: The degree of reliability attributed to a position. 1) **Surveyed** IHO Definition: The position(s) was(were) determined by the operation of making measurements for determining the relative position of points on, above or beneath the earth's surface. Survey implies a regular, controlled survey of any date. 2) **Unsurveyed** IHO Definition: Survey data does not exist or is very poor. 3) **Inadequately Surveyed** IHO Definition: Not surveyed to modern standards; or due to its age, scale, or positional or vertical uncertainties is not suitable to the type of navigation expected in the area. 4) **Approximate** IHO Definition: A position that is considered to be less than third-order accuracy, but is generally considered to be within 30.5 metres of its correct geographic location. Also may apply to an object whose position does not remain fixed. 5) **Position Doubtful** IHO Definition: Of uncertain position. The expression is used principally on charts to indicate that a wreck, shoal, etc., has been reported in various positions and not definitely determined in any. 6) **Unreliable** IHO Definition: A feature's position has been obtained from questionable or unreliable data. 7) **Reported (Not Surveyed)** IHO Definition: An object whose position has been reported and its position confirmed by some means other than a formal survey such as an independent report of the same object. 8) **Reported (Not Confirmed)** IHO Definition: An object whose position has been reported and its position has not been confirmed. 9) **Estimated** IHO Definition: The most probable position of an object determined from incomplete data or data of questionable accuracy. 10) **Precisely Known** IHO Definition: A position that is of a known value, such as the position of an anchor berth or other defined object. 11) **Calculated** IHO Definition: A position that is computed from data. Remarks: No remarks.

6.18 Positioning Device

Table 6-18 — Positioning Device

IHO Definition: - Remarks:

6.19 Change Types

Table 6-19 — Change Types

IHO Definition: -

1) **Advanced notice of changes**
   IHO Definition: -
2) **Discrepancy**
   IHO Definition: -
3) **Proposed changes**
   IHO Definition: -
4) **Temporary changes**
   IHO Definition: -

Remarks:

6.20 Height Length Units

Table 6-20 — Height Length Units

IHO Definition: Units of measure of waterway distances. (IHO Registry)

1) **Metres**
   IHO Definition: -
2) **Feet**
   IHO Definition: -
3) **Kilometres**
   IHO Definition: -
4) **Hectometres**
   IHO Definition: -
5) **Statute Miles**
   IHO Definition: -
6) **Nautical Miles**
   IHO Definition: -

Remarks:

6.21 Positioning Procedure

Table 6-21 — Positioning Procedure

IHO Definition: -

Remarks:

6.22 Horizontal Datum

Table 6-22 — Horizontal Datum

IHO Definition: Horizontal reference surface or the reference coordinate system used for geodetic control in the calculation of coordinates of points on the earth.

1) **WGS 72**
   IHO Definition: A standard for use in cartography, geodesy, and satellite navigation including GPS. This standard includes the definition of the coordinate system’s fundamental and derived constants, the ellipsoidal (normal) Earth Gravitational Model (EGM), a description of the associated World Magnetic Model (WMM), and a current list of local datum transformations. The WGS 72 is based on selected satellite, surface gravity and astrogeodetic data available through 1972.
2) **WGS 84**
   IHO Definition: A standard for use in cartography, geodesy, and satellite navigation including GPS. This standard includes the definition of the coordinate system’s fundamental and derived constants, the ellipsoidal (normal) Earth Gravitational Model (EGM), a description of the associated World Magnetic Model (WMM), and a current list of local datum transformations. WGS 84 is the reference coordinate system used by the Global Positioning System.

3) **European 1950**
IHO Definition: A geodetic datum first defined in 1950 suitable for use in Europe — west: Andorra; Cyprus; Denmark — onshore and offshore; Faroe Islands — onshore; France — offshore; Germany — offshore North Sea; Gibraltar; Greece — offshore; Israel — offshore; Italy including San Marino and Vatican City State; Ireland offshore; Malta; Netherlands — offshore; North Sea; Norway including Svalbard — onshore and offshore; Portugal — mainland — offshore; Spain — onshore; Turkey — onshore and offshore; United Kingdom UKCS offshore east of 6W including Channel Islands (Guernsey and Jersey). Egypt — Western Desert; Iraq — onshore; Jordan. European Datum 1950 references the International 1924 ellipsoid and the Greenwich prime meridian. European Datum 1950 origin is Fundamental point: Potsdam (Helmert Tower). Latitude: 5222’51.4456’’N, longitude: 1303’58.9283’’E (of Greenwich). European Datum 1950 is a geodetic datum for Topographic mapping, geodetic survey.
4) **Potsdam Datum**
IHO Definition: A geodetic datum first defined in 1990 suitable for use in Germany — Thuringen. Potsdam Datum/83 references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Potsdam Datum/83 origin is Fundamental point: Rauenberg. Latitude: 5227’12.021’’N, longitude: 1322’04.928’’E (of Greenwich). This station was destroyed in 1910 and the station at Potsdam substituted as the fundamental point. Potsdam Datum/83 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from BKG via EuroGeographics. http://crs.bkg.bund.de PD/83 is the realisation of DHDN in Thuringen. It is the resultant of applying a transformation derived at 13 points on the border between East and West Germany to **Pulkovo 1942**/83 points in Thuringen.
5) **Adindan**
IHO Definition: A geodetic datum first defined in 1958 suitable for use in Eritrea; Ethiopia; South Sudan; Sudan. Adindan references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Adindan origin is Fundamental point: Station 15; Adindan. Latitude: 2210’07.110’’N, longitude: 3129’21.608’’E (of Greenwich). Adindan is a geodetic datum for Topographic mapping. It was defined by information from US Coast and Geodetic Survey via Geophysical Reasearch vol 67 #11, October 1962. The 12th parallel traverse of 1966-70 (Point 58 datum, code 6620) is connected to the Blue Nile 1958 network in western Sudan. This has given rise to misconceptions that the Blue Nile network is used in west Africa.
6) **Afgooye**
IHO Definition: geodetic datum first defined in and suitable for use in Somalia — onshore. Afgooye references the Krassowsky 1940 ellipsoid and the Greenwich prime meridian. Afgooye is a geodetic datum for Topographic mapping.
7) **Ain el Abd 1970**
IHO Definition: A geodetic datum first defined in 1970 and suitable for use in Bahrain, Kuwait and Saudi Arabia — onshore. Ain el Abd 1970 references the International 1924 ellipsoid and the Greenwich prime meridian. Ain el Abd 1970 origin is Fundamental point: Ain El Abd. Latitude: 2814’06.171’’N, longitude: 4816’20.906’’E (of Greenwich). Ain el Abd 1970 is a geodetic datum for Topographic mapping.
8) **Anna 1 Astro 1965**
IHO Definition: A geodetic datum first defined in 1965 suitable for use in Cocos (Keeling) Islands — onshore. Cocos Islands 1965 references the Australian National Spheroid ellipsoid and the Greenwich prime meridian. Cocos Islands 1965 origin is Fundamental point: Anna 1. Cocos Islands 1965 is a geodetic datum for Military and topographic mapping It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
9) **Antigua Island Astro 1943**
IHO Definition: geodetic datum first defined in 1943 suitable for use in Antigua island — onshore. Antigua 1943 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Antigua 1943 origin is Fundamental point: station A14. Antigua 1943 is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey of Great Britain.
10) **Arc 1950**
IHO Definition: A geodetic datum first defined in 1950 suitable for use in Botswana; Malawi; Zambia; Zimbabwe. Arc 1950 references the Clarke 1880 (Arc) ellipsoid and the Greenwich prime meridian. Arc 1950 origin is Fundamental point: Buffelsfontein. Latitude: 3359’32.000’’S, longitude: 2530’44.622’’E (of Greenwich). Arc 1950 is a geodetic datum for Topographic mapping, geodetic survey.
11) **Arc 1960**
IHO Definition: A geodetic datum first defined in 1960 suitable for use in Kenya; Tanzania; Uganda. Arc 1960 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Arc 1960 origin is Fundamental point: Buffelsfontein. Latitude: 3359’32.000’’S, longitude: 2530’44.622’’E (of Greenwich). Arc 1960 is a geodetic datum for Topographic mapping, geodetic survey.
12) **Ascension Island 1958**
IHO Definition: A geodetic datum first defined in 1958 suitable for use in St Helena, Ascension and Tristan da Cunha — Ascension Island — onshore. Ascension Island 1958 references the International 1924 ellipsoid and the Greenwich prime meridian. Ascension Island 1958 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
13) **Astro Beacon ‘E’ 1945**
IHO Definition: Astro beacon ‘E’ 1945

14) **Astro DOS 71/4**
IHO Definition: A geodetic datum first defined in 1971 suitable for use in St Helena, Ascension and Tristan da Cunha — St Helena Island — onshore. Astro DOS 71 references the International 1924 ellipsoid and the Greenwich prime meridian. Astro DOS 71 origin is Fundamental point: DOS 71/4, Ladder Hill Fort, latitude: 1555'30''S, longitude: 543'25''W (of Greenwich). Astro DOS 71 is a geodetic datum for Geodetic control, military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000) and St. Helena Government, Environment and Natural Resources Directorate (ENRD).
15) **Astro Tern Island (FRIG) 1961**
IHO Definition: A geodetic datum first defined in 1961 suitable for use in United States (USA) — Hawaii — Tern Island and Sorel Atoll. Tern Island 1961 references the International 1924 ellipsoid and the Greenwich prime meridian. Tern Island 1961 origin is Fundamental point: station FRIG on tern island, station B4 on Sorol Atoll. Tern Island 1961 is a geodetic datum for Military and topographic mapping It was defined by information from DMA / NIMA / NGA TR8350.2 (original 1987 first edition and 3rd edition, Amendment 1, 3 January 2000). Two independent astronomic determinations considered to be consistent through adoption of common transformation to WGS 84 (see tfm code 15795).
16) **Astronomical Station 1952**
IHO Definition: Astronomical station 1952.
17) **Australian Geodetic 1966**
IHO Definition: A geodetic datum first defined in 1966 suitable for use in Australia — onshore and offshore. Papua New Guinea — onshore. Australian Geodetic Datum 1966 references the Australian National Spheroid ellipsoid and the Greenwich prime meridian. Australian Geodetic Datum 1966 origin is Fundamental point: Johnson Memorial Cairn. Latitude: 2556'54.5515''S, longitude: 13312'30.0771''E (of Greenwich). Australian Geodetic Datum 1966 is a geodetic datum for Topographic mapping. It was defined by information from Australian Map Grid Technical Manual. National Mapping Council of Australia Technical Publication 7; 1972.
18) **Australian Geodetic 1984**
IHO Definition: geodetic datum first defined in 1984 suitable for use in Australia — Queensland, South Australia, Western Australia, federal areas offshore west of 129E. Australian Geodetic Datum 1984 references the Australian National Spheroid ellipsoid and the Greenwich prime meridian. Australian Geodetic Datum 1984 origin is Fundamental point: Johnson Memorial Cairn. Latitude: 2556'54.5515''S, longitude: 13312'30.0771''E (of Greenwich). Australian Geodetic Datum 1984 is a geodetic datum for Topographic mapping. It was defined by information from 'GDA technical manual v2_2', Intergovernmental Committee on Surveying and Mapping. www.anzlic.org.au/icsm/gdtm/ Uses all data from 1966 adjustment with additional observations, improved software and a geoid model.
19) **Ayabelle Lighthouse**
IHO Definition: A geodetic datum suitable for use in Djibouti — onshore and offshore. Ayabelle Lighthouse references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Ayabelle Lighthouse origin is Fundamental point: Ayabelle Lighthouse. Ayabelle Lighthouse is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
20) **Bellevue (IGN)**
IHO Definition: A geodetic datum first defined in 1960 suitable for use in Vanuatu — southern islands — Aneityum, Efate, Erromango and Tanna. Bellevue references the International 1924 ellipsoid and the Greenwich prime meridian. Bellevue is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000). Datum covers all the major islands of Vanuatu in two different adjustment blocks, but practical usage is as given in the area of use.
21) **Bermuda 1957**
IHO Definition: A geodetic datum first defined in 1957 suitable for use in Bermuda — onshore. Bermuda 1957 references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Bermuda 1957 origin is Fundamental point: Fort George base. Latitude 3222'44.36''N, longitude 6440'58.11''W (of Greenwich). Bermuda 1957 is a geodetic datum for Topographic mapping. It was defined by information from Various oil industry sources.
22) **Bissau**
IHO Definition: A geodetic datum first defined in and is suitable for use in Guinea-Bissau — onshore. Bissau references the International 1924 ellipsoid and the Greenwich prime meridian. Bissau origin is Bissau is a geodetic datum for Topographic mapping. It was defined by information from NIMA TR8350.2 ftp://164.214.2.65/pub/gig/tr8350.2/changes.pdf.
23) **Bogota Observatory**
IHO Definition: A geodetic datum first defined in 1975 suitable for use in Colombia — mainland and offshore Caribbean. Bogota 1975 references the International 1924 ellipsoid and the Greenwich prime meridian. Bogota 1975 origin is Fundamental point: Bogota observatory. Latitude: 435'56.570''N, longitude: 7404'51.300''W (of Greenwich). Bogota 1975 is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico Agustin Codazzi (IGAC) special publication no. 1, 4th edition (1975) 'Geodesia: Resultados Definitvos de Parte de las Redes Geodesicas Establecidas en el Pais'. Replaces 1951 adjustment. Replaced by MAGNA-SIRGAS (datum code 6685).

24) **Bukit Rimpah**
IHO Definition: geodetic datum suitable for use in Indonesia — Banga and Belitung Islands. Bukit Rimpah references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Bukit Rimpah origin is 200'40.16''S, 10551'39.76''E (of Greenwich). Bukit Rimpah is a geodetic datum for Topographic mapping.
25) **Camp Area Astro**
IHO Definition: geodetic datum suitable for use in Antarctica — McMurdo Sound, Camp McMurdo area. Camp Area Astro references the International 1924 ellipsoid and the Greenwich prime meridian. Camp Area Astro is a geodetic datum for Geodetic and topographic survey. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
26) **Campo Inchauspe 1969**
IHO Definition: A geodetic datum suitable for use in Argentina — mainland onshore and Atlantic offshore Tierra del Fuego. Campo Inchauspe references the International 1924 ellipsoid and the Greenwich prime meridian. Campo Inchauspe origin is Fundamental point: Campo Inchauspe. Latitude: 3558'16.56''S, longitude: 6210'12.03''W (of Greenwich). Campo Inchauspe is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/
27) **Canton Astro 1966**
IHO Definition: A geodetic datum first defined in 1966 suitable for use in Kiribati — Phoenix Islands: Kanton, Orona, McKean Atoll, Birnie Atoll, Phoenix Seamounts. Phoenix Islands 1966 references the International 1924 ellipsoid and the Greenwich prime meridian. Phoenix Islands 1966 is a geodetic datum for Military and topographic mapping It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
28) **Cape Datum**
IHO Definition: A geodetic datum suitable for use in Botswana; Lesotho; South Africa — mainland; Swaziland. Cape references the Clarke 1880 (Arc) ellipsoid and the Greenwich prime meridian. Cape origin is Fundamental point: Buffelsfontein. Latitude: 3359'32.000''S, longitude: 2530'44.622''E (of Greenwich). Cape is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Private Communication, Directorate of Surveys and Land Information, Cape Town.
29) **Cape Canaveral**
IHO Definition: A geodetic datum first defined in 1963 suitable for use in North America — onshore — Bahamas and USA — Florida (east). Cape Canaveral references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Cape Canaveral origin is Fundamental point: Central 1950. Latitude: 28 29'32.36555''N, longitude 80 34'38.77362''W (of Greenwich). Cape Canaveral is a geodetic datum for US space and military operations. It was defined by information from US NGS and DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
30) **Carthage**
IHO Definition: A geodetic datum first defined in 1925 suitable for use in Tunisia — onshore and offshore. Carthage references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. Carthage origin is Fundamental point: Carthage. Latitude: 40.9464506g = 3651'06.50''N, longitude: 8.8724368g E of Paris = 1019'20.72''E (of Greenwich). Carthage is a geodetic datum for Topographic mapping. Fundamental point astronomic coordinates determined in 1878.
31) **Chatam Island Astro 1971**
IHO Definition: A geodetic datum first defined in 1971 suitable for use in New Zealand — Chatham Islands group — onshore. Chatham Islands Datum 1971 references the International 1924 ellipsoid and the Greenwich prime meridian. Chatham Islands Datum 1971 is a geodetic datum for Geodetic survey, topographic mapping, engineering survey. It was defined by information from Office of Surveyor General (OSG) Technical Report 14, June 2001. Replaced by Chatham Islands Datum 1979 (code 6673).
32) **Chua Astro**
IHO Definition: A geodetic datum suitable for use in Brazil — south of 18S and west of 54W, plus Distrito Federal. Paraguay — north. Chua references the International 1924 ellipsoid and the Greenwich prime meridian. Chua origin is Fundamental point: Chua. Latitude: 1945'41.160''S, longitude: 4806'07.560''W (of Greenwich). Chua is a geodetic datum for Geodetic survey. It was defined by information from NIMA http://earth-info.nima.mil/. The Chua origin and associated network is in Brazil with a connecting traverse through northern Paraguay. It was used in Brazil only as input into the Corrego Allegre adjustment and for government work in Distrito Federal.
33) **Corrego Alegre**
IHO Definition: A geodetic datum first defined in 1972 suitable for use in Brazil — onshore — west of 54W and south of 18S; also south of 15S between 54W and 42W; also east of 42W. Corrego Alegre 1970-72 references the International 1924 ellipsoid and the Greenwich prime meridian. Corrego Alegre 1970-72 origin is Fundamental point: Corrego Alegre. Latitude: 1950'14.91''S, longitude: 4857'41.98''W (of Greenwich). Corrego Alegre 1970-72 is a geodetic datum for Topographic mapping, geodetic survey. Superseded by SAD69. It was defined by information from IBGE. Replaces 1961 adjustment (datum code 1074). NIMA gives coordinates of origin as latitude: 1950'15.14''S, longitude: 4857'42.75''W; these may refer to 1961 adjustment.
34) **Dabola**

IHO Definition: A geodetic datum first defined in 1981 suitable for use in Guinea — onshore. Dabola 1981 references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. Dabola 1981 is a geodetic datum for Topographic mapping. It was defined by information from IGN Paris.

35) **Djakarta (Batavia)**
IHO Definition: A geodetic datum suitable for use in Indonesia — onshore Java and Bali. Batavia (Jakarta) references the Bessel 1841 ellipsoid and the Jakarta prime meridian. Batavia (Jakarta) origin is Fundamental point: Longitude at Batavia astronomical station. Latitude: 607'39.522''S, longitude: 000'00.0''E (of Jakarta). Latitude and azimuth at Genuk. Batavia (Jakarta) is a geodetic datum for Topographic mapping.

36) **DOS 1968**
IHO Definition: DOS 1968.

37) **Easter Island 1967**
IHO Definition: A geodetic datum first defined in 1967 suitable for use in Chile — Easter Island onshore. Easter Island 1967 references the International 1924 ellipsoid and the Greenwich prime meridian. Easter Island 1967 is a geodetic datum for Military and topographic mapping, +/- 25 meters in each component. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).

38) **European 1979**
IHO Definition: A geodetic datum first defined in 1979 suitable for use in Europe — west. European Datum 1979 references the International 1924 ellipsoid and the Greenwich prime meridian. European Datum 1979 origin is Fundamental point: Potsdam (Helmert Tower). Latitude: 5222'51.4456''N, longitude: 1303'58.9283''E (of Greenwich). European Datum 1979 is a geodetic datum for Scientific network. Replaced by 1987 adjustment.

39) **Fort Thomas 1955**
IHO Definition: Fort Thomas 1955 datum.

40) **Gan 1970**
IHO Definition: A geodetic datum first defined in 1970 suitable for use in Maldives — onshore. Gan 1970 references the International 1924 ellipsoid and the Greenwich prime meridian. Gan 1970 is a geodetic datum for Topographic mapping. It was defined by information from Various industry sources. In some references incorrectly named 'Gandajika 1970'.

41) **Geodetic Datum 1949**
IHO Definition: A geodetic datum first defined in 1949 suitable for use in New Zealand — North Island, South Island, Stewart Island — onshore and nearshore. New Zealand Geodetic Datum 1949 references the International 1924 ellipsoid and the Greenwich prime meridian. New Zealand Geodetic Datum 1949 origin is Fundamental point: Papatahi. Latitude: 4119' 8.900''S, longitude: 17502'51.000''E (of Greenwich). New Zealand Geodetic Datum 1949 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Land Information New Zealand. http://www.linz.govt.nz/rcs/linz/pub/web/root/core/SurveySystem/GeodeticInfo/GeodeticDatums/nzgd2000factsheet/index.jsp. Replaced by New Zealand Geodetic Datum 2000 (code 6167) from March 2000.

42) **Graciosa Base SW 1948**
IHO Definition: Graciosa Base SW 1948 datum.

43) **Guam 1963**
IHO Definition: A geodetic datum first defined in 1963 suitable for use in Guam — onshore. Guam 1963 references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Guam 1963 origin is Fundamental point: Tagcha. Latitude: 1322'38.49''N, longitude: 14445'51.56''E (of Greenwich). Guam 1963 is a geodetic datum for Topographic mapping. It was defined by information from US National Geospatial Intelligence Agency (NGA). http://earth-info.nga.mil/ Replaced by NAD83(HARN)

44) **Gunung Segara**
IHO Definition: A geodetic datum suitable for use in Indonesia — Kalimantan — onshore east coastal area including Mahakam delta coastal and offshore shelf areas. Gunung Segara references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Gunung Segara origin is Station P5 (Gunung Segara). Latitude 032'12.83''S, longitude 11708'48.47''E (of Greenwich). Gunung Segara is a geodetic datum for Topographic mapping. It was defined by information from TotalFinaElf.

45) **GUX 1 Astro**
IHO Definition: GUX 1 Astro datum.

46) **Herat North**
IHO Definition: A geodetic datum suitable for use in Afghanistan. Herat North references the International 1924 ellipsoid and the Greenwich prime meridian. Herat North origin is Fundamental point: Herat North. Latitude: 3423'09.08''N, longitude: 6410'58.94''E (of Greenwich). Herat North is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/.

47) **Hjorsey 1955**
IHO Definition: A geodetic datum first defined in 1955 suitable for use in Iceland — onshore. Hjorsey 1955 references the International 1924 ellipsoid and the Greenwich prime meridian. Hjorsey 1955 origin is Fundamental point: Latitude: 6431'29.26''N, longitude: 2222'05.84''W (of Greenwich). Hjorsey 1955 is a geodetic datum for 1/50,000 scale topographic mapping. It was defined by information from Landmaelingar Islands (National Survey of Iceland).

48) **Hong Kong 1963**
IHO Definition: A geodetic datum first defined in 1963 suitable for use in China — Hong Kong — onshore and offshore. Hong Kong 1963 references the Clarke 1858 ellipsoid and the Greenwich prime meridian. Hong Kong 1963 origin is Fundamental point: Trig 'Zero', 38.4 feet south along the transit circle of the Kowloon Observatory. Latitude 2218'12.82"N, longitude 11410'18.75"E (of Greenwich). Hong Kong 1963 is a geodetic datum for Topographic mapping and hydrographic charting. It was defined by information from Survey and Mapping Office, Lands Department. http://www.info.gov.hk/landsd/. Replaced by Hong Kong 1963(67) for military purposes only in 1967. Replaced by Hong Kong 1980.
49) **Hu-Tzu-Shan**
IHO Definition: A geodetic datum first defined in 1950 suitable for use in Taiwan, Republic of China — onshore — Taiwan Island, Penghu (Pescadores) Islands. Hu Tzu Shan 1950 references the International 1924 ellipsoid and the Greenwich prime meridian. Hu Tzu Shan 1950 origin is Fundamental point: Hu Tzu Shan. Latitude: 2358'32.34"N, longitude: 12058'25.975"E (of Greenwich). Hu Tzu Shan 1950 is a geodetic datum for Topographic mapping. It was defined by information from NIMA US NGA, http://earth-info.nga.mil/GandG/index.html
50) **Indian**
IHO Definition: Indian datum.
51) **Indian 1954**
IHO Definition: A geodetic datum first defined in 1954 suitable for use in Myanmar (Burma) — onshore; Thailand — onshore. Indian 1954 references the Everest 1830 (1937 Adjustment) ellipsoid and the Greenwich prime meridian. Indian 1954 origin is Extension of Kalianpur 1937 over Myanmar and Thailand. Indian 1954 is a geodetic datum for Topographic mapping.
52) **Indian 1975**
IHO Definition: A geodetic datum first defined in 1975 suitable for use in Thailand — onshore plus offshore Gulf of Thailand. Indian 1975 references the Everest 1830 (1937 Adjustment) ellipsoid and the Greenwich prime meridian. Indian 1975 origin is Fundamental point: Khau Sakaerang. Indian 1975 is a geodetic datum for Topographic mapping.
53) **Ireland 1965**
IHO Definition: A geodetic datum first defined in 1975 suitable for use in Ireland — onshore. United Kingdom (UK) — Northern Ireland (Ulster) — onshore. Ireland 1965 references the Airy Modified 1849 ellipsoid and the Greenwich prime meridian. Ireland 1965 origin is Adjusted to best mean fit 9 stations of the OSNI 1952 primary adjustment in Northern Ireland plus the 1965 values of 3 stations in the Republic of Ireland. Ireland 1965 is a geodetic datum for Geodetic survey, topographic mapping and engineering survey. It was defined by information from 'The Irish Grid — A Description of the Co-ordinate Reference System' published by **Ordnance Survey of Ireland**, Dublin and Ordnance Survey of Northern Ireland, Belfast. Differences from the 1965 adjustment (datum code 6299) are: average difference in Eastings 0.092m; average difference in Northings 0.108m; maximum vector difference 0.548m.
54) **ISTS 061 Astro 1968**
IHO Definition: A geodetic datum first defined in 1968 suitable for use in South Georgia and the South Sandwich Islands — South Georgia onshore. ISTS 061 Astro 1968 references the International 1924 ellipsoid and the Greenwich prime meridian. ISTS 061 Astro 1968 origin is Fundamental point: ISTS 061. ISTS 061 Astro 1968 is a geodetic datum for Military and topographic mapping It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
55) **ISTS 073 Astro 1969**
IHO Definition: A geodetic datum first defined in 1969 suitable for use in British Indian Ocean Territory — Chagos Archipelago — Diego Garcia. ISTS 073 Astro 1969 references the International 1924 ellipsoid and the Greenwich prime meridian. ISTS 073 Astro 1969 origin is Fundamental point: ISTS 073. ISTS 073 Astro 1969 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
56) **Johnston Island 1961**
IHO Definition: A geodetic datum first defined in 1961 suitable for use in United States Minor Outlying Islands — Johnston Island. Johnston Island 1961 references the International 1924 ellipsoid and the Greenwich prime meridian. Johnston Island 1961 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
57) **Kandawala**
IHO Definition: A geodetic datum first defined in 1930 suitable for use in Sri Lanka — onshore. Kandawala references the Everest 1830 (1937 Adjustment) ellipsoid and the Greenwich prime meridian. Kandawala origin is Fundamental point: Kandawala. Latitude: 714'06.838"N, longitude: 7952'36.670"E. Kandawala is a geodetic datum for Topographic mapping. It was defined by information from Abeyratne, Featherstone and Tantrigoda in Survey Review vol. 42 no. 317 (July 2010).
58) **Kerguelen Island 1949**
IHO Definition: A geodetic datum first defined in 1949 suitable for use in French Southern Territories — Kerguelen onshore. References the International 1924 ellipsoid and the Greenwich prime meridian. Origin is K0 1949. Is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from IGN Paris.
59) **Kertau 1968**

IHO Definition: A geodetic datum first defined in 1968 suitable for use in Malaysia — West Malaysia onshore and offshore east coast; Singapore — onshore and offshore. Kertau 1968 references the Everest 1830 Modified ellipsoid and the Greenwich prime meridian. Kertau 1968 origin is Fundamental point: Kertau. Latitude: 327’50.710”N, longitude: 10237’24.550”E (of Greenwich). Kertau 1968 is a geodetic datum for Geodetic survey, cadastre. It was defined by information from Defence Geographic Centre. Replaces MRT48 and earlier adjustments. Adopts metric conversion of 39.370113 inches per metre. Not used for 1969 metrication of RSO grid — see Kertau (RSO) (code 6751).

60) **Kusaie Astro 1951**
IHO Definition: A geodetic datum first defined in 1951 suitable for use in Federated States of Micronesia — Kosrae (Kusaie). Kusaie 1951 references the International 1924 ellipsoid and the Greenwich prime meridian. Kusaie 1951 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).

61) **L. C. 5 Astro 1961**
IHO Definition: L. C. 5 Astro 1961 datum.

62) **Leigon**
IHO Definition: A geodetic datum suitable for use in Ghana — onshore and offshore. Leigon references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Leigon origin is Fundamental point: GCS Station 121, Leigon. Latitude: 538’52.27”N, longitude: 011’46.08”W (of Greenwich). Leigon is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey International. Replaced Accra datum (code 6168) from 1978. Coordinates at Leigon fundamental point defined as Accra datum values for that point.

63) **Liberia 1964**
IHO Definition: A geodetic datum first defined in 1964 suitable for use in Liberia — onshore. Liberia 1964 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Liberia 1964 origin is Fundamental point: Robertsfield. Latitude: 613’53.02”N, longitude: 1021’35.44”W (of Greenwich). Liberia 1964 is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/.

64) **Luzon**
IHO Definition: A geodetic datum first defined in 1911 suitable for use in Philippines — onshore. Luzon references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Luzon origin is Fundamental point: Balacan. Latitude: 1333’41.000”N, longitude: 12152’03.000”E (of Greenwich). Luzon is a geodetic datum for Topographic mapping. It was defined by information from Coast and Geodetic Survey Replaced by Philippine Reference system of 1992 (datum code 6683).

65) **Mahe 1971**
IHO Definition: A geodetic datum first defined in 1971 suitable for use in Seychelles — Mahe Island. Mahe 1971 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Mahe 1971 origin is Fundamental point: Station SITE. Latitude: 440’14.644”S, longitude: 5528’44.488”E (of Greenwich). Mahe 1971 is a geodetic datum for US military survey. It was defined by information from Clifford Mugnier’s September 2007 PE&RS ‘Grids and Datums’ article on Seychelles (www.asprs.org/resources/grids/). South East Island 1943 (datum code 1138) used for topographic mapping, cadastral and hydrographic survey.

66) **Massawa**
IHO Definition: A geodetic datum suitable for use in Eritrea — onshore and offshore. Massawa references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Massawa is a geodetic datum for Topographic mapping.

67) **Merchich**
IHO Definition: A geodetic datum first defined in 1922 suitable for use in Morocco — onshore. Merchich references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. Merchich origin is Fundamental point: Merchich. Latitude: 3326’59.672”N, longitude: 733’27.295”W (of Greenwich). Merchich is a geodetic datum for Topographic mapping.

68) **Midway Astro 1961**
IHO Definition: A geodetic datum first defined in 1961 suitable for use in United States Minor Outlying Islands — Midway Islands — Sand Island and Eastern Island. Midway 1961 references the International 1924 ellipsoid and the Greenwich prime meridian. Midway 1961 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).

69) **Minna**
IHO Definition: A geodetic datum suitable for use in Nigeria — onshore and offshore. Minna references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Minna origin is Fundamental point: Minna base station L40. Latitude: 938’08.87”N, longitude: 630’58.76”E (of Greenwich). Minna is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/.

70) **Montserrat Island Astro 1958**
IHO Definition: A geodetic datum first defined in 1958 suitable for use in Montserrat — onshore. Montserrat 1958 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Montserrat 1958 origin is Fundamental point: station M36. Montserrat 1958 is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey of Great Britain.

71) **M’poraloko**

IHO Definition: A geodetic datum suitable for use in Gabon — onshore and offshore. M'poraloko references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. M'poraloko is a geodetic datum for Topographic mapping.

72) **Nahrwan**
IHO Definition: A geodetic datum first defined in 1934 suitable for use in Iraq — onshore; Iran — onshore northern Gulf coast and west bordering southeast Iraq. Nahrwan 1934 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Nahrwan 1934 origin is Fundamental point: Nahrwan south base. Latitude: 3319'10.87''N, longitude: 4443'25.54''E (of Greenwich). Nahrwan 1934 is a geodetic datum for Oil exploration and production. It was defined by information from Various industry sources. This adjustment later discovered to have a significant orientation error. In Iran replaced by FD58. In Iraq, replaced by Karbala 1979.

73) **Naparima, BWI**
IHO Definition: A geodetic datum first defined in 1972 suitable for use in Trinidad and Tobago — Tobago — onshore. Naparima 1972 references the International 1924 ellipsoid and the Greenwich prime meridian. Naparima 1972 origin is Fundamental point: Naparima. Latitude: 1016'44.860''N, longitude: 6127'34.620''W (of Greenwich). Naparima 1972 is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey International. Naparima 1972 is an extension of the Naparima 1955 network of Trinidad to include Tobago.

74) **North American 1927**
IHO Definition: A geodetic datum first defined in 1927 suitable for use in North and central America; Antigua and Barbuda; Bahamas; Belize; British Virgin Islandss. Usage shall be onshore only except that onshore and offshore shall apply to Canada east coast (New Brunswick; Newfoundland and Labrador; Prince Edward Island; Quebec). Cuba. Mexico (Gulf of Mexico and Caribbean coasts only). USA Alaska. USA Gulf of Mexico (Alabama; Florida; Louisiana; Mississippi; Texas). USA East Coast. Bahamas onshore plus offshore over internal continental shelf only. North American Datum 1927 references the Clarke 1866 ellipsoid and the Greenwich prime meridian. North American Datum 1927 origin is Fundamental point: Meade's Ranch. Latitude: 3913'26.686''N, longitude: 9832'30.506''W (of Greenwich). North American Datum 1927 is a geodetic datum for Topographic mapping. In United States (USA) and Canada, replaced by North American Datum 1983 (NAD83) (code 6269) ; in Mexico, replaced by Mexican Datum of 1993 (code 1042).

75) **North American 1983**
IHO Definition: A geodetic datum first defined in 1986 suitable for use in North America — onshore and offshore: Canada; **Puerto Rico**; United States (USA); US Virgin Islands; British Virgin Islands. North American Datum 1983 references the GRS 1980 ellipsoid and the Greenwich prime meridian. North American Datum 1983 origin is Origin at geocentre. North American Datum 1983 is a geodetic datum for Topographic mapping. Although the 1986 adjustment included connections to Greenland and Mexico, it has not been adopted there. In Canada and US, replaced NAD27.

76) **Observatorio Meteorologico 1939**
IHO Definition: A geodetic datum first defined in 1939 suitable for use in Portugal — western Azores onshore — Flores, Corvo. Azores Occidental Islands 1939 references the International 1924 ellipsoid and the Greenwich prime meridian. Azores Occidental Islands 1939 origin is Fundamental point: Observatario Meteorologico Flores. Azores Occidental Islands 1939 is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico e Cadastral Lisbon via EuroGeographics; http://crs.bkg.bund.de/crs-eu/.

77) **Old Egyptian 1907**
IHO Definition: A geodetic datum first defined in 1907 suitable for use in Egypt — onshore and offshore. Egypt 1907 references the Helmert 1906 ellipsoid and the Greenwich prime meridian. Egypt 1907 origin is Fundamental point: Station F1 (Venus). Latitude: 3001'42.86''N, longitude: 3116'33.60''E (of Greenwich). Egypt 1907 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey.

78) **Old Hawaiian**
IHO Definition: A geodetic datum suitable for use in United States (USA) — Hawaii — main islands onshore. Old Hawaiian references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Old Hawaiian origin is Fundamental point: Oahu West Base Astro. Latitude: 2118'13.89''N, longitude 15750'55.79''W (of Greenwich). Old Hawaiian is a geodetic datum for Topographic mapping. It was defined by information from http://www.ngs.noaa.gov/ (NADCON readme file). Hawaiian Islands were never on NAD27 but rather on Old Hawaiian Datum. NADCON conversion program provides transformation from Old Hawaiian Datum to NAD83 (original 1986 realization) but making the transformation appear to user as if from NAD27.

79) **Oman**
IHO Definition: A geodetic datum first defined in 2013 suitable for use in Oman — onshore and offshore. Oman National Geodetic Datum 2014 references the GRS 1980 ellipsoid and the Greenwich prime meridian. Oman National Geodetic Datum 2014 origin is 20 stations of the Oman primary network tied to ITRF2008 at epoch 2013.15. Oman National Geodetic Datum 2014 is a geodetic datum for Geodetic Survey. It was defined by information from National Survey Authority, Sultanate of Oman. Replaces WGS 84 (G874).

80) **Ordnance Survey of Great Britain 1936**

IHO Definition: A geodetic datum first defined in 1936 suitable for use in United Kingdom (UK) — offshore to boundary of UKCS within 4946'N to 6101'N and 733'W to 333'E; onshore Great Britain (England, Wales and Scotland). Isle of Man onshore. OSGB 1936 references the Airy 1830 ellipsoid and the Greenwich prime meridian. OSGB 1936 origin is Prior to 2002, fundamental point: Herstmonceux, Latitude: 5051'55.271''N, longitude: 020'45.882''E (of Greenwich). From April 2002 the datum is defined through the application of the OSTN transformation from ETRS89. OSGB 1936 is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey of Great Britain. The average accuracy of OSTN compared to the old triangulation network (down to 3rd order) is 0.1m. With the introduction of OSTN15, the area for OSGB 1936 has effectively been extended from Britain to cover the adjacent UK Continental Shelf.

81) **Pico de las Nieves**
IHO Definition: A geodetic datum suitable for use in Spain — Canary Islands onshore. Pico de las Nieves 1984 references the International 1924 ellipsoid and the Greenwich prime meridian. Pico de las Nieves 1984 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000). Replaces Pico de las Nieves 1968 (PN68). Replaced by REGCAN95.

82) **Pitcairn Astro 1967**
IHO Definition: A geodetic datum first defined in 1967 suitable for use in Pitcairn — Pitcairn Island. Pitcairn 1967 references the International 1924 ellipsoid and the Greenwich prime meridian. Pitcairn 1967 origin is Fundamental point: Pitcairn Astro. Latitude: 2504'06.87''S, longitude: 13006'47.83''W (of Greenwich). Pitcairn 1967 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000). Replaced by Pitcairn 2006.

83) **Point 58**
IHO Definition: A geodetic datum first defined in 1969 suitable for use in Senegal — central, Mali — southwest, Burkina Faso — central, Niger — southwest, Nigeria — north, Chad — central. All in proximity to the parallel of latitude of 12N. Point 58 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Point 58 origin is Fundamental point: Point 58. Latitude: 1252'44.045''N, longitude: 358'37.040''E (of Greenwich). Point 58 is a geodetic datum for Geodetic survey. It was defined by information from IGN Paris. Used as the basis for computation of the 12th Parallel traverse conducted 1966-70 from Senegal to Chad and connecting to the Blue Nile 1958 (Adindan) triangulation in Sudan.

84) **Pointe Noire 1948**
IHO Definition: Pointe Noire 1948 datum.

85) **Porto Santo 1936**
IHO Definition: A geodetic datum first defined in 1936 suitable for use in Portugal — Madeira, Porto Santo and Desertas islands — onshore. Porto Santo 1936 references the International 1924 ellipsoid and the Greenwich prime meridian. Porto Santo 1936 origin is SE Base on Porto Santo island. Porto Santo 1936 is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico e Cadastral Lisbon http://www.igeo.pt Replaced by 1995 adjustment (datum code 6663). For Selvagens see Selvagem Grande (code 6616).

86) **Provisional South American 1956**
IHO Definition: A geodetic datum first defined in 1956 suitable for use in Aruba — onshore; Bolivia; Bonaire — onshore; Brazil — offshore — Amazon Cone shelf; Chile — onshore north of 4330'S; Curacao — onshore; Ecuador — mainland onshore; Guyana — onshore; Peru — onshore; Venezuela — onshore. Provisional South American Datum 1956 references the International 1924 ellipsoid and the Greenwich prime meridian. Provisional South American Datum 1956 origin is Fundamental point: La Canoa. Latitude: 834'17.170''N, longitude: 6351'34.880''W (of Greenwich). Provisional South American Datum 1956 is a geodetic datum for Topographic mapping. Same origin as La Canoa datum.

87) **Provisional South Chilean 1963**
IHO Definition: A geodetic datum first defined in 1963 suitable for use in Argentina and Chile — Tierra del Fuego, onshore. Hito XVIII 1963 references the International 1924 ellipsoid and the Greenwich prime meridian. Hito XVIII 1963 origin is Chile-Argentina boundary survey. Hito XVIII 1963 is a geodetic datum for Geodetic survey. It was defined by information from Various oil company records. Used in Tierra del Fuego.

88) **Puerto Rico**
IHO Definition: A geodetic datum first defined in 1901 suitable for use in Puerto Rico, US Virgin Islands and British Virgin Islands — onshore. Puerto Rico references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Puerto Rico origin is Fundamental point: Cardona Island Lighthouse. Latitude:1757'31.40''N, longitude: 6638'07.53''W (of Greenwich). Puerto Rico is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey of Great Britain and http://www.ngs.noaa.gov/ (NADCON readme file). NADCON conversion program provides transformation from Puerto Rico Datum to NAD83 (original 1986 realization) but making the transformation appear to user as if from NAD27.

89) **Qatar National**
IHO Definition: A geodetic datum first defined in 1995 suitable for use in Qatar — onshore. Qatar National Datum 1995 references the International 1924 ellipsoid and the Greenwich prime meridian. Qatar National Datum 1995 origin is defined by transformation from WGS 84 — see coordinate operation code 1840.

Qatar National Datum 1995 is a geodetic datum for Topographic mapping. It was defined by information from Qatar Centre for Geographic Information.

90) **Qornoq**
IHO Definition: A geodetic datum first defined in 1927 suitable for use in Greenland — west coast onshore. Qornoq 1927 references the International 1924 ellipsoid and the Greenwich prime meridian. Qornoq 1927 origin is Fundamental point: Station 7008. Latitude: 6431'06.27"N, longitude: 5112'24.86"W (of Greenwich). Qornoq 1927 is a geodetic datum for Topographic mapping. It was defined by information from Kort & Matrikelstyrelsen, Copenhagen. Origin coordinates from NIMA http://earth-info.nima.mil/.

91) **Reunion**
IHO Definition: A geodetic datum first defined in 1947 suitable for use in Reunion — onshore. Reunion 1947 references the International 1924 ellipsoid and the Greenwich prime meridian. Reunion 1947 origin is Fundamental point: Piton des Neiges (Borne). Latitude: 2105'13.119"S, longitude: 5529'09.193"E (of Greenwich). Reunion 1947 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from IGN Paris. Replaced by RGR92 (datum code 6627).

92) **Rome 1940**
IHO Definition: A geodetic datum first defined in and is suitable for use in Italy — onshore and offshore; San Marino, Vatican City State. Monte Mario (Rome) references the International 1924 ellipsoid and the Rome prime meridian. Monte Mario (Rome) origin is Fundamental point: Monte Mario. Latitude: 4155'25.51"N, longitude: 000' 00.00"E (of Rome). Monte Mario (Rome) is a geodetic datum for Topographic mapping. Replaced Genova datum, Bessel 1841 ellipsoid, from 1940.

93) **Santo (DOS) 1965**
IHO Definition: A geodetic datum first defined in 1965 suitable for use in Vanuatu — northern islands — Aese, Ambrym, Aoba, Epi, Espiritu Santo, Maewo, Malo, Malkula, Paama, Pentecost, Shepherd and Tutuba. Santo 1965 references the International 1924 ellipsoid and the Greenwich prime meridian. Santo 1965 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000). Datum covers all the major islands of Vanuatu in two different adjustment blocks, but practical usage is as given in the area of use.

94) **Sao Braz**
IHO Definition: A geodetic datum first defined in 1995 suitable for use in Portugal — eastern Azores onshore — Sao Miguel, Santa Maria, Formigas. Azores Oriental Islands 1995 references the International 1924 ellipsoid and the Greenwich prime meridian. Azores Oriental Islands 1995 origin is Fundamental point: Forte de So Bras. Origin and orientation constrained to those of the 1940 adjustment. Azores Oriental Islands 1995 is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico e Cadastral Lisbon; http://www.igeo.pt/ Classical and GPS observations. Replaces 1940 adjustment (datum code 6184).

95) **Sapper Hill 1943**
IHO Definition: A geodetic datum first defined in 1943 suitable for use in Falkland Islands (Malvinas) — onshore. Sapper Hill 1943 references the International 1924 ellipsoid and the Greenwich prime meridian. Sapper Hill 1943 is a geodetic datum for Topographic mapping.

96) **Schwarzeck**
IHO Definition: A geodetic datum suitable for use in Namibia — onshore and offshore. Schwarzeck references the Bessel Namibia (GLM) ellipsoid and the Greenwich prime meridian. Schwarzeck origin is Fundamental point: Schwarzeck. Latitude: 2245'35.820"S, longitude: 1840'34.549"E (of Greenwich). Fixed during German South West Africa-British Bechuanaland boundary survey of 1898-1903. Schwarzeck is a geodetic datum for Topographic mapping. It was defined by information from Private Communication, Directorate of Surveys and Land Information, Cape Town.

97) **Selvagem Grande 1938**
IHO Definition: A geodetic datum suitable for use in Portugal — Selvagens islands (Madeira province) — onshore. Selvagem Grande references the International 1924 ellipsoid and the Greenwich prime meridian. Selvagem Grande is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico e Cadastral Lisbon http://www.igeo.pt.

98) **South American 1969**
IHO Definition: A geodetic datum first defined in 1969 suitable for use in Brazil — onshore and offshore. In rest of South America — onshore north of approximately 45S and Tierra del Fuego. South American Datum 1969 references the GRS 1967 Modified ellipsoid and the Greenwich prime meridian. South American Datum 1969 origin is Fundamental point: Chua. Geodetic latitude: 1945'41.6527"S; geodetic longitude: 4806'04.0639"W (of Greenwich). (Astronomic coordinates: Latitude 1945'41.34"S +/- 0.05", longitude 4806'07.80"W +/- 0.08"). South American Datum 1969 is a geodetic datum for Topographic mapping. It was defined by information from DMA 1974. SAD69 uses GRS 1967 ellipsoid but with 1/f to exactly 2 decimal places. In Brazil only, replaced by SAD69(96) (datum code 1075).

99) **South Asia**
IHO Definition: South Asia datum.

100) **Tananarive Observatory 1925**
IHO Definition: A geodetic datum first defined in 1925 suitable for use in Madagascar — onshore and nearshore. Tananarive 1925 references the International 1924 ellipsoid and the Greenwich prime meridian. Tananarive 1925 origin is Fundamental point: Tananarive observatory. Latitude: 1855'02.10"S,

longitude: 4733’06.75’’E (of Greenwich). Tananarive 1925 is a geodetic datum for Topographic mapping. It was defined by information from IGN Paris.

101) **Timbalai 1948**

IHO Definition: A geodetic datum first defined in 1948 suitable for use in Brunei — onshore and offshore; Malaysia — East Malaysia (Sabah; Sarawak) — onshore and offshore. Timbalai 1948 references the Everest 1830 (1967 Definition) ellipsoid and the Greenwich prime meridian. Timbalai 1948 origin is Fundamental point: Station P85 at Timbalai. Latitude: 517’ 3.548’’N, longitude: 11510’56.409’’E (of Greenwich). Timbalai 1948 is a geodetic datum for Topographic mapping. It was defined by information from Defence Geographic Centre. In 1968, the original adjustment was densified in Sarawak and extended to Sabah.

102) **Tokyo**

IHO Definition: A geodetic datum first defined in 1918 suitable for use in Japan — onshore; North Korea — onshore; South Korea — onshore. Tokyo references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Tokyo origin is Fundamental point: Nikon-Keido-Genten. Latitude: 3539’17.5148’’N, longitude: 13944’40.5020’’E (of Greenwich). Longitude derived in 1918. Tokyo is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Geographic Survey Institute; Japan; Bulletin 40 (March 1994). Also http://vldb.gsi.go.jp/sokuchi/datum/tokyodatum.html. In Japan, replaces Tokyo 1892 (code 1048) and replaced by Japanese Geodetic Datum 2000 (code 6611). In Korea used only for geodetic applications before being replaced by Korean 1985 (code 6162).

103) **Tristan Astro 1968**

IHO Definition: A geodetic datum first defined in 1968 suitable for use in St Helena, Ascension and Tristan da Cunha — Tristan da Cunha island group including Tristan, Inaccessible, Nightingale, Middle and Stoltenhoff Islands. Tristan 1968 references the International 1924 ellipsoid and the Greenwich prime meridian. Tristan 1968 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).

104) **Viti Levu 1916**

IHO Definition: A geodetic datum first defined in 1912 suitable for use in Fiji — Viti Levu island. Viti Levu 1912 references the Clarke 1880 (international foot) ellipsoid and the Greenwich prime meridian. Viti Levu 1912 Latitude origin was obtained astronomically at station Monavatu = 1753’28.285’’S, longitude origin was obtained astronomically at station Suva = 17825’35.835’’E. Viti Levu 1912 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Clifford J. Mugnier in Photogrammetric Engineering and Remote Sensing, October 2000, www.asprs.org. For topographic mapping, replaced by Fiji 1956. For other purposes, replaced by Fiji 1986.

105) **Wake-Eniwetok 1960**

IHO Definition: A geodetic datum first defined in 1960 suitable for use in Marshall Islands — onshore. Wake atoll onshore. Marshall Islands 1960 references the Hough 1960 ellipsoid and the Greenwich prime meridian. Marshall Islands 1960 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).

106) **Wake Island Astro 1952**

IHO Definition: A geodetic datum first defined in 1952 suitable for use in Wake atoll — onshore. Wake Island 1952 references the International 1924 ellipsoid and the Greenwich prime meridian. Wake Island 1952 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).

107) **Yacare**

IHO Definition: A geodetic datum suitable for use in Uruguay — onshore. Yacare references the International 1924 ellipsoid and the Greenwich prime meridian. Yacare origin is Fundamental point: Yacare. Latitude: 3035’53.68’’S, longitude: 5725’01.30’’W (of Greenwich). Yacare is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/

108) **Zanderij**

IHO Definition: A geodetic datum suitable for use in Suriname — onshore and offshore. Zanderij references the International 1924 ellipsoid and the Greenwich prime meridian. Zanderij is a geodetic datum for Topographic mapping.

109) **American Samoa 1962**

IHO Definition: A geodetic datum first defined in 1962 suitable for use in American Samoa — Tutuila, Aunu’u, Ofu, Olesega and Ta’u islands. American Samoa 1962 references the Clarke 1866 ellipsoid and the Greenwich prime meridian. American Samoa 1962 origin is Fundamental point: Betty 13 eccentric. Latitude: 1420’08.34’’S, longitude: 17042’52.25’’W (of Greenwich). American Samoa 1962 is a geodetic datum for Topographic mapping. It was defined by information from NIMA TR8350.2 revision of January 2000. Oil industry sources for origin description details.

110) **Deception Island**

IHO Definition: A geodetic datum suitable for use in Antarctica — South Shetland Islands — Deception Island. Deception Island references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Deception Island is a geodetic datum for Military and scientific mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).

111) **Indian 1960**

IHO Definition: A geodetic datum suitable for use in Cambodia — onshore; Vietnam — onshore and offshore Cuu Long basin. Indian 1960 references the Everest 1830 (1937 Adjustment) ellipsoid and the Greenwich prime meridian. Indian 1960 origin is DMA extension over IndoChina of the Indian 1954 network adjusted to better fit local geoid. Indian 1960 is a geodetic datum for Topographic mapping. Also known as Indian (DMA Reduced).

112)**Indonesian 1974**

IHO Definition: A geodetic datum first defined in 1974 suitable for use in Indonesia — onshore. Indonesian Datum 1974 references the Indonesian National Spheroid ellipsoid and the Greenwich prime meridian. Indonesian Datum 1974 origin is Fundamental point: Padang. Latitude: 056'38.414''S, longitude: 10022' 8.804''E (of Greenwich). Ellipsoidal height 3.190m, gravity-related height 14.0m above mean sea level. Indonesian Datum 1974 is a geodetic datum for Topographic mapping. It was defined by information from Bakosurtanal 1979 paper by Jacob Rais. Replaced by DGN95.

113)**North Sahara 1959**

IHO Definition: A geodetic datum first defined in 1959 suitable for use in Algeria — onshore and offshore. Nord Sahara 1959 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Nord Sahara 1959 origin is Coordinates of primary network readjusted on ED50 datum and then transformed conformally to Clarke 1880 (RGS) ellipsoid. Nord Sahara 1959 is a geodetic datum for Topographic mapping. It was defined by information from 'Le System Geodesique Nord-Sahara'; IGN Paris Adjustment includes Morocco and Tunisia but use only in Algeria. Within Algeria the adjustment is north of 32N but use has been extended southwards in many disconnected projects, some based on independent astro stations rather than the geodetic network.

114)**Pulkovo 1942**

IHO Definition: A geodetic datum first defined in 1942 suitable for use in Armenia; Azerbaijan; Belarus; Estonia — onshore; Georgia — onshore; Kazakhstan; Kyrgyzstan; Latvia — onshore; Lithuania — onshore; Moldova; Russian Federation — onshore; Tajikistan; Turkmenistan; Ukraine — onshore; Uzbekistan. Pulkovo 1942 references the Krassowsky 1940 ellipsoid and the Greenwich prime meridian. Pulkovo 1942 origin is Fundamental point: Pulkovo observatory. Latitude: 5946'18.550''N, longitude: 3019'42.090''E (of Greenwich). Pulkovo 1942 is a geodetic datum for Topographic mapping.

116)**S-JTSK**

IHO Definition: A geodetic datum suitable for use in Czech Republic; Slovakia. System Jednotne Trigonometricke Site Katastralni references the Bessel 1841 ellipsoid and the Greenwich prime meridian. System Jednotne Trigonometricke Site Katastralni origin is Modification of Austrian MGI datum, code 6312. System Jednotne Trigonometricke Site Katastralni is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Research Institute for Geodesy Topography and Cartography (VUGTK); Prague. S-JTSK = System of the Unified Trigonometrical Cadastral Network.

117)**Voirol 1950**

IHO Definition: Voirol 1950 datum.

118)**Average Terrestrial System 1977**

IHO Definition: A geodetic datum first defined in 1977 suitable for use in Canada — New Brunswick; Nova Scotia; Prince Edward Island. Average Terrestrial System 1977 references the Average Terrestrial System 1977 ellipsoid and the Greenwich prime meridian. Average Terrestrial System 1977 is a geodetic datum for Topographic mapping. It was defined by information from New Brunswick Geographic Information Corporation land and water information standards manual. In use from 1979.

119)**Compensation Geodesique du Quebec 1977**

IHO Definition: Compensation Geodesique du Quebec 1977.

120)**Finnish (KKJ)**

IHO Definition: A geodetic datum first defined in 1966 suitable for use in Finland — onshore. Kartastokoordinaattijarjestelma (1966) references the International 1924 ellipsoid and the Greenwich prime meridian. Kartastokoordinaattijarjestelma (1966) origin is Adjustment with fundamental point SF31 based on ED50 transformed to best fit the older VVJ adjustment. Kartastokoordinaattijarjestelma (1966) is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from National Land Survey of Finland; http://www.maanmittauslaitos.fi. Adopted in 1970.

121)**Ordnance Survey of Ireland**

IHO Definition: A geodetic datum first defined in 1952 suitable for use in United Kingdom (UK) — Northern Ireland (Ulster) — onshore. OSNI 1952 references the Airy 1830 ellipsoid and the Greenwich prime meridian. OSNI 1952 origin is Position fixed to the coordinates from the 19th century Principle Triangulation of station Divis. Scale and orientation controlled by position of Principle Triangulation stations Knocklayd and Trostan. OSNI 1952 is a geodetic datum for Geodetic survey and topographic mapping. It was defined by information from Ordnance Survey of Northern Ireland. Replaced by Geodetic Datum of 1965 alias 1975 Mapping Adjustment or TM75 (datum code 6300).

122)**Revised Kertau**

IHO Definition: A geodetic datum first defined in 1969 suitable for use in Malaysia — West Malaysia; Singapore. Kertau (RSO) references the Everest 1830 (RSO 1969) ellipsoid and the Greenwich prime meridian. Kertau (RSO) is a geodetic datum for Metrication of RSO grid. It was defined by information from Defence Geographic Centre. Adopts metric conversion of 0.914398 metres per yard exactly. This is a truncation of the Sears 1922 ratio.

123) **Revised Nahrwan**
IHO Definition: A geodetic datum first defined in 1967 suitable for use in Arabian Gulf; Qatar — offshore; United Arab Emirates (UAE) — Abu Dhabi; Dubai; Sharjah; Ajman; Fujairah; Ras Al Kaimah; Umm Al Qaiwain — onshore and offshore. Nahrwan 1967 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Nahrwan 1967 origin is Fundamental point: Nahrwan south base. Latitude: 3319’10.87’’N, longitude: 4443’25.54’’E (of Greenwich). Nahrwan 1967 is a geodetic datum for Topographic mapping. In Iraq, replaces Nahrwan 1934.

124) **GGRS 76 (Greece)**
IHO Definition: A geodetic datum first defined in 1987 suitable for use in Greece — onshore. Greek Geodetic Reference System 1987 references the GRS 1980 ellipsoid and the Greenwich prime meridian. Greek Geodetic Reference System 1987 origin is Fundamental point: Dionysos. Latitude 3804’33.8’’N, longitude 2355’51.0’’E of Greenwich; geoid height 7.0 m. Greek Geodetic Reference System 1987 is a geodetic datum for Topographic mapping. It was defined by information from L. Portokalakis; Public Petroleum Corporation of Greece. Replaced (old) Greek datum. Oil industry work based on ED50.

125) **Nouvelle Triangulation de France**
IHO Definition: A geodetic datum first defined in 1895 suitable for use in France — onshore — mainland and Corsica. Nouvelle Triangulation Francaise references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. Nouvelle Triangulation Francaise origin is Fundamental point: Pantheon. Latitude: 4850’46.522’’N, longitude: 220’48.667’’E (of Greenwich). Nouvelle Triangulation Francaise is a geodetic datum for Topographic mapping.

126) **RT 90 (Sweden)**
IHO Definition: A geodetic datum first defined in 1982 suitable for use in Sweden — onshore and offshore. Rikets koordinatsystem 1990 references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Rikets koordinatsystem 1990 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from National Land Survey of Sweden Replaces RT38 adjustment (datum code 6308).

127) **Geocentric Datum of Australia**
IHO Definition: A geodetic datum first defined in 1994 suitable for use in Australia including Lord Howe Island, Macquarie Islands, Ashmore and Cartier Islands, Christmas Island, Cocos (Keeling) Islands, Norfolk Island. All onshore and offshore. Geocentric Datum of Australia 1994 references the GRS 1980 ellipsoid and the Greenwich prime meridian. Geocentric Datum of Australia 1994 origin is ITRF92 at epoch 1994.0. Geocentric Datum of Australia 1994 is a geodetic datum for Topographic mapping, geodetic survey. It was defined by information from Australian Surveying and Land Information Group Internet WWW page. http://www.auslig.gov.au/geodesy/datums/gda.htm#specs Coincident with WGS84 to within 1 metre.

128) **BJZ54 (A954 Beijing Coordinates)**
IHO Definition: A geodetic datum first defined in 1954 suitable for use in China — onshore. Beijing 1954 references the Krassowsky 1940 ellipsoid and the Greenwich prime meridian. Beijing 1954 origin is Pulkovo, transferred through Russian triangulation. Beijing 1954 is a geodetic datum for Topographic mapping. It was defined by information from Chinese Science Bulletin, 2009, 54:2714-2721 Scale determined through three baselines in northeast China. Discontinuities at boundaries of adjustment blocks. From 1982 replaced by Xian 1980 and New Beijing.

129) **Modified BJZ54**
IHO Definition: Modified BJZ54 datum.

130) **GDZ80**
IHO Definition: GDZ80 datum.

131) **Local Datum**
IHO Definition: An arbitrary datum defined by a local harbour authority, from which levels and tidal heights are measured by this authority.

Remarks: All necessary information for conversion of geographic coordinates from most of the Geodetic Datums in the above list to WGS-84 is contained in the ‘User’s Handbook on Datum Transformations involving WGS-84’, prepared by the US Defense Mapping Agency and which is available from the IHB as IHO Publication S-60 (English and French Versions), along with an associated standard datum transformation software on floppy disk called ‘MADTRAN’.

6.23 Reference Point

Table 6-23 — Reference Point

IHO Definition: -
Remarks:

6.24 Signal Status

Table 6-24 — Signal Status

IHO Definition: The indication of an element of a signal sequence being a period of light/sound or eclipse/silence.

1) **Lit/Sound**
   IHO Definition: The indication of an element of a signal sequence being a period of light or sound.
2) **Eclipsed/Silent**
   IHO Definition: The indication of an element of a signal sequence being a period of eclipse or silence.

Remarks: No remarks.

6.25 Signal Duration

Table 6-25 — Signal Duration

IHO Definition: The time occupied by a single instance of light/sound or eclipse/silence in a signal sequence.

Remarks: No remarks.

6.26 Sector Line Length

Table 6-26 — Sector Line Length

IHO Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector line length specifies the displayed length of the line, in ground units, defining the limit of the sector.

Remarks: No remarks.

6.27 Sector Bearing

Table 6-27 — Sector Bearing

IHO Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector bearing specifies the limit of the sector.

Remarks: The values given to the common limits of adjacent sectors should be identical. The orientation of bearing is from seaward to the central object. This conforms with the method used in 'List of Lights' publications. A generic term such as 'to shore' cannot be used; a specific bearing must be encoded. Where a light sector limit is defined as 'to the shore', it should be encoded using a value that ensures that, when the limit is drawn, it will fall entirely on land.

6.28 Wave Length Value

Table 6-28 — Wave Length Value

IHO Definition: The distance between two successive peaks (or other points of identical phase) on an electromagnetic wave.

Remarks: No remarks.

6.29 Radar Band

Table 6-29 — Radar Band

IHO Definition: The band code character of the electromagnetic spectrum within which radar wave lengths lie.

Remarks: No remarks.

6.30 Number of Features

Table 6-30 — Number of Features

IHO Definition: The number of features of identical character that exist as a co-located group.

Remarks: No remarks.

6.31 Multiplicity Known

Table 6-31 — Multiplicity Known

IHO Definition: The number of features of identical character that exist as a co-located group is or is not known.

Remarks: No remarks.

6.32 Sector Extension

Table 6-32 — Sector Extension

IHO Definition: Distance in screen millimetres (mm) by which a sector arc is extended beyond the default.

Remarks: The sector extension is calculated by ENC production software systems.

6.33 Moire Effect

Table 6-33 — Moire Effect

IHO Definition: A short range (up to 2km) type of directional light. Sodium lighting gives a yellow background to a screen on which a vertical black line will be seen by an observer on the centre line.

Remarks: No remarks.

6.34 Category of Cable

Table 6-34 — Category of Cable

IHO Definition: Classification of the cable based on the services provided.

1) **Power Line**
   IHO Definition: A cable that transmits or distributes electrical power.
3) **Transmission Line**
   IHO Definition: Multiple un-insulated cables usually supported by steel lattice towers. Such features are generally more prominent than normal power lines.
4) **Telephone**
   IHO Definition: A cable that transmits telephone signals.
5) **Telegraph**
   IHO Definition: An apparatus, system or process for communication at a distance by electric transmission over wire.
6) **Mooring Cable**
   IHO Definition: A chain or very strong fibre or wire rope used to anchor or moor vessels or buoys.
7) **Ferry**
   IHO Definition: A vessel for transporting passengers, vehicles, and/or goods across a stretch of water, especially as a regular service.
8) **Fibre Optic Cable**

IHO Definition: A cable made of glass or plastic fiber designed to guide light along its length, fibre optic cables are widely used in fiber-optic communication, which permits transmission over longer distances and at higher data rates than other forms of communication.
Remarks: No remarks.

6.35 Shackle Type

Table 6-35 — Shackle Type

IHO Definition: Types of shackle.

1) **forelock shackles**
   IHO Definition: -
2) **clenching shackles**
   IHO Definition: -
3) **bolt shackles**
   IHO Definition: -
4) **screw pin shackles**
   IHO Definition: -
5) **kenter shackle**
   IHO Definition: -
6) **quick release link**
   IHO Definition: -

Remarks:

6.36 Category of Pile

Table 6-36 — Category of Pile

IHO Definition: Classification of pile, driven into the earth as a foundation or support for a structure.

1) **Stake**
   IHO Definition: An elongated wood or metal pole embedded in the seabed to serve as a marker or support.
3) **Post**
   IHO Definition: A vertical piece of timber, metal or concrete forced into the earth or sea bed.
4) **Tripodal**
   IHO Definition: A single structure comprising 3 or more piles held together (sections of heavy timber, steel or concrete), and forced into the earth or sea bed.
5) **Piling**
   IHO Definition: A number of piles, usually in a straight line, and usually connected or bolted together.
6) **Area of Piles**
   IHO Definition: A number of piles, usually in a straight line, but not connected by structural members.
7) **Pipe**
   IHO Definition: A vertical hollow cylinder of metal, wood, or other material forced into the earth or seabed.

Remarks: No remarks.

6.37 Category of Silo/Tank

Table 6-37 — Category of Silo/Tank

IHO Definition: Classification based on the product for which a silo or tank is used.

1) **Silo in General**
   IHO Definition: A large storage structure used for storing loose materials.
2) **Tank in General**
   IHO Definition: A fixed structure for storing liquids.
3) **Grain Elevator**
   IHO Definition: A storage building for grain. Usually a tall frame, metal or concrete structure with an especially compartmented interior.
4) **Water Tower**
   IHO Definition: A tower supporting an elevated storage tank of water.

Remarks: No remarks.

6.38 Building Shape

Table 6-38 — Building Shape

IHO Definition: The specific shape of the building.

5) **High-Rise Building**
IHO Definition: A building having many storeys.
6) **Pyramid**
IHO Definition: A polyhedron of which one face is a polygon of any number of sides, and the other faces are triangles with a common vertex.
7) **Cylindrical**
IHO Definition: Shaped like a cylinder, which is a solid geometrical figure generated by straight lines fixed in direction and describing with one of its points a closed curve, especially a circle.
8) **Spherical**
IHO Definition: Shaped like a sphere, which is a body the surface of which is at all points equidistant from the centre.
9) **Cubic**
IHO Definition: A shape the sides of which are six equal squares; a regular hexahedron.

Remarks: No remarks.

6.39 Product

Table 6-39 — Product

IHO Definition: The various substances which are transported, stored or exploited.

1) **Oil**
IHO Definition: A thick, slippery liquid that will not dissolve in water, usually petroleum based in the context of storage tanks.
2) **Gas**
IHO Definition: A substance with particles that can move freely, usually a fuel substance in the context of storage tanks.
3) **Water**
IHO Definition: A colourless, odourless, tasteless liquid that is a compound of hydrogen and oxygen.
4) **Stone**
IHO Definition: A general term for rock and rock fragments ranging in size from pebbles and gravel to boulders or large rock masses.
5) **Coal**
IHO Definition: A hard black mineral that is burned as fuel.
6) **Ore**
IHO Definition: A solid rock or mineral from which metal is obtained.
7) **Chemicals**
IHO Definition: Any substance obtained by or used in a chemical process.
8) **Drinking Water**
IHO Definition: Water that is suitable for human consumption.
9) **Milk**
IHO Definition: A white fluid secreted by female mammals as food for their young.
10) **Bauxite**
IHO Definition: A mineral from which aluminum is obtained.
11) **Coke**
IHO Definition: A solid substance obtained after gas and tar have been extracted from coal, used as a fuel.
12) **Iron Ingots**
IHO Definition: An oblong lump of cast iron metal.
13) **Salt**
IHO Definition: Sodium chloride obtained from mines or by the evaporation of sea water.
14) **Sand**
IHO Definition: Loose material consisting of small but easily distinguishable, separate grains, between 0.0625 and 2.000 millimetres in diameter.
15) **Timber**
IHO Definition: Wood prepared for use in building or carpentry.

16) **Sawdust/Wood Chips**
IHO Definition: Powdery fragments of wood made in sawing timber or coarse chips produced for use in manufacturing pressed board.
17) **Scrap Metal**
IHO Definition: Discarded metal suitable for being reprocessed.
18) **Liquefied Natural Gas**
IHO Definition: Natural gas that has been liquefied for ease of transport by cooling the gas to -162 Celsius.
19) **Liquefied Petroleum Gas**
IHO Definition: A compressed gas consisting of flammable light hydrocarbons and derived from petroleum.
20) **Wine**
IHO Definition: The fermented juice of grapes.
21) **Cement**
IHO Definition: A substance made of powdered lime and clay, mixed with water.
22) **Grain**
IHO Definition: A small hard seed, especially that of any cereal plant such as wheat, rice, corn, rye etc.
23) **Electricity**
IHO Definition: Electric charge or current.
24) **Ice**
IHO Definition: The solid form of water.
25) **Clay**
IHO Definition: (Particles of less than 0.002mm); stiff, sticky earth that becomes hard when baked.

Remarks: No remarks.

6.40 Category of Offshore Platform

Table 6-40 — Category of Offshore Platform

IHO Definition: Classification of an offshore raised structure.

1) **Oil Rig**
IHO Definition: A temporary mobile structure, either fixed or floating, used in the exploration stages of oil and gas fields.
2) **Production Platform**
IHO Definition: A term used to indicate a permanent offshore structure equipped to control the flow of oil or gas. It does not include entirely submarine structures.
3) **Observation/Research Platform**
IHO Definition: A platform from which one's surroundings or events can be observed, noted or recorded such as for scientific study.
4) **Articulated Loading Platform**
IHO Definition: A metal lattice tower, buoyant at one end and attached at the other by a universal joint to a concrete filled base on the sea bed. The platform may be fitted with a helicopter platform, emergency accommodation and hawser/hose retrieval.
5) **Single Anchor Leg Mooring**
IHO Definition: A rigid frame or tube with a buoyancy device at its upper end , secured at its lower end to a universal joint on a large steel or concrete base resting on the sea bed, and at its upper end to a mooring buoy by a chain or wire.
6) **Mooring Tower**
IHO Definition: A platform secured to the sea bed and surmounted by a turntable to which ships moor.
7) **Artificial Island**
IHO Definition: A man-made structure usually built for the exploration or exploitation of marine resources, marine scientific research, tidal observations, etc.
8) **Floating Production, Storage and Off-Loading Vessel**
IHO Definition: An offshore oil/gas facility consisting of a moored tanker/barge by which the product is extracted, stored and exported.
9) **Accommodation Platform**
IHO Definition: A platform used primarily for eating, sleeping and recreation purposes.
10) **Navigation, Communication and Control Buoy**
IHO Definition: A floating structure with control room, power and storage facilities, attached to the sea bed by a flexible pipeline and cables.
11) **Floating Oil Tank**
IHO Definition: A floating structure, anchored to the seabed, for storing oil.

Remarks: No remarks.

6.41 Horizontal Width

Table 6-41 — Horizontal Width

IHO Definition: A measurement of the shorter of two linear axis. Remarks: No remarks.

6.42 Horizontal Length

Table 6-42 — Horizontal Length

IHO Definition: A measurement of the longer of two linear axis. Remarks: No remarks.

6.43 Category of Cardinal Mark

Table 6-43 — Category of Cardinal Mark

IHO Definition: The four quadrants (north, east, south and west) are bounded by the true bearings NW-NE, NE-SE, SE-SW and SW-NW taken from the point of interest. A cardinal mark is named after the quadrant in which it is placed. The name of the cardinal mark indicates that it should be passed to the named side of the mark. 1) **North Cardinal Mark** IHO Definition: Quadrant bounded by the true bearing NW-NE taken from the point of interest; it should be passed to the north side of the mark. 2) **East Cardinal Mark** IHO Definition: Quadrant bounded by the true bearing NE-SE taken from the point of interest. It should be passed to the east side of the mark. 3) **South Cardinal Mark** IHO Definition: Quadrant bounded by the true bearing SE-SW taken from the point of interest; it should be passed to the south side of the mark. 4) **West Cardinal Mark** IHO Definition: Quadrant bounded by the true bearing SW-NW taken from the point of interest; it should be passed to the west side of the mark. Remarks: Cardinal marks are used in conjunction with the compass to indicate where a mariner will find safe navigable water. Cardinal marks do not have a distinctive shape but are normally pillar or spar. They are always painted in yellow and black horizontal bands and their distinctive double cone top-marks are always black. (Note that such top-marks are encoded as separate TOPMAR objects). Cardinal marks may also have a special system of flashing white lights and if such lights are fitted they are encoded as separate LIGHTS objects.

6.44 Type of Light

Table 6-44 — Type of Light

IHO Definition: Type of light equipment. Remarks: For example bulb type (250mm, 300mm, 400mm), LED type (integral type, 200, 200HI, 250, 300, 350).

6.45 NMEAString

Table 6-45 — NMEAString

IHO Definition: NMEA string captured from the positioning device.(proposed by CCG) Remarks:

6.46 Light Characteristic

Table 6-46 — Light Characteristic

IHO Definition: The distinct character, such as **fixed**, **flashing**, or **occulting**, which is given to each light to avoid confusion with neighbouring ones.

1) **Fixed**
IHO Definition: A signal light that shows continuously, in any given direction, with constant luminous intensity and colour.
2) **Flashing**
IHO Definition: A rhythmic light in which the total duration of light in a period is clearly shorter than the total duration of darkness and all the appearances of light are of equal duration.
3) **Long-Flashing**
IHO Definition: A single-flashing light in which a single flash of not less than two seconds duration is regularly repeated.
4) **Quick-Flashing**
IHO Definition: A rhythmic light in which flashes are repeated at a rate of not less than 50 flashes per minutes but less than 80 flashes per minutes. It may be:
— Continuous quick-flashing: A quick-flashing light in which a flash is regularly repeated.
— Group quick-flashing: A quick-flashing light in which a group of two or more flashes, which are specified in number, is regularly repeated.
5) **Very Quick-Flashing**
IHO Definition: A rhythmic light in which flashes are repeated at a rate of not less than 80 flashes per minute but less than 160 flashes per minute. It may be:
— Continuous very quick-flashing: A very quick-flashing light in which a flash is regularly repeated.
— Group very quick-flashing: A very quick-flashing light in which a group of two or more flashes, which are specified in number, is regularly repeated.
6) **Continuous Ultra Quick-Flashing**
IHO Definition: A rhythmic light in which flashes are regularly repeated at a rate of not less than 160 flashes per minute.
7) **Isophased**
IHO Definition: A light with all durations of light and darkness equal.
8) **Occulting**
IHO Definition: A rhythmic light in which the total duration of light in a period is clearly longer than the total duration of darkness and all the eclipses are of equal duration. It may be:
— Single-occulting: An occulting light in which an eclipse is regularly repeated.
— Group-occulting: An occulting light in which a group of two or more eclipses, which are specified in number, is regularly repeated.
— Composite group-occulting: An occulting light in which a sequence of groups of one or more eclipses, which are specified in number, is regularly repeated, and the groups comprise different numbers of eclipses.
— Composite group-occulting: An occulting light in which a sequence of groups of one or more eclipses, which are specified in number, is regularly repeated, and the groups comprise different numbers of eclipses.
9) **Interrupted Quick Flashing**
IHO Definition: A quick light in which the sequence of flashes is interrupted by regularly repeated eclipses of constant and long duration.
10) **Interrupted Very Quick Flashing**
IHO Definition: A light in which the very rapid alterations of light and darkness are interrupted at regular intervals by eclipses of long duration.
11) **Interrupted Ultra Quick-Flashing**
IHO Definition: A light in which the ultra quick flashes (160 or more per minute) are interrupted at regular intervals by eclipses of long duration.
12) **Morse**
IHO Definition: A rhythmic light in which appearances of light of two clearly different durations are grouped to represent a character or characters in the Morse code.
13) **Fixed and Flash**
IHO Definition: A rhythmic light in which a fixed light is combined with a flashing light of higher luminous intensity.
14) **Flash and Long-Flash**
IHO Definition: A rhythmic light in which a flashing light is combined with a long-flashing light of higher luminous intensity.
15) **Occulting and Flash**
IHO Definition: A rhythmic light in which an occulting light is combined with a flashing light of higher luminous intensity.
16) **Fixed and Long-Flash**

IHO Definition: A rhythmic light in which a fixed light is combined with a long-flashing light of higher luminous intensity.

17) **Occulting Alternating**
IHO Definition: An alternating light in which the total duration of light in each period is clearly longer than the total duration of darkness and in which the intervals of darkness (occultations) are all of equal duration.
18) **Long-Flash Alternating**
IHO Definition: An alternating single-flashing light in which an appearance of light of not less than two seconds duration is regularly repeated.
19) **Flash Alternating**
IHO Definition: An alternating rhythmic light in which the total duration of light in a period is clearly shorter than the total duration of darkness and all the appearances of light are of equal duration.
20) **Group Alternating**
IHO Definition: Occulting light in which the occultations are combined in groups, each group including the same number of occultations, and in which the groups are repeated at regular intervals.
25) **Quick-Flash Plus Long-Flash**
IHO Definition: A rhythmic light in which a group of quick flashes is followed by one or more long flashes in a regularly repeated sequence with a regular periodicity.
26) **Very Quick-Flash Plus Long-Flash**
IHO Definition: A rhythmic light in which a group of very quick flashes is followed by one or more long flashes in a regularly repeated sequence with a regular periodicity.
27) **Ultra Quick-Flash Plus Long-Flash**
IHO Definition: A rhythmic light in which a group of ultra quick flashes is followed by one or more long flashes in a regularly repeated sequence with a regular periodicity.
28) **Alternating**
IHO Definition: A signal light that shows, in any given direction, two or more colours in a regularly repeated sequence with a regular periodicity.
29) **Fixed and Alternating Flashing**
IHO Definition: A rhythmic light in which a fixed light is combined with a flashing light of higher luminous intensity and different colour.

Remarks: No remarks.

6.47 Category Of Power Source

Table 6-47 — Category Of Power Source

IHO Definition:

1) **battery**
IHO Definition:
2) **generator**
IHO Definition:
3) **solar panel**
IHO Definition:
4) **electrical service**
IHO Definition:

Remarks:

6.48 Category Of Synthetic AIS Aid To Navigation

Table 6-48 — Category Of Synthetic AIS Aid To Navigation

IHO Definition:

1) **predicted**
IHO Definition:
2) **monitored**
IHO Definition:

Remarks:

6.49 Category Of Physical AIS Aid To Navigation

Table 6-49 — Category Of Physical AIS Aid To Navigation

IHO Definition:

1) **Physical AIS Type 1**
   IHO Definition:
2) **Physical AIS Type 2**
   IHO Definition:
3) **Physical AIS Type 3**
   IHO Definition:

Remarks:

6.50 Virtual AIS Aid to Navigation Type

Table 6-50 — Virtual AIS Aid to Navigation Type

IHO Definition: A purpose of a virtual AIS Aid to Navigation.

1) **North Cardinal**
   IHO Definition: Indicates that it should be passed to the north side of the aid.
2) **East Cardinal**
   IHO Definition: Indicates that it should be passed to the east side of the aid.
3) **South Cardinal**
   IHO Definition: Indicates that it should be passed to the south side of the aid.
4) **West Cardinal**
   IHO Definition: Indicates that it should be passed to the west side of the aid.
5) **Port Lateral**
   IHO Definition: Indicates the port boundary of a navigational channel or suggested route when proceeding in the conventional direction of buoyage.
6) **Starboard Lateral**
   IHO Definition: Indicates the starboard boundary of a navigational channel or suggested route when proceeding in the conventional direction of buoyage.
7) **Preferred Channel to Port**
   IHO Definition: At a point where a channel divides, when proceeding in the conventional direction of buoyage, the preferred channel (or primary route) is indicated by a modified port-hand lateral mark.
8) **Preferred Channel to Starboard**
   IHO Definition: At a point where a channel divides, when proceeding in the conventional direction of buoyage, the preferred channel (or primary route) is indicated by a modified starboard-hand lateral mark.
9) **Isolated Danger**
   IHO Definition: A mark used alone to indicate a dangerous reef or shoal. The mark may be passed on either hand.
10) **Safe Water**
    IHO Definition: Indicates that there is navigable water around the mark.
11) **Special Purpose**
    IHO Definition: A special purpose aid is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notice to Mariners.
12) **New Danger Marking**
    IHO Definition: A mark used to indicate the existence of a recently identified new danger, such as a wreck.

Remarks: No remarks.

6.51 MMSI Code

Table 6-51 — MMSI Code

IHO Definition: The Maritime Mobile Service Identity (MMSI) Code is formed of a series of nine digits which are transmitted over the radio path in order to uniquely identify ship stations, ship earth stations, coast stations, coast earth stations, and group calls. These identities are formed in such a way that the identity or part thereof can be used by telephone and telex subscribers connected to the general telecommunications network principally to call ships automatically.

Remarks: No remarks.

6.52 Category of Radar Transponder Beacon

Table 6-52 — Category of Radar Transponder Beacon

IHO Definition: Classification of radar transponder beacon based on functionality.

1) **Ramark, Radar Beacon Transmitting Continuously**
   IHO Definition: A radar marker beacon which continuously transmits a signal appearing as a radial line on a radar screen, the line indicating the direction of the beacon. Ramarks are intended primarily for marine use. The name 'ramark' is derived from the words radar marker.
2) **Racon, Radar Transponder Beacon**
   IHO Definition: A radar beacon which returns a coded signal which provides identification of the beacon, as well as range and bearing. The range and bearing are indicated by the location of the first character received on the radar screen. The name 'racon' is derived from the words radar beacon.
3) **Leading Racon/Radar Transponder Beacon**
   IHO Definition: A radar beacon that may be used (in conjunction with at least one other radar beacon) to indicate a leading line.

Remarks: No remarks.

6.53 Topmark/Daymark Shape

Table 6-53 — Topmark/Daymark Shape

IHO Definition: The shape a topmark or daymark exhibits.

1) **Cone (Point Up)**
   IHO Definition: Is where the vertex points up. A cone is a solid figure generated by straight lines drawn from a fixed point (the vertex) to a circle in a plane not containing the vertex. Cones are commonly used as International Association of Lighthouse Authorities — IALA topmarks, lateral.
2) **Cone (Point Down)**
   IHO Definition: Is where the vertex points down. A cone is a solid figure generated by straight lines drawn from a fixed point (the vertex) to a circle in a plane not containing the vertex. Cones are commonly used as International Association of Lighthouse Authorities — IALA topmarks, lateral.
3) **Sphere**
   IHO Definition: A curved surface all points of which are equidistant from a fixed point within, called the centre.
4) **2 Spheres**
   IHO Definition: Two spheres, one above the other. Two black spheres are commonly used as an International Association of Lighthouse Authorities — IALA topmark (isolated danger).
5) **Cylinder**
   IHO Definition: A solid geometrical figure generated by straight lines fixed in direction and describing with one of point a closed curve, especially a circle (in which case the figure is circular cylinder, it's ends being parallel circles). Cylinders are commonly used as International Association of Lighthouse Authorities — IALA topmarks lateral.
6) **Board**
   IHO Definition: Usually of rectangular shape, made from timber or metal and used to provide a contrast with the natural background of a daymark. The actual daymark is often painted on to this board.
7) **X-Shaped**
   IHO Definition: Having a shape or a cross-section like the capital letter X. An x-shape as an International Association of Lighthouse Authorities — IALA topmark should be 3 dimensional in shape. It is made of at least three crossed bars.
8) **Upright Cross**
   IHO Definition: A cross with one vertical member and one horizontal member; that is, similar in shape to the character '+'.
9) **Cube (Point Up)**
   IHO Definition: A cube standing on one of its vertexes. A cube is a solid contained by six equal squares, a regular hexahedron.
10) **2 Cones (Point to Point)**
    IHO Definition: 2 cones, one above the other, with their vertices together in the centre.
11) **2 Cones (Base to Base)**
    IHO Definition: 2 cones, one above the other, with their bases together in the centre and their vertices pointing up and down.
12) **Rhombus**
    IHO Definition: A plane figure having four equal sides and equal opposite angles (two acute and two obtuse); an oblique equilateral parallelogram.
13) **2 Cones (Points Upward)**

IHO Definition: 2 cones, one above the other, with their vertices pointing up.
14) **2 Cones (Points Downward)**
IHO Definition: 2 cones, one above the other, with their vertices pointing down.
15) **Besom (Point Up)**
IHO Definition: Besom: A bundle of rods or twigs. Perch: A staff placed on top of a buoy, rock or shoal as a mark for navigation. A besom, point up is where the thicker (untied) end of the besom is at the bottom.
16) **Besom (Point Down)**
IHO Definition: Besom: A bundle of rods or twigs. Perch: A staff placed on top of a buoy, rock or shoal as a mark for navigation. A besom, point down is where the thinner (tied) end of the besom is at the bottom.
17) **Flag**
IHO Definition: A flag mounted on a short pole.
18) **Sphere Over a Rhombus**
IHO Definition: A sphere located above a rhombus.
19) **Square**
IHO Definition: A plane figure with four right angles and four equal straight sides.
20) **Rectangle (Horizontal)**
IHO Definition: Where the two longer opposite sides are standing horizontally. A rectangle is a plane figure with four right angles and four straight sides, opposite sides being parallel and equal in length.
21) **Rectangle (Vertical)**
IHO Definition: Where the two longer opposite sides are standing vertically. A rectangle is a plane figure with four right angles and four straight sides, opposite sides being parallel and equal in length.
22) **Trapezium (Up)**
IHO Definition: A quadrilateral having one pair of opposite sides parallel, and which stands on its longer parallel side.
23) **Trapezium (Down)**
IHO Definition: A quadrilateral having one pair of opposite sides parallel, and which stands on its shorter parallel side.
24) **Triangle (Point Up)**
IHO Definition: A figure having three angles and three sides, and which has a vertex at the top.
25) **Triangle (Point Down)**
IHO Definition: A figure having three angles and three sides, and which has a side at the top.
26) **Circle**
IHO Definition: A perfectly round plane figure whose circumference is everywhere equidistant from its centre.
27) **Two Upright Crosses (One Over the Other)**
IHO Definition: Two upright crosses, generally vertically disposed one above the other.
28) **T-Shape**
IHO Definition: Having a shape like the capital letter T.
29) **Triangle Pointing Up Over a Circle**
IHO Definition: A triangle, vertex uppermost, located above a circle.
30) **Upright Cross Over a Circle**
IHO Definition: An upright cross located above a circle.
31) **Rhombus Over a Circle**
IHO Definition: A rhombus located above a circle.
32) **Circle Over a Triangle Pointing Up**
IHO Definition: A circle located over a triangle, vertex uppermost.
33) **Other Shape (See Shape Information)**
IHO Definition: An uncommon and/or non-standardized shape as textually described using an associated attribute.
34) **Tubular**
IHO Definition: Having the form of or consisting of a tube.

Remarks: No remarks.

6.54 Orientation Value

Table 6-54 — Orientation Value

IHO Definition: The angular distance measured from true north to the major axis of the feature.

Remarks: No remarks.

6.55 Category of Special Purpose Mark

Table 6-55 — Category of Special Purpose Mark

IHO Definition: Classification of an aid to navigation which signifies some special purpose.

1) **Firing Danger Mark**
IHO Definition: A mark used to indicate a firing danger area, usually at sea.
2) **Target Mark**
IHO Definition: Any object toward which something is directed. The distinctive marking or instrumentation of a ground point to aid its identification on a photograph.
3) **Marker Ship Mark**
IHO Definition: A mark marking the position of a ship which is used as a target during some military exercise.
4) **Degaussing Range Mark**
IHO Definition: A mark used to indicate a degaussing range.
5) **Barge Mark**
IHO Definition: A mark of relevance to barges.
6) **Cable Mark**
IHO Definition: A mark used to indicate the position of submarine cables or the point at which they run on to the land.
7) **Spoil Ground Mark**
IHO Definition: A mark used to indicate the limit of a spoil ground.
8) **Outfall Mark**
IHO Definition: A mark used to indicate the position of an outfall or the point at which it leaves the land.
9) **ODAS**
IHO Definition: Ocean Data Acquisition System.
10) **Recording Mark**
IHO Definition: A mark used to record data for scientific purposes.
11) **Seaplane Anchorage Mark**
IHO Definition: A mark used to indicate a seaplane anchorage.
12) **Recreation Zone Mark**
IHO Definition: A mark used to indicate a recreation zone.
13) **Private Mark**
IHO Definition: A privately maintained mark.
14) **Mooring Mark**
IHO Definition: A mark indicating a mooring or moorings.
15) **LANBY**
IHO Definition: A large buoy designed to take the place of a lightship where construction of an offshore light station is not feasible.
16) **Leading Mark**
IHO Definition: Aids to navigation or other indicators so located as to indicate the path to be followed. Leading marks identify a leading line when they are in transit.
17) **Measured Distance Mark**
IHO Definition: A mark forming part of a transit indicating one end of a measured distance.
18) **Notice Mark**
IHO Definition: A notice board or sign indicating information to the mariner.
19) **TSS Mark**
IHO Definition: A mark indicating a Traffic Separation Scheme.
20) **Anchoring Prohibited Mark**
IHO Definition: A mark indicating an anchoring prohibited area.
21) **Berthing Prohibited Mark**
IHO Definition: A mark indicating that berthing is prohibited.
22) **Overtaking Prohibited Mark**
IHO Definition: A mark indicating that overtaking is prohibited.
23) **Two-Way Traffic Prohibited Mark**
IHO Definition: A mark indicating a one-way route.
24) **Reduced Wake Mark**
IHO Definition: A mark indicating that vessels must not generate excessive wake.
25) **Speed Limit Mark**
IHO Definition: A mark indicating that a speed limit applies.
26) **Stop Mark**
IHO Definition: A mark indicating the place where the bow of a ship must stop when traffic lights show red.
27) **General Warning Mark**
IHO Definition: A mark indicating that special caution must be exercised in the vicinity of the mark.
28) **Sound Ship's Siren Mark**
IHO Definition: A mark indicating that a ship should sound its siren or horn.

29) **Restricted Vertical Clearance Mark**
IHO Definition: A mark indicating the minimum vertical space available for passage.
30) **Maximum Vessel's Draught Mark**
IHO Definition: A mark indicating the maximum draught of vessel permitted.
31) **Restricted Horizontal Clearance Mark**
IHO Definition: A mark indicating the minimum horizontal space available for passage.
32) **Strong Current Warning Mark**
IHO Definition: A mark warning of strong currents.
33) **Berthing Permitted Mark**
IHO Definition: A mark indicating that berthing is allowed.
34) **Overhead Power Cable Mark**
IHO Definition: A mark indicating an overhead power cable.
35) **Channel Edge Gradient Mark**
IHO Definition: A mark indicating the gradient of the slope of a dredge channel edge.
36) **Telephone Mark**
IHO Definition: A mark indicating the presence of a telephone.
37) **Ferry Crossing Mark**
IHO Definition: A mark indicating that a ferry route crosses the ship route; often used with a 'sound ship's siren' mark.
39) **Pipeline Mark**
IHO Definition: A mark used to indicate the position of submarine pipelines or the point at which they run on to the land.
40) **Anchorage Mark**
IHO Definition: A mark indicating an anchorage area.
41) **Clearing Mark**
IHO Definition: A mark used to indicate a clearing line.
42) **Control Mark**
IHO Definition: A mark indicating the location at which a restriction or requirement exists.
43) **Diving Mark**
IHO Definition: A mark indicating that diving may take place in the vicinity.
44) **Refuge Beacon**
IHO Definition: A mark providing or indicating a place of safety.
45) **Foul Ground Mark**
IHO Definition: A mark indicating a foul ground.
46) **Yachting Mark**
IHO Definition: A mark installed for use by yachtsmen.
47) **Heliport Mark**
IHO Definition: A mark indicating an area where helicopters may land.
48) **GNSS Mark**
IHO Definition: A mark indicating a location at which a GNSS position has been accurately determined.
49) **Seaplane Landing Mark**
IHO Definition: A mark indicating an area where sea-planes land.
50) **Entry Prohibited Mark**
IHO Definition: A mark indicating that entry is prohibited.
51) **Work in Progress Mark**
IHO Definition: A mark indicating that work (generally construction) is in progress.
52) **Mark With Unknown Purpose**
IHO Definition: A mark whose detailed characteristics are unknown.
53) **Wellhead Mark**
IHO Definition: A mark indicating a borehole that produces or is capable of producing oil or natural gas.
54) **Channel Separation Mark**
IHO Definition: A mark indicating the point at which a channel divides separately into two channels.
55) **Marine Farm Mark**
IHO Definition: A mark indicating the existence of a fish, mussel, oyster or pearl farm/culture.
56) **Artificial Reef Mark**
IHO Definition: A mark indicating the existence or the extent of an artificial reef.
57) **Ice Mark**
IHO Definition: A mark, used year round, that may be submerged when ice passes through the area.
58) **Nature Reserve Mark**
IHO Definition: A mark used to define the boundary of a nature reserve.
59) **Fish Aggregating Device**
IHO Definition: A fish aggregating (or aggregation) device (FAD) is a man-made object used to attract ocean going pelagic fish such as marlin, tuna and mahi-mahi (dolphin fish). They usually consist of buoys or floats tethered to the ocean floor with concrete blocks.
60) **Wreck Mark**
IHO Definition: A mark used to indicate the existence of a wreck.
61) **Customs Mark**

IHO Definition: A mark used to indicate the existence of a customs checkpoint.
62) **Causeway Mark**
IHO Definition: A mark used to indicate the existence of a causeway.
63) **Wave Recorder**
IHO Definition: A surface following buoy used to measure wave activity.
64) **Jetski Prohibited**
IHO Definition: A mark indicating a jetski prohibited area.

Remarks: A mark may be a beacon, a buoy, a signpost or may take another form.

6.56 Estimated Range of Transmission

Table 6-56 — Estimated Range of Transmission

IHO Definition: The estimated range of a non-optical electromagnetic transmission.

Remarks: The estimated range (distance) assumes 'in vacuo' transmission and a standard antenna height of 5 metres. Thus it gives a hint to the mariner whether they are likely to receive transmission at a certain distance from an object.

6.57 Category of Radio Station

Table 6-57 — Category of Radio Station

IHO Definition: Classification of radio services offered by a radio station.

1) **Circular (Non-Directional) Marine or Aero-Marine Radiobeacon**
IHO Definition: A radio station which need not necessarily be manned, the emissions of which, radiated around the horizon, enable its bearing to be determined by means of the radio direction finder of a ship.
2) **Directional Radiobeacon**
IHO Definition: A special type of radiobeacon station the emissions of which are intended to provide a definite track for guidance.
3) **Rotating Pattern Radiobeacon**
IHO Definition: A special type of radiobeacon station emitting a beam of waves to which a uniform turning movement is given, the bearing of the station being determined by means of an ordinary listening receiver and a stop watch. Also referred to as a rotating loop radiobeacon.
4) **Consol Beacon**
IHO Definition: A type of long range position fixing beacon.
5) **Radio Direction-Finding Station**
IHO Definition: A radio station intended to determine only the direction of other stations by means of transmission from the latter.
6) **Coast Radio Station Providing QTG Service**
IHO Definition: A radio station which is prepared to provide QTG service; that is to say, to transmit upon request from a ship a radio signal, the bearing of which can be taken by that ship.
7) **Aeronautical Radiobeacon**
IHO Definition: A radio beacon designed for aeronautical use.
8) **Decca**
IHO Definition: The Decca Navigator System is a high accuracy, short to medium range radio navigational aid intended for coastal and landfall navigation.
9) **Loran C**
IHO Definition: A low frequency electronic position fixing system using pulsed transmissions at 100 Khz.
10) **Differential GNSS**
IHO Definition: A radiobeacon transmitting DGPS correction signals.
11) **Toran**
IHO Definition: An electronic position fixing system used mainly by aircraft.
12) **Omega**
IHO Definition: A long-range radio navigational aid which operates within the VLF frequency band. The system comprises eight land based stations.
13) **Syledis**
IHO Definition: A ranging position fixing system operating at 420-450 MHz over a range of up to 400 Km.
14) **Chaika**
IHO Definition: Chaika is a low frequency electronic position fixing system using pulsed transmissions at 100 Khz.
19) **Radio Telephone Station**

IHO Definition: The equipment needed at one station to carry on two way voice communication by radio waves only.

20) **AIS Base Station**
IHO Definition: An onshore AIS unit that monitors traffic in the waterways.

Remarks: A radiobeacon is a radio transmitter which emits a distinctive or characteristic signal on which a bearing may be taken.

6.58 Type of Environmental Observation Equipment

Table 6-58 — Type of Environmental Observation Equipment

IHO Definition: Type of sensor used to observe the environment. For example Anemometer, fog monitor, etc.

Remarks: No remarks.

6.59 Value of Maximum Range

Table 6-59 — Value of Maximum Range

IHO Definition: The extreme distance at which an feature can be seen or a signal detected.

Remarks: Does not apply to lights where the ‘value of nominal range’ should be used.

6.60 Signal Period

Table 6-60 — Signal Period

IHO Definition: The time occupied by an entire cycle of intervals of light and eclipse.

Remarks: No remarks.

6.61 Signal Output

Table 6-61 — Signal Output

IHO Definition: Strength of signal output.

Remarks: No remarks.

6.62 Signal Group

Table 6-62 — Signal Group

IHO Definition: The number of signals, the combination of signals or the morse character(s) within one period of full sequence.

Remarks: No remarks.

6.63 Signal Frequency

Table 6-63 — Signal Frequency

IHO Definition: The frequency of a signal.

Remarks: No remarks.

6.64 Category of Fog Signal

Table 6-64 — Category of Fog Signal

IHO Definition: Classification of the various means of generating the fog signal.

1) **Explosive**
IHO Definition: A signal produced by the firing of explosive charges.
2) **Diaphone**
IHO Definition: A diaphone uses compressed air and generally emits a powerful low-pitched sound, which often concludes with a brief sound of suddenly lowered pitch, termed the 'grunt'.
3) **Siren**
IHO Definition: A type of fog signal apparatus which produces sound by virtue of the passage of air through slots or holes in a revolving disk.
4) **Nautophone**
IHO Definition: A horn having a diaphragm oscillated by electricity.
5) **Reed**
IHO Definition: [1] A reed uses compressed air and emits a weak, high pitched sound.[2] Any of various water or marsh plants with a firm stem. (Concise Oxford English Dictionary)
6) **Tyfon**
IHO Definition: A diaphragm horn which operates under the influence of compressed air or steam.
7) **Bell**
IHO Definition: A ringing sound with a short range.
8) **Whistle**
IHO Definition: A distinctive sound made by a jet of air passing through an orifice. The apparatus may be operated automatically, by hand or by air being forced up a tube by waves acting on a buoy.
9) **Gong**
IHO Definition: A sound produced by vibration of a disc when struck.
10) **Horn**
IHO Definition: A horn uses compressed air or electricity to vibrate a diaphragm and exists in a variety of types which differ greatly in their sound and power.

Remarks: The classification 'horn' is the generic term for fog signals 'nautophone', 'reed' and 'tyfon'.

6.65 Value of Nominal Range

Table 6-65 — Value of Nominal Range

IHO Definition: The luminous range of a light in a homogenous atmosphere in which the meteorological visibility is 10 sea miles.

Remarks: No remarks.

6.66 Value of Luminous Range

Table 6-66 — Value of Luminous Range

IHO Definition: The greatest distance at which a light can be seen merely as a function of its luminous intensity, the meteorological visibility, and the sensitivity of the observer's eye.

Remarks: No remarks.

6.67 Value of Geographic Range

Table 6-67 — Value of Geographic Range

IHO Definition: The greatest distance at which a light can be seen as a function of the curvature of the earth and the heights of the light source and the observer.

Remarks: No remarks.

6.68 Major Light

Table 6-68 — Major Light

IHO Definition: A statement expressing if a light is considered to be a major light in terms of ECDIS display in a particular area.

Remarks: Major light is only intended to provide an indication to the ECDIS that the light is considered to be an important light in terms of its display. As such this is a cartographic attribute to aid the compiler in determining the most appropriate display for a light; it is not intended to be used as a formal classification method for lights.

6.69 Light Visibility

Table 6-69 — Light Visibility

IHO Definition: The specific visibility of a light, with respect to the light's intensity and ease of recognition.

1) **High Intensity**
   IHO Definition: Non-marine lights with a higher power than marine lights and visible from well off shore (often 'Aero' lights).
2) **Low Intensity**
   IHO Definition: Non-marine lights with lower power than marine lights.
3) **Faint**
   IHO Definition: A decrease in the apparent intensity of a light which may occur in the case of partial obstructions.
4) **Intensified**
   IHO Definition: A light in a sector is intensified (that is, has longer range than other sectors).
5) **Unintensified**
   IHO Definition: A light in a sector is unintensified (that is, has shorter range than other sectors).
6) **Visibility Deliberately Restricted**
   IHO Definition: A light sector is deliberately reduced in intensity, for example to reduce its effect on a built-up area.
7) **Obscured**
   IHO Definition: Said of the arc of a light sector designated by its limiting bearings in which the light is not visible from seaward.
8) **Partially Obscured**
   IHO Definition: This value specifies that parts of the sector are obscured.
9) **Visible in Line of Range**
   IHO Definition: Lights that must in line to be visible.

Remarks: No remarks.

6.70 Signal Generation

Table 6-70 — Signal Generation

IHO Definition: The mechanism used to generate a fog or light signal.

1) **Automatically**
   IHO Definition: Signal generation is initiated by a self regulating mechanism such as a timer or light sensor.
2) **By Wave Action**
   IHO Definition: The signal is generated by the motion of the sea surface such as a bell in a buoy.
3) **By Hand**
   IHO Definition: The signal is generated by a manually operated mechanism such as a hand cranked siren.
4) **By Wind**
   IHO Definition: The signal is generated by the motion of air such as a wind driven whistle.
5) **Radio Activated**
   IHO Definition: Activated by radio signal.
6) **Call Activated**
   IHO Definition: Activated by making a call to a manned station.

Remarks: No remarks.

6.71 Exhibition Condition of Light

Table 6-71 — Exhibition Condition of Light

IHO Definition: The outward display of the light. 1) **Light Shown Without Change of Character** IHO Definition: A light shown throughout the 24 hours without change of character. 2) **Daytime Light** IHO Definition: A light which is only exhibited by day. 3) **Fog Light** IHO Definition: A light which is exhibited in fog or conditions of reduced visibility. 4) **Night Light** IHO Definition: A light which is only exhibited at night. Remarks: No remarks.

6.72 Category of Light

Table 6-72 — Category of Light

IHO Definition: Classification of different light types. 1) **Directional Function** IHO Definition: A light illuminating a sector of very narrow angle and intended to mark a direction to follow. 4) **Leading Light** IHO Definition: A light associated with other lights so as to form a leading line to be followed. 5) **Aero Light** IHO Definition: An aero light is established for aeronautical navigation and may be of higher power than marine lights and visible from well offshore. 6) **Air Obstruction Light** IHO Definition: A light marking an obstacle which constitutes a danger to air navigation. 8) **Flood Light** IHO Definition: A broad beam light used to illuminate a structure or area. 9) **Strip Light** IHO Definition: A light whose source has a linear form generally horizontal, which can reach a length of several metres. 10) **Subsidiary Light** IHO Definition: A light placed on or near the support of a main light and having a special use in navigation. 11) **Spotlight** IHO Definition: A powerful light focused so as to illuminate a small area. 12) **Front** IHO Definition: Term used with leading lights to describe the position of the light on the lead as viewed from seaward. 13) **Rear** IHO Definition: Term used with leading lights to describe the position of the light on the lead as viewed from seaward. 14) **Lower** IHO Definition: Term used with leading lights to describe the position of the light on the lead as viewed from seaward. 15) **Upper** IHO Definition: Term used with leading lights to describe the position of the light on the lead as viewed from seaward. 17) **Emergency** IHO Definition: A light available as a backup to a main light which will be illuminated should the main light fail. 18) **Bearing Light** IHO Definition: A light which enables its approximate bearing to be obtained without the use of a compass. 19) **Horizontally Disposed** IHO Definition: A group of lights of identical character and almost identical position, that are disposed horizontally. 20) **Vertically Disposed** IHO Definition: A group of lights of identical character and almost identical position, that are disposed vertically. Remarks: All lights are considered to be marine lights unless the category of light indicates otherwise.

6.73 Traffic Flow

Table 6-73 — Traffic Flow

IHO Definition: Direction of vessels passing a reference point.

1) **Inbound**
   IHO Definition: Traffic flow in a general direction toward a port or similar destination.
2) **Outbound**
   IHO Definition: Traffic flow in a general direction away from a port or similar point of origin.
3) **One-Way**
   IHO Definition: Traffic flow in one general direction only.
4) **Two-Way**
   IHO Definition: Traffic flow in two generally opposite directions.

Remarks: No remarks.

6.74 Technique of Vertical Measurement

Table 6-74 — Technique of Vertical Measurement

IHO Definition: Survey method used to obtain depth information.

1) **Found by Echo Sounder**
   IHO Definition: The depth was determined by using an instrument that determines depth of water by measuring the time interval between emission of a sonic or ultrasonic signal and return of its echo from the bottom.
2) **Found by Side Scan Sonar**
   IHO Definition: The depth was computed from a record produced by active sonar in which fixed acoustic beams are directed into the water perpendicularly to the direction of travel to scan the seabed and generate a record of the seabed configuration.
3) **Found by Multi Beam**
   IHO Definition: The depth was determined by using a wide swath echo sounder that uses multiple beams to measure depths directly below and transverse to the ship's track.
4) **Found by Diver**
   IHO Definition: The depth was determined by a person skilled in the practice of diving.
5) **Found by Lead Line**
   IHO Definition: The depth was determined by using a line, graduated with attached marks and fastened to a sounding lead.
6) **Swept by Wire-Drag**
   IHO Definition: The given area was determined to be free from navigational dangers to a certain depth by towing a buoyed wire at the desired depth by two launches, or a least depth was identified using the same technique.
7) **Found by Laser**
   IHO Definition: The depth was determined by using an instrument that measures distance by emitting timed pulses of laser light and measuring the time between emission and reception of the reflected pulses.
8) **Swept by Vertical Acoustic System**
   IHO Definition: The given area has been swept using a system comprised of multiple echo sounder transducers attached to booms deployed from the survey vessel.
9) **Found by Electromagnetic Sensor**
   IHO Definition: The depth was determined by using an instrument that compares electromagnetic signals.
10) **Photogrammetry**
    IHO Definition: The science or art of obtaining reliable measurements from photographs.
11) **Satellite Imagery**
    IHO Definition: The depth was determined by using instruments placed aboard an artificial satellite.
12) **Found by Levelling**
    IHO Definition: The depth was determined by using levelling techniques to find the elevation of the point relative to a datum.
13) **Swept by Side Scan Sonar**
    IHO Definition: The given area was determined to be free from navigational dangers to a certain depth by towing a side scan sonar.
14) **Computer Generated**
    IHO Definition: The sounding was determined from a bottom model constructed using a computer.
15) **Found by LIDAR**
    IHO Definition: The depth was measured by using an instrument that measures distance by emitting timed pulses of laser light and measuring the time between emission and reception of the reflected pulses.
16) **Synthetic Aperture Radar**

IHO Definition: A radar with a synthetic aperture antenna which is composed of a large number of elementary transducing elements. The signals are electronically combined into a resulting signal equivalent to that of a single antenna of a given aperture in a given direction.

17) **Hyperspectral Imagery**
IHO Definition: Term used to describe the imagery derived from subdividing the electromagnetic spectrum into very narrow bandwidths. These narrow bandwidths may be combined with or subtracted from each other in various ways to form images useful in precise terrain or target analysis.

Remarks: -

6.75 Quality of Vertical Measurement

Table 6-75 — Quality of Vertical Measurement

IHO Definition: The reliability of the value of a sounding.

1) **Depth Known**
IHO Definition: The depth from the chart datum to the seabed (or to the top of a drying feature) is known.
2) **Depth or Least Depth Unknown**
IHO Definition: The depth from chart datum to the seabed, or the shoalest depth of the feature is unknown.
3) **Doubtful Sounding**
IHO Definition: A depth that may be less than indicated.
4) **Unreliable Sounding**
IHO Definition: A depth that is considered to be an unreliable value.
5) **No Bottom Found at Value Shown**
IHO Definition: Upon investigation the bottom was not found at this depth.
6) **Least Depth Known**
IHO Definition: The shoalest depth over a feature is of known value.
7) **Least Depth Unknown, Safe Clearance at Value Shown**
IHO Definition: The least depth over a feature is unknown, but there is considered to be safe clearance at this depth.
8) **Value Reported (Not Surveyed)**
IHO Definition: Depth value obtained from a report, but not fully surveyed.
9) **Value Reported (Not Confirmed)**
IHO Definition: Depth value obtained from a report, which it has not been possible to confirm.
10) **Maintained Depth**
IHO Definition: The depth at which a channel is kept by human influence, usually by dredging.
11) **Not Regularly Maintained**
IHO Definition: Depths may be altered by human influence, but will not be routinely maintained.

Remarks: -

6.76 Maximal Permitted Draught

Table 6-76 — Maximal Permitted Draught

IHO Definition: The maximal permitted draught of a vessel or convoy according to the particular article/clause of the applicable law/regulation.

Remarks: -

6.77 Depth Range Minimum Value

Table 6-77 — Depth Range Minimum Value

IHO Definition: The minimum (shoalest) value of a depth range.

Remarks: Where the area dries, the value is negative.

6.78 Based On Fixed Marks

Table 6-78 — Based On Fixed Marks

IHO Definition: A straight route (known as a recommended track, range or leading line), which comprises: a. at least two structures (usually beacons or daymarks) and/or natural features, which may carry lights and/or top-marks. The structures/features are positioned so that when observed to be in line, a vessel can follow a known bearing with safety. (Adapted from International Association of Lighthouse Authorities — IALA Aids to Navigation Guide, 1990); or b. a single structure or natural feature, which may carry lights and/or a topmark, and a specified bearing which can be followed with safety. (S-57 Edition 3.1, Appendix A Chapter 2, Page 2.72, November 2000, as amended). Remarks: No remarks.

6.79 Category of Navigation Line

Table 6-79 — Category of Navigation Line

IHO Definition: Classification of route guidance given to vessels. 1) **Clearing Line** IHO Definition: A straight line that marks the boundary between a safe and a dangerous area or that passes clear of a navigational danger. 2) **Transit Line** IHO Definition: A line passing through one or more fixed marks. 3) **Leading Line Bearing a Recommended Track** IHO Definition: A line passing through one or more clearly defined objects, along the path of which a vessel can approach safely up to a certain distance off. Remarks: No remarks.

6.80 Category of Lateral Mark

Table 6-80 — Category of Lateral Mark

IHO Definition: Classification of lateral marks in the IALA Buoyage System. 1) **Port-Hand Lateral Mark** IHO Definition: Indicates the port boundary of a navigational channel or suggested route when proceeding in the “conventional direction of buoyage”. 2) **Starboard-Hand Lateral Mark** IHO Definition: Indicates the starboard boundary of a navigational channel or suggested route when proceeding in the “conventional direction of buoyage”. 3) **Preferred Channel to Starboard Lateral Mark** IHO Definition: At a point where a channel divides, when proceeding in the “conventional direction of buoyage”, the preferred channel (or primary route) is indicated by a modified port-hand lateral mark. 4) **Preferred Channel to Port Lateral Mark** IHO Definition: At a point where a channel divides, when proceeding in the “conventional direction of buoyage”, the preferred channel (or primary route) is indicated by a modified starboard-hand lateral mark. 5) **Right-Hand Side of the Waterway** IHO Definition: Indicates the right-hand side of the inland waterway. 6) **Left-Hand Side of the Waterway** IHO Definition: Indicates the left-hand side of the inland waterway. 7) **Right-Hand Side of the Channel** IHO Definition: Indicates the right-hand side of a channel of an inland waterway. 8) **Left-Hand Side of the Channel** IHO Definition: Indicates the left-hand side of a channel of an inland waterway. 9) **Bifurcation of the Waterway** IHO Definition: Indicates a bifurcation of the inland waterway. 10) **Bifurcation of the Channel** IHO Definition: Indicates a bifurcation of a channel of an inland waterway. 11) **Channel Near the Right Bank** IHO Definition: Indicates that the channel is near the right bank. 12) **Channel Near the Left Bank** IHO Definition: Indicates that the channel is near the left bank.

13) **Channel Cross-Over to the Right Bank**
IHO Definition: Indicates that the channel crosses from the left to the right bank.
14) **Channel Cross-Over to the Left Bank**
IHO Definition: Indicates that the channel crosses from the right to the left bank.
15) **Danger Point or Obstacles at the Right-Hand Side**
IHO Definition: Indicates a danger point or obstacles at the right-hand side.
16) **Danger Point or Obstacles at the Left-Hand Side**
IHO Definition: Indicates a danger point or obstacles at the left-hand side.
17) **Turn Off at the Right-Hand Side**
IHO Definition: Indicates a turn off at the right-hand side.
18) **Turn Off at the Left-Hand Side**
IHO Definition: Indicates a turn off at the left-hand side.
19) **Junction at the Right-Hand Side**
IHO Definition: Indicates a junction at the right-hand side.
20) **Junction at the Left-Hand Side**
IHO Definition: Indicates a junction at the left-hand side.
21) **Harbour Entry at the Right-Hand Side**
IHO Definition: Indicates a harbour entry at the right-hand side.
22) **Harbour Entry at the Left-Hand Side**
IHO Definition: Indicates a harbour entry at the left-hand side.
23) **Bridge Pier Mark**
IHO Definition: Indicates a bridge pier in an inland waterway.
24) **Entry From a Lake to a Narrower Waterway, Right Bank**
IHO Definition: Indicates the right bank of the entry from a lake or a lake-like expansion to a section of the waterway which is narrower.
25) **Entry From a Lake to a Narrower Waterway, Left Bank**
IHO Definition: Indicates the left bank of the entry from a lake or a lake-like expansion to a section of the waterway which is narrower.
26) **Change Bank**
IHO Definition: Change bank.
27) **Continue Along Bank**
IHO Definition: Continue along bank.

Remarks: There are two international buoyage regions, A and B, between which lateral marks differ. When top-marks, retro reflectors and/or lights are fitted to these marks, they are encoded as separate features.

6.81 Manned Structure

Table 6-81 — Manned Structure

IHO Definition: An expression of the feature being permanently manned or not.

Remarks: No remarks.

6.82 Vertical Datum

Table 6-82 — Vertical Datum

IHO Definition: The reference level used for expressing the vertical measurements of points on the earth's surface. Also called datum level, reference plane, levelling datum, datum for sounding reduction, datum for heights.

1) **Mean Low Water Springs**
IHO Definition: The average height of the low waters of spring tides. This level is used as a tidal datum in some areas. Also called spring low water.
2) **Mean Lower Low Water Springs**
IHO Definition: The average height of lower low water springs at a place.
3) **Mean Sea Level**
IHO Definition: The average height of the surface of the sea at a tide station for all stages of the tide over a 19-year period, usually determined from hourly height readings measured from a fixed predetermined reference level.
4) **Lowest Low Water**
IHO Definition: An arbitrary level conforming to the lowest tide observed at a place, or somewhat lower.
5) **Mean Low Water**
IHO Definition: The average height of all low waters at a place over a 19-year period.

6) **Lowest Low Water Springs**
IHO Definition: An arbitrary level conforming to the lowest water level observed at a place at spring tides during a period of time shorter than 19 years.
7) **Approximate Mean Low Water Springs**
IHO Definition: An arbitrary level, usually within 0.3m from that of Mean Low Water Springs (MLWS).
8) **Indian Spring Low Water**
IHO Definition: An arbitrary tidal datum approximating the level of the mean of the lower low water at spring tides. It was first used in waters surrounding India.
9) **Low Water Springs**
IHO Definition: An arbitrary level, approximating that of mean low water springs (MLWS).
10) **Approximate Lowest Astronomical Tide**
IHO Definition: An arbitrary level, usually within 0.3m from that of Lowest Astronomical Tide (LAT).
11) **Nearly Lowest Low Water**
IHO Definition: An arbitrary level approximating the lowest water level observed at a place, usually equivalent to the Indian Spring Low Water (ISLW).
12) **Mean Lower Low Water**
IHO Definition: The average height of the lower low waters at a place over a 19-year period.
13) **Low Water**
IHO Definition: The lowest level reached at a place by the water surface in one oscillation. Also called low tide.
14) **Approximate Mean Low Water**
IHO Definition: An arbitrary level, usually within 0.3m from that of Mean Low Water (MLW).
15) **Approximate Mean Lower Low Water**
IHO Definition: An arbitrary level, usually within 0.3m from that of Mean Lower Low Water (MLLW).
16) **Mean High Water**
IHO Definition: The average height of all high waters at a place over a 19-year period.
17) **Mean High Water Springs**
IHO Definition: The average height of the high waters of spring tides. Also called spring high water.
18) **High Water**
IHO Definition: The highest level reached at a place by the water surface in one oscillation.
19) **Approximate Mean Sea Level**
IHO Definition: An arbitrary level, usually within 0.3m from that of Mean Sea Level (MSL).
20) **High Water Springs**
IHO Definition: An arbitrary level, approximating that of mean high water springs (MHWS).
21) **Mean Higher High Water**
IHO Definition: The average height of higher high waters at a place over a 19-year period.
22) **Equinoctial Spring Low Water**
IHO Definition: The level of low water springs near the time of an equinox.
23) **Lowest Astronomical Tide**
IHO Definition: The lowest tide level which can be predicted to occur under average meteorological conditions and under any combination of astronomical conditions.
24) **Local Datum**
IHO Definition: An arbitrary datum defined by a local harbour authority, from which levels and tidal heights are measured by this authority.
25) **International Great Lakes Datum 1985**
IHO Definition: A vertical reference system with its zero based on the **mean water level** at Rimouski/Pointe-au-Pere, Quebec, over the period 1970 to 1988.
26) **Mean Water Level**
IHO Definition: The average of all hourly water levels over the available period of record.
27) **Lower Low Water Large Tide**
IHO Definition: The average of the lowest low waters, one from each of 19 years of observations.
28) **Higher High Water Large Tide**
IHO Definition: The average of the highest high waters, one from each of 19 years of observations.
29) **Nearly Highest High Water**
IHO Definition: An arbitrary level approximating the highest water level observed at a place, usually equivalent to the high water springs.
30) **Highest Astronomical Tide**
IHO Definition: The highest tidal level which can be predicted to occur under average meteorological conditions and under any combination of astronomical conditions.
31) **Local Low Water Reference Level**
IHO Definition: Low water reference level of the local area.
32) **Local High Water Reference Level**
IHO Definition: High water reference level of the local area.
33) **Local Mean Water Reference Level**
IHO Definition: Mean water reference level of the local area.
34) **Equivalent Height of Water (German GIW)**

IHO Definition: A low water level which is the result of a defined low water discharge — called “equivalent discharge”.
35) **Highest Shipping Height of Water (German HSW)**
IHO Definition: Upper limit of water levels where navigation is allowed.
36) **Reference Low Water Level According to Danube Commission**
IHO Definition: The water level at a discharge, which is exceeded 94 % of the year within a period of 30 years.
37) **Highest Shipping Height of Water According to Danube Commission**
IHO Definition: The water level at a discharge, which is exceeded 1% of the year within a period of 30 years.
38) **Dutch River Low Water Reference Level (OLR)**
IHO Definition: The water level at a discharge, which is exceeded 95% of the year within a period of 20 years.
39) **Russian Project Water Level**
IHO Definition: Conditional low water level with established probability.
40) **Russian Normal Backwater Level**
IHO Definition: Highest water level derived from the upper backwater stream in watercourse or reservoir under the normal operational conditions.
41) **Ohio River Datum**
IHO Definition: The Ohio River datum.
43) **Dutch High Water Reference Level**
IHO Definition: Dutch High Water Reference Level.
44) **Baltic Sea Chart Datum 2000**
IHO Definition: The datum refers to each Baltic country’s realization of the European Vertical Reference System (EVRS) with land-uplift epoch 2000, which is connected to the Normaal Amsterdams Peil (NAP).
45) **Dutch Estuary Low Water Reference Level (OLW)**
IHO Definition: Dutch Estuary Low Water Reference Level (OLW).

Remarks: No remarks.

6.83 Function

Table 6-83 — Function

IHO Definition: A specific role that describes a feature.
2) **Harbour-Masters Office**
IHO Definition: A local official who has charge of mooring and berthing of vessels, collecting harbour fees, etc.
3) **Customs Office**
IHO Definition: Serves as a government office where customs duties are collected, the flow of goods are regulated and restrictions enforced, and shipments or vehicles are cleared for entering or leaving a country.
4) **Health Office**
IHO Definition: The office which is charged with the administration of health laws and sanitary inspections.
5) **Hospital**
IHO Definition: An institution or establishment providing medical or surgical treatment for the ill or wounded.
6) **Post Office**
IHO Definition: The public department, agency or organisation responsible primarily for the collection, transmission and distribution of mail.
7) **Hotel**
IHO Definition: An establishment, especially of a comfortable or luxurious kind, where paying visitors are provided with accommodation, meals and other services.
8) **Railway Station**
IHO Definition: A building with platforms where trains arrive, load, discharge and depart.
9) **Police Station**
IHO Definition: The headquarters of a local police force and that is where those under arrest are first charged.
10) **Water-Police Station**
IHO Definition: The headquarters of a local water-police force.
11) **Pilot Office**
IHO Definition: The office or headquarters of pilots; the place where the services of a pilot may be obtained.
12) **Pilot Lookout**

IHO Definition: A distinctive structure or place on shore from which personnel keep watch upon events at sea or along the coast.

13) **Bank Office**
IHO Definition: An office for custody, deposit, loan, exchange or issue of money.
14) **Headquarters for District Control**
IHO Definition: The quarters of an executive officer (director, manager, etc.) with responsibility for an administrative area.
15) **Transit Shed/Warehouse**
IHO Definition: A building or part of a building for storage of wares or goods.
16) **Factory**
IHO Definition: A building or buildings with equipment for manufacturing; a workshop.
17) **Power Station**
IHO Definition: A stationary plant containing apparatus for large scale conversion of some form of energy (such as hydraulic, steam, chemical or nuclear energy) into electrical energy.
18) **Administrative**
IHO Definition: A building for the management of affairs.
19) **Educational Facility**
IHO Definition: A building concerned with education (for example school, college, university, etc).
20) **Church**
IHO Definition: A building for public Christian worship.
21) **Chapel**
IHO Definition: A place for Christian worship other than a parish, cathedral or church, especially one attached to a private house or institution.
22) **Temple**
IHO Definition: A building for public Jewish worship.
23) **Pagoda**
IHO Definition: A Hindu or Buddhist temple or sacred building.
24) **Shinto Shrine**
IHO Definition: A building for public Shinto worship.
25) **Buddhist Temple**
IHO Definition: A building for public Buddhist worship.
26) **Mosque**
IHO Definition: A Muslim place of worship.
27) **Marabout**
IHO Definition: A shrine marking the burial place of a Muslim holy man.
28) **Lookout**
IHO Definition: Keeping a watch upon events at sea or along the coast.
29) **Communication**
IHO Definition: Transmitting and/or receiving electronic communication signals.
30) **Television**
IHO Definition: A system for reproducing on a screen visual images transmitted (usually with sound) by radio signals.
31) **Radio**
IHO Definition: Transmitting and/or receiving radio-frequency electromagnetic waves as a means of communication.
32) **Radar**
IHO Definition: A method, system or technique of using beamed, reflected, and timed radio waves for detecting, locating, or tracking objects, and for measuring altitudes.
33) **Light Support**
IHO Definition: A structure serving as a support for one or more lights.
34) **Microwave**
IHO Definition: Broadcasting and receiving signals using microwaves.
35) **Cooling**
IHO Definition: Generation of chilled liquid and/or gas for cooling purposes.
36) **Observation**
IHO Definition: A place from which the surroundings can be observed but at which a watch is not habitually maintained.
37) **Timeball**
IHO Definition: A visual time signal in the form of a ball.
38) **Clock**
IHO Definition: Instrument for measuring time and recording hours.
39) **Control**
IHO Definition: Used to control the flow of traffic within a specified range of an installation.
40) **Airship Mooring**
IHO Definition: Equipment or structure to secure an airship.
41) **Stadium**
IHO Definition: An arena for holding and viewing events.

42) **Bus Station**
IHO Definition: A building where buses and coaches regularly stop to take on and/or let off passengers, especially for long-distance travel.
43) **Passenger Terminal Building**
IHO Definition: A building within a terminal for the loading and unloading of passengers.
44) **Sea Rescue Control**
IHO Definition: A unit responsible for promoting efficient organization of search and rescue services and for coordinating the conduct of search and rescue operations within a search and rescue region.
45) **Observatory**
IHO Definition: A building designed and equipped for making observations of astronomical, meteorological, or other natural phenomena.
46) **Ore Crusher**
IHO Definition: A building or structure used to crush ore.
47) **Boathouse**
IHO Definition: A building or shed, usually built partly over water, for sheltering a boat or boats.
48) **Pumping Station**
IHO Definition: A facility to move solids, liquids or gases by means of pressure or suction.

Remarks: No remarks.

6.84 Category of Landmark

Table 6-84 — Category of Landmark

IHO Definition: Classification of prominent cultural and natural features in the landscape.

1) **Cairn**
IHO Definition: A mound of stones, usually conical or pyramidal, raised as a landmark or to designate a point of importance in surveying.
2) **Cemetery**
IHO Definition: A site and associated structures devoted to the burial of the dead.
3) **Chimney**
IHO Definition: A vertical structure containing a passage or flue for discharging smoke and gases of combustion.
4) **Dish Aerial**
IHO Definition: A parabolic aerial for the receipt and transmission of high frequency radio signals.
5) **Flagstaff**
IHO Definition: A staff or pole on which flags are raised.
6) **Flare Stack**
IHO Definition: A tall structure used for burning-off waste oil or gas.
7) **Mast**
IHO Definition: A relatively tall structure usually held vertical by guy lines.
8) **Windsock**
IHO Definition: A tapered fabric sleeve mounted so as to catch and swing with the wind, thus indicating the wind direction.
9) **Monument**
IHO Definition: A structure erected and/or maintained as a memorial to a person and/or event.
10) **Column/Pillar**
IHO Definition: A cylindrical or slightly tapering body of considerably greater length than diameter erected vertically.
11) **Memorial Plaque**
IHO Definition: A slab of metal, usually ornamented, erected as a memorial to a person or event.
12) **Obelisk**
IHO Definition: A tapering shaft usually of stone or concrete, square or rectangular in section, with a pyramidal apex.
13) **Statue**
IHO Definition: A representation of a living being, sculptured, moulded, or cast in a variety of materials (for example: marble, metal, or plaster).
14) **Cross**
IHO Definition: A monument, or other structure in form of a cross.
15) **Dome**
IHO Definition: A landmark comprising a hemispherical or spheroidal shaped structure.
16) **Radar Scanner**
IHO Definition: A device used for directing a radar beam through a search pattern.
17) **Tower**

IHO Definition: A relatively tall, narrow structure that may either stand alone or may form part of another structure.

18) **Windmill**
IHO Definition: A system of vanes attached to a tower and driven by wind (excluding wind turbines).
19) **Windmotor**
IHO Definition: A modern structure for the use of wind power.
20) **Spire/Minaret**
IHO Definition: A tall conical or pyramid-shaped structure often built on the roof or tower of a building, especially a church or mosque.
21) **Large Rock or Boulder on Land**
IHO Definition: An isolated rocky formation or a single large stone.
22) **Triangulation Mark**
IHO Definition: A recoverable point on the earth, whose geographic position has been determined by angular methods with geodetic instruments. A triangulation point is a selected point, which has been marked with a station mark, or it is a conspicuous natural or artificial feature.
23) **Boundary Mark**
IHO Definition: A marker identifying the location of a surveyed boundary line.
24) **Observation Wheel**
IHO Definition: Wheels with passenger cars mounted external to the rim and independently rotated by electric motors.
25) **Torii**
IHO Definition: A form of decorative gateway or portal, consisting of two upright wooden posts connected at the top by two horizontal crosspieces, commonly found at the entrance to Shinto temples.
26) **Bridge**
IHO Definition: (1) An elevated structure extending across or over the weather deck of a vessel, or part of such a structure. The term is sometimes modified to indicate the intended use, such as navigating bridge or signal bridge. (2) A structure erected over a depression or an obstacle such as a body of water, railroad, etc., to provide a roadway for vehicles or pedestrians.
27) **Dam**
IHO Definition: A barrier to check or confine anything in motion; particularly one constructed to hold back water and raise its level to form a reservoir, or to prevent flooding.

Remarks: No remarks.

6.85 Type of Buoy

Table 6-85 — Type of Buoy

IHO Definition: Type equipment used as a buoy in a particular installation.

Remarks: Types of light buoy; for example LANBY-100, LS-35, LL-30, LL-28, LL-26, LL-26(M), LL-24, LS-24, LSP-24, LT-10. Types of buoy e.g. U-17C(P), U-17S(P), U-17C(S), U-17S(S), UR-17C(P), UR-17S(P), UR-17C(S), UR-17.

6.86 Buoy Shape

Table 6-86 — Buoy Shape

IHO Definition: The principal shape and/or design of a buoy.

1) **Conical**
IHO Definition: The upper part of the body above the water-line, or the greater part of the superstructure, has approximately the shape or the appearance of a pointed cone with the point upwards.
2) **Can**
IHO Definition: The upper part of the body above the water-line, or the greater part of the superstructure, has the shape of a cylinder, or a truncated cone that approximates to a cylinder, with a flat end uppermost.
3) **Spherical**
IHO Definition: Shaped like a sphere, which is a body the surface of which is at all points equidistant from the centre.
4) **Pillar**
IHO Definition: The upper part of the body above the water-line, or the greater part of the superstructure is a narrow vertical structure, pillar or lattice tower.
5) **Spar**
IHO Definition: The upper part of the body above the water-line, or the greater part of the superstructure, has the form of a pole, or of a very long cylinder, floating upright.

6) **Barrel**
IHO Definition: The upper part of the body above the water-line, or the greater part of the superstructure, has the form of a barrel or cylinder floating horizontally.
7) **Superbuoy**
IHO Definition: A very large buoy designed to carry a signal light of high luminous intensity at a high elevation.
8) **Ice Buoy**
IHO Definition: A specially constructed shuttle shaped buoy which is used in ice conditions.

Remarks: The principal shapes are those recommended in the International Association of Lighthouse Authorities — IALA System.

6.87 Visual Prominence

Table 6-87 — Visual Prominence

IHO Definition: The extent to which a feature, either natural or artificial, is visible from seaward.

1) **Visually Conspicuous**
IHO Definition: Term applied to a feature either natural or artificial which is distinctly and notably visible from seaward.
2) **Not Visually Conspicuous**
IHO Definition: An object that may be visible from seaward, but cannot be used as a fixing mark and is not conspicuous.
3) **Prominent**
IHO Definition: Objects which are easily identifiable, but do not justify being classed as conspicuous.

Remarks: No remarks.

6.88 Vertical Length

Table 6-88 — Vertical Length

IHO Definition: The total vertical length of a feature.

Remarks: For floating objects: the vertical distance from the surface of water to the highest point of that object.

For fixed objects: the vertical distance from seabed or ground to the highest point of that object.

For objects on top of other objects: the vertical distance from the lowest to the highest point of that object.

Vertical length measurements do not require a datum.

6.89 Status

Table 6-89 — Status

IHO Definition: The condition of an object at a given instant in time.

1) **Permanent**
IHO Definition: Intended to last or function indefinitely.
2) **Occasional**
IHO Definition: Acting on special occasions; happening irregularly.
3) **Recommended**
IHO Definition: Presented as worthy of confidence, acceptance, use, etc.
4) **Not in Use**
IHO Definition: Use has ceased, but the facility still exists intact; disused.
5) **Periodic/Intermittent**
IHO Definition: Recurring at intervals.
6) **Reserved**
IHO Definition: Set apart for some specific use.
7) **Temporary**
IHO Definition: Meant to last only for a time.
8) **Private**
IHO Definition: Administered by an individual or corporation, rather than a State or a public body.
9) **Mandatory**

IHO Definition: Compulsory; enforced.

11) **Extinguished**
IHO Definition: No longer lit.
12) **Illuminated**
IHO Definition: Lit by floodlights, strip lights, etc.
13) **Historic**
IHO Definition: Famous in history; of historical interest.
14) **Public**
IHO Definition: Belonging to, available to, used or shared by, the community as a whole and not restricted to private use.
15) **Synchronized**
IHO Definition: Occur at a time, coincide in point of time, be contemporary or simultaneous.
16) **Watched**
IHO Definition: Looked at or observed over a period of time especially so as to be aware of any movement or change.
17) **Unwatched**
IHO Definition: Usually automatic in operation, without any permanently-stationed personnel to superintend it.
18) **Existence Doubtful**
IHO Definition: A feature that has been reported but has not been definitely determined to exist.
19) **On Request**
IHO Definition: When you ask for it.
20) **Drop Away**
IHO Definition: To become lower in level.
21) **Rising**
IHO Definition: To become higher in level.
22) **Increasing**
IHO Definition: Becoming larger in magnitude.
23) **Decreasing**
IHO Definition: Becoming smaller in magnitude.
24) **Strong**
IHO Definition: Not easily broken or destroyed.
25) **Good**
IHO Definition: In a satisfactory condition to use.
26) **Moderately**
IHO Definition: Fairly but not very.
27) **Poor**
IHO Definition: Not as good as it could be or should.
28) **Buoyed**
IHO Definition: Marked by buoys.
29) **Fully Operational**
IHO Definition: Entire observation platform is operating in accordance with, or exceeding, manufacturer specifications.
30) **Partially Operational**
IHO Definition: At least one instrument that is part of an observation platform is not operating to manufacturer specification.
31) **Drifting**
IHO Definition: Floating platform at the mercy of environmental elements, whether intentional or not.
32) **Broken**
IHO Definition: Fractured or in pieces.
33) **Offline**
IHO Definition: Observation platform is intentionally not reporting an environmental observation.
34) **Discontinued**
IHO Definition: Observation station, suite of instruments, or an individual instrument, for a particular location, has been removed and is no longer at the particular location.
35) **Manual Observation**
IHO Definition: Observations made by a human observer.
36) **Unknown Status**
IHO Definition: Status of an observation platform, suite of instruments, or individual instrument is not known or unspecified.
37) **Confirmed**
IHO Definition: Made certain as to truth, accuracy, validity, availability, etc.
38) **Candidate**
IHO Definition: Item selected for an action.
39) **Under Modification**
IHO Definition: Item that is in the process of being modified.
41) **Under Removal / Deletion**

IHO Definition: Item in the process of being removed or deleted.
42) **Removed / Deleted**
IHO Definition: Item that has been removed or deleted.
43) **Candidate for Modification**
IHO Definition: Item selected for modification.
Remarks: No remarks.

6.90 Radar Conspicuous

Table 6-90 — Radar Conspicuous

IHO Definition: A feature which returns a strong radar echo.

Remarks: No remarks.

6.91 Nature of Construction

Table 6-91 — Nature of Construction

IHO Definition: The building's primary construction material.

1) **Masonry**
IHO Definition: Constructed of stones or bricks, usually quarried, shaped, and mortared.
2) **Concreted**
IHO Definition: Constructed of concrete, a material made of sand and gravel that is united by cement into a hardened mass used for roads, foundations, etc.
3) **Loose Boulders**
IHO Definition: Constructed from large stones or blocks of concrete, often placed loosely for protection against waves or water turbulence.
4) **Hard Surfaced**
IHO Definition: Constructed with a surface of hard material, usually a term applied to roads surfaced with asphalt or concrete.
5) **Unsurfaced**
IHO Definition: Constructed with no extra protection, usually a term applied to roads not surfaced with a hard material.
6) **Wooden**
IHO Definition: Constructed from wood.
7) **Metal**
IHO Definition: Constructed from metal.
8) **Glass Reinforced Plastic**
IHO Definition: Constructed from a plastic material strengthened with fibres of glass.
9) **Painted**
IHO Definition: The application of paint to some other construction or natural feature.
10) **Framework**
IHO Definition: Constructed from a lattice framework of, often diagonal, intersecting struts.
11) **Latticed**
IHO Definition: A structure of crossed wooden or metal strips usually arranged to form a diagonal pattern of open spaces between the strips.
12) **Glass**
IHO Definition: Any artificial or natural substance having similar properties and composition, as fused borax, obsidian, or the like.
13) **Fiberglass**
IHO Definition: Constructed from fiberglass.
14) **Plastic**
IHO Definition: Constructed from plastic.

Remarks: No remarks.

6.92 Marks Navigational — System Of

Table 6-92 — Marks Navigational — System Of

IHO Definition: The system of navigational buoyage a region complies with.

1) **IALA A**
IHO Definition: Navigational aids conform to the International Association of Lighthouse Authorities — IALA A system.
2) **IALA B**
IHO Definition: Navigational aids conform to the International Association of Lighthouse Authorities — IALA B system.
9) **No System**
IHO Definition: Navigational aids do not conform to any defined system.
10) **Other System**
IHO Definition: Navigational aids conform to a defined system other than International Association of Lighthouse Authorities — IALA.
11) **CEVNI**
IHO Definition: CEVNI (European Code for Navigation on Inland Waterways) is the European code for rivers, canals land lakes in most of Europe.
12) **Russian Inland Waterway Regulations**
IHO Definition: Navigational aids conform to the Russian inland waterway regulations.
13) **Brazilian National Inland Waterway Regulations — Two Sides**
IHO Definition: Navigational aids conform to the Brazilian national inland waterway regulations for two sides.
14) **Brazilian National Inland Waterway Regulations — Side Independent**
IHO Definition: Navigational aids conform to the Brazilian national inland waterway regulations — side independent.
15) **Paraguay-Parana Waterway — Brazilian Complementary Aids**
IHO Definition: Navigational aids conform to the Brazilian complementary aids on the Paraguay-Parana waterway.

Remarks: No remarks.

6.93 Height

Table 6-93 — Height

IHO Definition: The value of the vertical distance to the highest point of the object, measured from a specified vertical datum.

Remarks: Height must not be used for floating objects.

6.94 Elevation

Table 6-94 — Elevation

IHO Definition: The altitude of the ground level of an object, measured from a specified vertical datum.

Remarks: No remarks.

6.95 Colour Pattern

Table 6-95 — Colour Pattern

IHO Definition: A regular repeated design containing more than one colour.

1) **Horizontal Stripes**
IHO Definition: Straight bands or stripes of differing colours oriented horizontally.
2) **Vertical Stripes**
IHO Definition: Straight bands or stripes of differing colours oriented vertically.
3) **Diagonal Stripes**
IHO Definition: Straight bands or stripes of differing colours oriented diagonally (that is, not horizontally or vertically).
4) **Squared**
IHO Definition: Often referred to as checker plate, where alternate colours are used to create squares similar to a chess or draught board. The pattern may be straight or diagonal.
5) **Stripes (Direction Unknown)**
IHO Definition: Straight bands or stripes of differing colours oriented in an unknown direction.

6) **Border Stripe**
IHO Definition: A band or stripe of colour which is displayed around the outer edge of the object, which may also form a border to an inner pattern or plain colour.
7) **Single Colour**
IHO Definition: One solid colour of uniform coverage.
8) **Rectangle**
IHO Definition: A four-sided shape that is made up of two pairs of parallel lines and that has four right angles, on a different coloured background.
9) **Triangle**
IHO Definition: A shape that is made up of three lines and three angles, on a different coloured background.

Remarks: No remarks.

6.96 Colour

Table 6-96 — Colour

IHO Definition: The property possessed by an object of producing different sensations on the eye as a result of the way it reflects or emits light.

1) **White**
IHO Definition: The achromatic object colour of greatest lightness characteristically perceived to belong to objects that reflect diffusely nearly all incident energy throughout the visible spectrum.
2) **Black**
IHO Definition: The achromatic color of least lightness characteristically perceived to belong to objects that neither reflect nor transmit light.
3) **Red**
IHO Definition: A color whose hue resembles that of blood or of the ruby or is that of the long-wave extreme of the visible spectrum.
4) **Green**
IHO Definition: Of the color green.
5) **Blue**
IHO Definition: A color whose hue is that of the clear sky or that of the portion of the color spectrum lying between green and violet.
6) **Yellow**
IHO Definition: A color whose hue resembles that of ripe lemons or sunflowers or is that of the portion of the spectrum lying between green and orange.
7) **Grey**
IHO Definition: Of the color grey.
8) **Brown**
IHO Definition: Any of a group of colors between red and yellow in hue, of medium to low lightness, and of moderate to low saturation.
9) **Amber**
IHO Definition: A variable color averaging a dark orange yellow.
10) **Violet**
IHO Definition: Any of a group of colors of reddish-blue hue, low lightness, and medium saturation.
11) **Orange**
IHO Definition: Any of a group of colors that are between red and yellow in hue.
12) **Magenta**
IHO Definition: A deep purplish red.
13) **Pink**
IHO Definition: Any of a group of colors bluish red to red in hue, of medium to high lightness, and of low to moderate saturation.

Remarks: No remarks.

6.97 Beacon Shape

Table 6-97 — Beacon Shape

IHO Definition: Describes the characteristic geometric form of the beacon.

1) **Stake, Pole, Perch, Post**
IHO Definition: An elongated wood or metal pole, driven into the ground or seabed, which serves as a navigational aid or a support for a navigational aid.

2) **Withy**
IHO Definition: A tree without roots stuck or spoiled into the bottom of the sea to serve as a navigational aid.
3) **Beacon Tower**
IHO Definition: A solid structure of the order of 10 metres in height used as a navigational aid.
4) **Lattice Beacon**
IHO Definition: A structure consisting of strips of metal or wood crossed or interlaced to form a structure to serve as an aid to navigation or as a support for an aid to navigation.
5) **Pile Beacon**
IHO Definition: A long heavy timber(s) or section(s) of steel, wood, concrete, etc., forced into the seabed to serve as an aid to navigation or as a support for an aid to navigation.
6) **Cairn**
IHO Definition: A mound of stones, usually conical or pyramidal, raised as a landmark or to designate a point of importance in surveying.
7) **Buoyant Beacon**
IHO Definition: A tall spar-like beacon fitted with a permanently submerged buoyancy chamber, the lower end of the body is secured to seabed sinker either by a flexible joint or by a cable under tension.

Remarks: No remarks.

6.98 Date Start

Table 6-98 — Date Start

IHO Definition: The earliest date on which an object (for example a buoy) will be present.

Remarks: The Date Start should be encoded using 4 digits for the calendar year (YYYY), 2 digits for the month (MM) (for example April = 04) and 2 digits for the day (DD).

When no specific month and/or day is required/known, indication of the month and/or day is omitted, and replaced with dashes (-).

When no specific year is required (that is, the event or date range ends at the same time each year) the following two cases may be considered:

- same day each year: ——MMDD
- same month each year: ——MM—

This conforms to ISO 8601: 2004. Date Start indicates the earliest date of an event or the start of a date range. It is used to indicate the start of a fixed date range, the start of a periodic date range, or the deployment or implementation of a feature at a specific date in the future.

6.99 Date End

Table 6-99 — Date End

IHO Definition: The latest date on which an object (for example a buoy) will be present.

Remarks: The Date End should be encoded using 4 digits for the calendar year (YYYY), 2 digits for the month (MM) (for example April = 04) and 2 digits for the day (DD).

When no specific month and/or day is required/known, indication of the month and/or day is omitted, and replaced with dashes (-).

When no specific year is required (that is, the event or date range ends at the same time each year) the following two cases may be considered:

- same day each year: ——MMDD
- same month each year: ——MM—

This conforms to ISO 8601: 2004. Date End indicates the latest date of an event or the end of a date range. It is used to indicate the end of a fixed date range, the end of a periodic date range, or the removal or cancellation of a feature at a specific date in the future.

6.100 Installation Date

Table 6-100 — Installation Date

IHO Definition: The date when an item was installed.
Remarks: No remarks.

6.101 AtoN Number

Table 6-101 — AtoN Number

IHO Definition: Identifier from a list of Aids to Navigation publication, such as List of Lights.
Remarks: No remarks.

6.102 Interoperability Identifier

Table 6-102 — Interoperability Identifier

IHO Definition: A common unique identifier for entities which describe a single real-world feature, and which is used to identify instances of the feature in end-user systems where the feature may be included in multiple data product types.
Remarks: No remarks.

6.103 ID Code

Table 6-103 — ID Code

IHO Definition: Identification code as specified in predefined system. Also called identification number.
Remarks: No remarks.

6.104 Scale Minimum

Table 6-104 — Scale Minimum

IHO Definition: The minimum scale at which the feature may be used for example for ECDIS presentation.
Remarks: The modulus of the scale is indicated, that is 1:1 250 000 is encoded as 1250000.

6.105 Source Date

Table 6-105 — Source Date

IHO Definition: The production date of the source; for example the date of measurement.
Remarks: No remarks.

6.106 Source

Table 6-106 — Source

IHO Definition: The publication, document, or reference work from which information comes or is acquired.

Remarks: May be populated with the corresponding paper chart Notice to Mariners numbers, although other references are permitted.

6.107 Pictorial Representation

Table 6-107 — Pictorial Representation

IHO Definition: Indicates whether a pictorial representation of the feature is available. Remarks: The 'pictorial representation' could be a drawing or a photo. The string encodes the file name of an external graphic file (pixel/vector).

6.108 Inspection Frequency

Table 6-108 — Inspection Frequency

IHO Definition: A statement of how frequently an item is inspected. Remarks: No remarks.

6.109 Inspection Requirements

Table 6-109 — Inspection Requirements

IHO Definition: A statement of what requirements are in place for how inspection of an item is carried out. Remarks: No remarks.

6.110 AtoN Maintenance Record

Table 6-110 — AtoN Maintenance Record

IHO Definition: A reference following the Uniform Resource Identifier (URI) principles to a record of maintenance. Remarks: No remarks.

6.111 Display Name

Table 6-111 — Display Name

IHO Definition: A statement expressing if a feature name is to be displayed in certain system display settings or not. Remarks: Where it is allowable to encode multiple instances of feature name for a single feature instance, only one feature name instance can indicate that the name is to be displayed (display name set to True).

6.112 Name

Table 6-112 — Name

IHO Definition: The individual name of a feature. Remarks: No remarks.

6.113 Aid Availability Category

Table 6-113 — Aid Availability Category

IHO Definition: A Category denoting the significance of an Aid to Navigation, expressed in terms of the probability that an AtoN or system of AtoN, as defined by the Competent Authority, is performing its specified function at any randomly chosen time. 1) **Category 1** IHO Definition: An AtoN or system of AtoN that is considered by the Competent Authority to be of vital navigational significance. 2) **Category 2** IHO Definition: An AtoN or system of AtoN that is considered by the Competent Authority to be of important navigational significance. 3) **Category 3** IHO Definition: An AtoN or system of AtoN that is considered by the Competent Authority to be of necessary navigational significance. Remarks: No remarks.

6.114 Condition

Table 6-114 — Condition

IHO Definition: The various conditions of buildings and other constructions. 1) **Under Construction** IHO Definition: Being built but not yet capable of function. 2) **Ruined** IHO Definition: A structure in a decayed or deteriorated condition resulting from neglect or disuse, or a damaged structure in need of repair. 3) **Under Reclamation** IHO Definition: An area of the sea, a lake or the navigable part of a river that is being reclaimed as land, usually by the dumping of earth and other material. 4) **Wingless** IHO Definition: A windmill or wind turbine from which the vanes or turbine blades are missing. 5) **Planned Construction** IHO Definition: Detailed planning has been completed but construction has not been initiated. Remarks: The default ‘condition’ should be considered to be completed, undamaged and working normally.

6.115 Remote Monitoring System

Table 6-115 — Remote Monitoring System

IHO Definition: Classification or name of system used to remotely monitor a feature. Remarks: No remarks.

6.116 Delivery Point

Table 6-116 — Delivery Point

IHO Definition: Details of where post can be delivered such as the apartment, name and/or number of a street, building or PO Box. Remarks: No remarks.

6.117 City Name

Table 6-117 — City Name

IHO Definition: The name of a town or city. Remarks: No remarks.

6.118 Administrative Division

Table 6-118 — Administrative Division

IHO Definition: A generic term for an administrative region within a country at a level below that of the sovereign state. Remarks: No remarks.

6.119 Country Name

Table 6-119 — Country Name

IHO Definition: The name of a nation. Remarks: No remarks.

6.120 Postal Code

Table 6-120 — Postal Code

IHO Definition: Known in various countries as a postcode, or ZIP code, the postal code is a series of letters and/or digits that identifies each postal delivery area. Remarks: No remarks.

6.121 Category Of Aggregation

Table 6-121 — Category Of Aggregation

IHO Definition: Named aggregations between two or more aids to navigation and/or navigationally relevant features. 1) **leading line** IHO Definition: - 3) **measured distance** IHO Definition: - 2) **range system** IHO Definition: - Remarks: No remarks.

6.122 Text

Table 6-122 — Text

IHO Definition: A non-formatted digital text string. Remarks: Should be used, for example, to hold the information that is for short cautionary or explanatory notes.

Therefore, text populated in text must not exceed 300 characters. Text may be in English, or in a national language. No formatting of text is possible within text. If formatted text, or text strings exceeding 300 characters, is required, then an alternate concept should be used.

6.123 Language

Table 6-123 — Language

IHO Definition: The method of human communication, either spoken or written, consisting of the use of words in a structured and conventional way. Remarks: The language is encoded by a 3 character code following ISO 639-2/T.

6.124 Headline

Table 6-124 — Headline

IHO Definition: Words set at the head of a passage or page to introduce or categorize. Remarks: No remarks.

6.125 File Reference

Table 6-125 — File Reference

IHO Definition: The file name of an externally referenced text file. Remarks: No remarks.

6.126 File Locator

Table 6-126 — File Locator

IHO Definition: The location of a fragment of text or other information in a support file. Remarks: Application schemas must describe how the associated file is identified. The associated file will commonly be named in a file reference co-attribute of the same complex attribute. Each DCEG must specify requirements for the format of the associated file and the semantics of file locator. For example, the value of file locator may be an HTML ID in an HTML file, line number in a text file, or a bookmark in a PDF file.

6.127 Uncertainty Variable Factor

Table 6-127 — Uncertainty Variable Factor

IHO Definition: The factor to be applied to the variable component of an uncertainty equation so as to provide the best estimate of the variable horizontal or vertical accuracy component for positions, depths, heights, vertical distances and vertical clearances. Remarks: No remarks.

6.128 Uncertainty Fixed

Table 6-128 — Uncertainty Fixed

IHO Definition: The best estimate of the fixed horizontal or vertical accuracy component for positions, depths, heights, vertical distances and vertical clearances. Remarks: No remarks.

6.129 Information in National Language

Table 6-129 — Information in National Language

IHO Definition: Textual information in national language characters. Remarks: Encodes any textual information about an object using a specified national language. The textual information could be, for example, a list, a table or a text. This should be used, for example, to hold the information that is shown on paper charts by cautionary and explanatory notes.

6.130 Horizontal Distance Uncertainty

Table 6-130 — Horizontal Distance Uncertainty

IHO Definition: The best estimate of the horizontal accuracy of horizontal clearances and distances. Remarks: The error is assumed to be positive and negative. The plus/minus character must not be encoded.

6.131 Orientation Uncertainty

Table 6-131 — Orientation Uncertainty

IHO Definition: The best estimate of the accuracy of a bearing. Remarks: No remarks.

6.132 Category of Temporal Variation

Table 6-132 — Category of Temporal Variation

IHO Definition: An assessment of the likelihood of change over time. 1) **Extreme Event** IHO Definition: Indication of the possible impact of a significant event (for example hurricane, earthquake, volcanic eruption, landslide, etc), which is considered likely to have changed the seafloor or landscape significantly. 2) **Likely to Change and Significant Shoaling Expected** IHO Definition: Continuous or frequent change (for example river siltation, sand waves, seasonal storms, icebergs, etc) that is likely to result in new significant shoaling. 3) **Likely to Change But Significant Shoaling Not Expected** IHO Definition: Continuous or frequent change (for example sand wave shift, seasonal storms, icebergs, etc) that is not likely to result in new significant shoaling. 4) **Likely to Change** IHO Definition: Continuous or frequent change to non-bathymetric features (for example river siltation, glacier creep/recession, sand dunes, buoys, marine farms, etc). 5) **Unlikely to Change** IHO Definition: Significant change to the seafloor is not expected. 6) **Unassessed** IHO Definition: Not having been assessed.

Remarks: No remarks.

6.133 Maximum Display Scale

Table 6-133 — Maximum Display Scale

IHO Definition: The value considered by the Data Producer to be the maximum (largest) scale at which the data is to be displayed before it can be considered to be 'grossly overscaled'.

Remarks: No remarks.

6.134 Minimum Display Scale

Table 6-134 — Minimum Display Scale

IHO Definition: The smallest intended viewing scale for the data.

Remarks: No remarks.

6.135 Aton Commissioning

Table 6-135 — Aton Commissioning

IHO Definition: Different stages of the establishment of aids to navigation.

1) **Buoy establishment**
   IHO Definition:
2) **Light establishment**
   IHO Definition:
3) **Beacon establishment**
   IHO Definition:
4) **Audible signal establishment**
   IHO Definition:
5) **Fog signal establishment**
   IHO Definition:
6) **AIS transmitter establishment**
   IHO Definition:
7) **V-AIS establishment**
   IHO Definition:
8) **RACON establishment**
   IHO Definition:
9) **DGPS station establishment**
   IHO Definition:
10) **eLORAN station establishment**
    IHO Definition:
11) **DGLONASS station establishment**
    IHO Definition:
12) **e-Chayka station establishment**
    IHO Definition:
13) **EGNOS establishment**
    IHO Definition:

Remarks: No remarks.

6.136 Aton Removal

Table 6-136 — Aton Removal

IHO Definition: Different stages of removal or temporary removal of aids to navigation.

1) **Buoy removal**
   IHO Definition:

2) **Buoy temporary removal**
   IHO Definition:
3) **Light removal**
   IHO Definition:
4) **Light temporary removal**
   IHO Definition:
5) **Beacon removal**
   IHO Definition:
6) **Beacon temporary removal**
   IHO Definition:
7) **Fog signal removal**
   IHO Definition:
8) **Fog signal temporary removal**
   IHO Definition:
9) **Audible signal removal**
   IHO Definition:
10) **Audible signal temporary removal**
   IHO Definition:
11) **V-AIS removal**
   IHO Definition:
12) **V-AIS temporary removal**
   IHO Definition:
13) **RACON signal removal**
   IHO Definition:
14) **RACON temporary removal**
   IHO Definition:
15) **DGPS removal**
   IHO Definition:
16) **DGPS temporary removal**
   IHO Definition:
17) **EGNOS removal**
   IHO Definition:
18) **EGNOS temporary removal**
   IHO Definition:
19) **LORAN C station removal**
   IHO Definition:
20) **LORAN C station temporary removal**
   IHO Definition:
21) **eLORAN removal**
   IHO Definition:
22) **eLORAN temporary removal**
   IHO Definition:
23) **Chayka station removal**
   IHO Definition:
24) **Chayka station temporary removal**
   IHO Definition:
25) **e-Chayka station removal**
   IHO Definition:
26) **e-Chayka station temporary removal**
   IHO Definition:

Remarks: No remarks.

6.137 Aton Replacement

Table 6-137 — Aton Replacement

IHO Definition: Different stages of replacing aids to navigation.

1) **Buoy change**
   IHO Definition:
2) **Buoy temporary change**
   IHO Definition:
3) **Light change**
   IHO Definition:
4) **Light temporary change**

IHO Definition:
5) **Sector light change**
IHO Definition:
6) **Sector light temporary change**
IHO Definition:
7) **Beacon change**
IHO Definition:
8) **Beacon temporary change**
IHO Definition:
9) **Fog signal change**
IHO Definition:
10) **Fog signal temporary change**
IHO Definition:
11) **Audible signal change**
IHO Definition:
12) **Audible signal temporary change**
IHO Definition:
13) **V-AIS change**
IHO Definition:
14) **V-AIS temporary change**
IHO Definition:
15) **RACON signal change**
IHO Definition:
16) **RACON temporary change**
IHO Definition:
Remarks: No remarks.

6.138 Fixed Aton Change

Table 6-138 — Fixed Aton Change

IHO Definition: Different types of changes in fixed aids to navigation.
1) **Beacon missing**
IHO Definition:
2) **Beacon damaged**
IHO Definition:
3) **Light beacon Unlit**
IHO Definition:
4) **Light beacon Unreliable**
IHO Definition:
5) **Light beacon Not synchronized**
IHO Definition:
6) **Light beacon damaged**
IHO Definition:
7) **Beacon topmark missing**
IHO Definition:
8) **Beacon topmark damaged**
IHO Definition:
9) **Beacon daymark unreliable**
IHO Definition:
10) **Floodlit beacon Unlit**
IHO Definition:
11) **Beacon restored to normal**
IHO Definition:
Remarks: No remarks.

6.139 Floating Aton Change

Table 6-139 — Floating Aton Change

IHO Definition: Different types of changes in floating aids to navigation.
1) **Buoy adrift**

IHO Definition:
2) **Buoy damaged**
IHO Definition:
3) **Buoy daymark unreliable**
IHO Definition:
4) **Buoy destroyed**
IHO Definition:
5) **Buoy missing**
IHO Definition:
6) **Buoy move**
IHO Definition:
7) **Buoy off position**
IHO Definition:
8) **Buoy re-establishment**
IHO Definition:
9) **Buoy restored to normal**
IHO Definition:
10) **Buoy topmark damaged**
IHO Definition:
11) **Buoy topmark missing**
IHO Definition:
12) **Buoy will be withdrawn**
IHO Definition:
13) **Buoy withdrawn**
IHO Definition:
14) **Decommissioned for winter**
IHO Definition:
15) **Lifted for Winter**
IHO Definition:
16) **Light buoy Light damaged**
IHO Definition:
17) **Light buoy Light not synchronized**
IHO Definition:
18) **Light buoy Light unlit**
IHO Definition:
19) **Light buoy Light unreliable**
IHO Definition:
20) **Marine Aids to Navigation unreliable**
IHO Definition:
21) **Recommissioned for navigation season**
IHO Definition:
22) **Replaced by Winter Spar**
IHO Definition:
23) **Seasonal decommissioning complete**
IHO Definition:
24) **Seasonal decommissioning in progress**
IHO Definition:
25) **Seasonal recommissioning complete**
IHO Definition:
26) **Seasonal recommissioning in progress**
IHO Definition:

Remarks: No remarks.

6.140 Audible Signal Aton Change

Table 6-140 — Audible Signal Aton Change

IHO Definition: Changes related to the operational status of audible and fog signals.

1) **Audible signal out of service**
IHO Definition:
2) **Fog signal out of service**
IHO Definition:
3) **Audible signal operating properly**
IHO Definition:

4) **Fog signal operating properly**
IHO Definition:
Remarks: No remarks.

6.141 Lighted Aton Change

Table 6-141 — Lighted Aton Change

IHO Definition: Changes related to the operational status of lighted aids to navigation.

1) **Light unlit**
IHO Definition:
2) **Light unreliable**
IHO Definition:
3) **Light re-establishment**
IHO Definition:
4) **Light range reduced**
IHO Definition:
5) **Light without rhythm**
IHO Definition:
6) **Light out of synchronization**
IHO Definition:
7) **Light daymark unreliable**
IHO Definition:
8) **Light operating properly**
IHO Definition:
9) **Sector light Sector obscured**
IHO Definition:
10) **Front leading/range light Unlit**
IHO Definition:
11) **Rear leading/range light Unlit**
IHO Definition:
12) **Front leading/range light Unreliable**
IHO Definition:
13) **Rear leading/range light Unreliable**
IHO Definition:
14) **Front leading/range light Light range reduced**
IHO Definition:
15) **Rear leading/range light Light range reduced**
IHO Definition:
16) **Front leading/range light without rhythm**
IHO Definition:
17) **Rear leading/range light without rhythm**
IHO Definition:
18) **Leading/range lights out of synchronization**
IHO Definition:
19) **Front leading/range beacon Unreliable**
IHO Definition:
20) **Rear leading/range beacon Unreliable**
IHO Definition:
21) **Front leading/range light is operating properly**
IHO Definition:
22) **Rear leading/range light is operating properly**
IHO Definition:
23) **Front leading/range beacon restored to normal**
IHO Definition:
24) **Rear leading/range beacon restored to normal**
IHO Definition:

Remarks: No remarks.

6.142 Electronic Aton Change

Table 6-142 — Electronic Aton Change

IHO Definition:

1) **AIS transmitter out of service**
   IHO Definition:
2) **AIS transmitter unreliable**
   IHO Definition:
3) **AIS transmitter operating properly**
   IHO Definition:
4) **V-AIS out of service**
   IHO Definition:
5) **V-AIS unreliable**
   IHO Definition:
6) **V-AIS operating properly**
   IHO Definition:
7) **RACON out of service**
   IHO Definition:
8) **RACON unreliable**
   IHO Definition:
9) **RACON operating properly**
   IHO Definition:
10) **DGPS out of service**
    IHO Definition:
11) **DGPS operating properly**
    IHO Definition:
12) **DGPS unreliable**
    IHO Definition:
13) **LORAN C operating properly**
    IHO Definition:
14) **LORAN C unreliable**
    IHO Definition:
15) **LORAN C out of service**
    IHO Definition:
16) **eLORAN operating properly**
    IHO Definition:
17) **eLORAN unreliable**
    IHO Definition:
18) **eLORAN out of service**
    IHO Definition:
19) **DGLOANSS operating properly**
    IHO Definition:
20) **DGLOANSS unreliable**
    IHO Definition:
21) **DGLOANSS out of service**
    IHO Definition:
22) **Chayka operating properly**
    IHO Definition:
23) **Chayka unreliable**
    IHO Definition:
24) **Chayka out of service**
    IHO Definition:
25) **e-Chayka operating properly**
    IHO Definition:
26) **e-Chayka unreliable**
    IHO Definition:
27) **e-Chayka out of service**
    IHO Definition:
28) **EGNOS operating properly**
    IHO Definition:
29) **EGNOS unreliable**
    IHO Definition:
30) **EGNOS out of service**
    IHO Definition:

Remarks: No remarks.

6.143 Positioning Equipment

Table 6-143 — Positioning Equipment

IHO Definition: 1) **DGPS Receiver** IHO Definition: 2) **GLONASS Receiver** IHO Definition: 3) **GPS Receiver** IHO Definition: 4) **GPS/WAAS Receiver** IHO Definition: Remarks: No remarks.

7 Complex Attributes

7.1 Contact Address

Table 7-1 — Contact Address

IHO Definition: Direction or superscription of a letter, package, etc., specifying the name of the place to which it is directed, and optionally a contact person or organisation who should receive it. Sub-attributes: • **Delivery Point** (see Clause 6.116) • **City Name** (see Clause 6.117) • **Administrative Division** (see Clause 6.118) • **Country Name** (see Clause 6.119) • **Postal Code** (see Clause 6.120) Remarks: No remarks.

7.2 Directional Character

Table 7-2 — Directional Character

IHO Definition: A directional light is a light illuminating a sector of very narrow angle and intended to mark a direction to follow. Sub-attributes: • **Moire Effect** (see Clause 6.33) • **Orientation** (see Clause 7.7) Remarks: No remarks.

7.3 Feature Name

Table 7-3 — Feature Name

IHO Definition: Provides the name of an entity, defines the national language of the name, and provides the option to display the name at various system display settings. Sub-attributes: • **Display Name** (see Clause 6.111) • **Language** (see Clause 6.123) • **Name** (see Clause 6.112) Remarks: No remarks.

7.4 Fixed Date Range

Table 7-4 — Fixed Date Range

IHO Definition: An active period of a single fixed event or occurrence, as the date range between discrete start and end dates.

Sub-attributes:

- **Date End** (see Clause 6.99)
- **Date Start** (see Clause 6.98)

Remarks: Dates must be encoded in the format YYYYMMDD; using 4 digits for the calendar year (YYYY) and, optionally, 2 digits for the month (MM) (for example April = 04) and 2 digits for the day (DD). When no specific month and/or day is required/known, the values are replaced with dashes (-). The date range of a recurring event or occurrence must be encoded using periodicDateRange.

7.5 Light Sector

Table 7-5 — Light Sector

IHO Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference.

Sub-attributes:

- **Colour** (see Clause 6.96)
- **Directional Character** (see Clause 7.2)
- **Light Visibility** (see Clause 6.69)
- **Sector Limit** (see Clause 7.13)
- **Value of Nominal Range** (see Clause 6.65)
- **Sector Information** (see Clause 7.12)
- **Sector Extension** (see Clause 6.32)

Remarks: No remarks.

7.6 Multiplicity of Features

Table 7-6 — Multiplicity of Features

IHO Definition: The number of features of identical character that exist as a colocated group.

Sub-attributes:

- **Multiplicity Known** (see Clause 6.31)
- **Number of Features** (see Clause 6.30)

Remarks: No remarks.

7.7 Orientation

Table 7-7 — Orientation

IHO Definition:

1) The angular distance measured from true north to the major axis of the feature.
2) In ECDIS, the mode in which information on the ECDIS is being presented. Typical modes include: north-up — as shown on a nautical chart, north is at the top of the display; Ships head-up — based on the actual heading of the ship, (e.g. Ships gyrocompass); course-up display — based on the course or route being taken.

Sub-attributes:

- **Orientation Uncertainty** (see Clause 6.131)
- **Orientation Value** (see Clause 6.54)

Remarks: No remarks.

7.8 Periodic Date Range

Table 7-8 — Periodic Date Range

IHO Definition: The active period of a recurring event or occurrence. Sub-attributes: • **Date End** (see Clause 6.99) • **Date Start** (see Clause 6.98) Remarks: No remarks.

7.9 Radar Wave Length

Table 7-9 — Radar Wave Length

IHO Definition: The distance between two successive peaks (or other points of identical phase) on an electromagnetic wave in the radar band of the electromagnetic spectrum. Sub-attributes: • **Radar Band** (see Clause 6.29) • **Wave Length Value** (see Clause 6.28) Remarks: No remarks.

7.10 Rhythm of Light

Table 7-10 — Rhythm of Light

IHO Definition: The sequence of times occupied by intervals of light/sound and eclipse/silence for all light characteristics or sound signals. Sub-attributes: • **Light Characteristic** (see Clause 6.46) • **Signal Group** (see Clause 6.62) • **Signal Period** (see Clause 6.60) • **Signal Sequence** (see Clause 7.17) Remarks: No remarks.

7.11 Sector Characteristics

Table 7-11 — Sector Characteristics

IHO Definition: Describes the characteristics of a light sector. Sub-attributes: • **Light Characteristic** (see Clause 6.46) • **Light Sector** (see Clause 7.5) • **Signal Group** (see Clause 6.62) • **Signal Period** (see Clause 6.60) • **Signal Sequence** (see Clause 7.17) Remarks: No remarks.

7.12 Sector Information

Table 7-12 — Sector Information

IHO Definition: Additional textual information about a light sector. Sub-attributes:

• **Language** (see Clause 6.123) • **Text** (see Clause 6.122) Remarks: No remarks.

7.13 Sector Limit

Table 7-13 — Sector Limit

IHO Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. The sector limit specifies the limits of the sector In a clockwise direction around the central feature (for example a light). Sub-attributes: • **Sector Limit One** (see Clause 7.14) • **Sector Limit Two** (see Clause 7.15) Remarks: No remarks.

7.14 Sector Limit One

Table 7-14 — Sector Limit One

IHO Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector limit one specifies the first limit of the sector. The order of sector limit one and sector limit two is clockwise around the central feature (for example a light). Sub-attributes: • **Sector Bearing** (see Clause 6.27) • **Sector Line Length** (see Clause 6.26) Remarks: No remarks.

7.15 Sector Limit Two

Table 7-15 — Sector Limit Two

IHO Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector limit two specifies the second limit of the sector. The order of sector limit one and sector limit two is clockwise around the central feature (for example a light). Sub-attributes: • **Sector Bearing** (see Clause 6.27) • **Sector Line Length** (see Clause 6.26) Remarks: No remarks.

7.16 Shape Information

Table 7-16 — Shape Information

IHO Definition: Textual information about the shape of a non-standard topmark. Sub-attributes: • **Language** (see Clause 6.123) • **Text** (see Clause 6.122) Remarks: No formatting of text is possible within shape information. If formatted text is required, then an associated text file referenced by the complex attribute textual description must be used.

7.17 Signal Sequence

Table 7-17 — Signal Sequence

IHO Definition: The sequence of times occupied by intervals of light and eclipse for all light characteristics.

Sub-attributes:

- **Signal Duration** (see Clause 6.25)
- **Signal Status** (see Clause 6.24)

Remarks: No remarks.

7.18 Spatial Accuracy

Table 7-18 — Spatial Accuracy

IHO Definition: Provides an indication of the vertical and horizontal positional uncertainty of bathymetric data, optionally within a specified date range.

Sub-attributes:

- **Fixed Date Range** (see Clause 7.4)
- **Horizontal Position Uncertainty** (see Clause 7.24)
- **Vertical Uncertainty** (see Clause 7.27)

Remarks: No remarks.

7.19 Cable Dimensions

Table 7-19 — Cable Dimensions

IHO Definition: The dimensions of a cable to give its length and diameter.

Sub-attributes:

- **Cable Length** (see Clause 6.9)
- **Height Length Units** (see Clause 6.20)
- **Diameter** (see Clause 6.10)

Remarks:

7.20 Change Details

Table 7-20 — Change Details

IHO Definition: -

Sub-attributes:

- **Aton Commissioning** (see Clause 6.135)
- **Aton Removal** (see Clause 6.136)
- **Aton Replacement** (see Clause 6.137)
- **Fixed Aton Change** (see Clause 6.138)
- **Floating Aton Change** (see Clause 6.139)
- **Audible Signal Aton Change** (see Clause 6.140)
- **Lighted Aton Change** (see Clause 6.141)
- **Electronic Aton Change** (see Clause 6.142)

Remarks:

7.21 Obscured Sector

Table 7-21 — Obscured Sector

IHO Definition: -

Sub-attributes:
- **Sector Limit** (see Clause 7.13)
- **Sector Information** (see Clause 7.12)

Remarks:

7.22 Sinker Dimensions

Table 7-22 — Sinker Dimensions

IHO Definition: The dimensions of a sinker/anchor to give its three dimensional shape measurements.

Sub-attributes:
- **Height Length Units** (see Clause 6.20)
- **Horizontal Length** (see Clause 6.42)
- **Horizontal Width** (see Clause 6.41)
- **Vertical Length** (see Clause 6.88)

Remarks:

7.23 Positioning Method

Table 7-23 — Positioning Method

IHO Definition: A description of the method used to obtain a position.(proposed by CCG)

Sub-attributes:
- **Positioning Equipment** (see Clause 6.143)
- **NMEAString** (see Clause 6.45)

Remarks:

7.24 Horizontal Position Uncertainty

Table 7-24 — Horizontal Position Uncertainty

IHO Definition: The best estimate of the accuracy of a position.

Sub-attributes:
- **Uncertainty Fixed** (see Clause 6.128)
- **Uncertainty Variable Factor** (see Clause 6.127)

Remarks: The expected input is the maximum of the two-dimensional error. The error is assumed to be positive and negative.

7.25 Information

Table 7-25 — Information

IHO Definition: Textual information about the feature. The information may be provided as a string of text or as a file name of a single external text file that contains the text.

Sub-attributes:
- **File Locator** (see Clause 6.126)
- **File Reference** (see Clause 6.125)
- **Headline** (see Clause 6.124)
- **Language** (see Clause 6.123)
- **Text** (see Clause 6.122)

Remarks: At least one of the sub-attributes file reference or text must be populated.The sub-attribute file reference is generally used for long text strings or those that require formatting, however, there is no restriction on the type of text (except for lexical level) that can be held in files referenced by sub-attribute file reference.

7.26 Textual Description

Table 7-26 — Textual Description

IHO Definition: Encodes the file name of a single external text file that contains the text in a defined language, which provides additional textual information that cannot be provided using other allowable attributes for the feature. Sub-attributes: • **File Reference** (see Clause 6.125) • **Language** (see Clause 6.123) Remarks: No remarks.

7.27 Vertical Uncertainty

Table 7-27 — Vertical Uncertainty

IHO Definition: The best estimate of the vertical accuracy of depths, heights, vertical distances and vertical clearances. Sub-attributes: • **Uncertainty Fixed** (see Clause 6.128) • **Uncertainty Variable Factor** (see Clause 6.127) Remarks: Encodes the vertical uncertainty associated with any vertical measurement.

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